

THE
HAND-BOOK
OF
ARTILLERY.

BY CAPT. JOSEPH ROBERTS,
Fourth Regt. Artillery U. S. Army.

RICHMOND:
PRINTED BY RITCHIE & DUNNAVANT.
1861.

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*** START OF THE PROJECT GUTENBERG EBOOK THE HAND-
BOOK OF ARTILLERY ***

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PROCEEDINGS.

The following Report was made by the Committee appointed at a meeting of the staff of the Artillery School at Fort Monroe, Va., to whom the commanding officer of the School had referred this work:

Your Committee to which has been referred the consideration of the work of Captain Roberts, proposed as a text-book for the Artillery School, beg leave to submit the following Report, viz:

The work submitted by Captain Roberts, and entitled “Hand-book of Artillery,” embraces sections on the following subjects.

[For subjects see [Table of Contents, page 7.](#)]

Under each of these heads, except the last, the work contains a number of questions and answers. Your Committee have carefully examined each of these questions and their corresponding answers, and find that the answers have been principally drawn from the following sources, viz: Gibbon’s Artillerist’s Manual, Light and Heavy Artillery Tactics, and the Ordnance Manual, all of which works have been authorized by the War Department. Wherever the prescribed authorities furnish the means of answering the questions, they appear to have been followed as closely as possible.

The idea of the arrangement, and a few of the questions and answers, appear to have been taken from “Burns’ Questions and Answers on Artillery;” but that work has been so far deviated from, as fairly to entitle the present work to be considered as an original compilation.

In the opinion of your Committee, the arrangement of the subjects and the selection of the several questions and answers have been judicious. The work is one which may be advantageously used for reference by the officers, and is admirably adapted to the instruction of non-commissioned officers and privates of Artillery.

Your Committee do therefore recommend that it be substituted as a text-book in place of “Burns’ Questions and Answers on Artillery.”

(Signed) I. VOGDES,
CAPT. 1ST ART'Y.

(Signed) E. O. C. ORD,
CAPT. 3D ART'Y.

(Signed) J. A. HASKIN,
BVT. MAJ. AND CAPT. 1ST ART'Y.

The preceding Report was adopted, and the Staff recommended this work as a book of instruction at the Artillery School, in lieu of "Burns' Questions and Answers on Artillery."

PREFACE.

The following compilation originated in an attempt to adapt Lieut. Col. Burns' "Questions and Answers on Artillery" to the United States service. The British Artillery being very different from ours, it was found necessary to omit many of Burns' questions, and to introduce others.

The compiler is under great obligations to several of his brother officers at Fort Monroe (especially to Major Haskin, 1st Artillery), for their kindness in assisting him in the compilation of this little volume, and for important suggestions in the revision of many of the "answers."

Fort Monroe, Va., 1860.

CONTENTS.

The pages refer to the sidenotes in the margin of the text.

	<i>Page.</i>
Preface,	<u>5</u>
General Table of Contents,	<u>7</u>
Part I., Section 1. On Artillery in General,	<u>9</u>
“ “ 2. On Guns,	<u>26</u>
“ “ 3. On Howitzers,	<u>28</u>
“ “ 4. On Columbiads,	<u>30</u>
“ “ 5. On Mortars,	<u>31</u>
“ “ 6. On Sea-coast Artillery,	<u>35</u>
“ “ 7. Siege Artillery,	<u>36</u>
“ “ 8. Field Guns and Field Batteries,	<u>40</u>
Part II. “ 1. Pointing Guns and Howitzers,	<u>51</u>
“ “ 2. Pointing Mortars,	<u>56</u>
Part III. Charges,	<u>60</u>
IV. Ranges,	<u>63</u>
V. Ricochet,	<u>73</u>
VI. Recoil,	<u>77</u>
VII. Windage,	<u>80</u>
VIII. Gunpowder,	<u>83</u>
IX. Projectiles,	<u>90</u>
X. Laboratory Stores,	<u>109</u>
XI. Platforms,	<u>118</u>
XII. Artillery Carriages and Machines,	<u>123</u>
XIII. Practical Gunnery,	<u>147</u>
XIV. Miscellaneous,	<u>151</u>
Appendix—Rifle Cannon,	<u>163</u>
Index,	<u>169</u>

**THE HAND-BOOK OF
ARTILLERY.**

PART I. SECTION I. ARTILLERY IN GENERAL.

1. What is understood by the term *Artillery*?

9]

Heavy fire-arms of every description.

2. How many kinds of Artillery are employed in the land service of the United States?

Four, viz.: Guns, Howitzers, Columbiads, and Mortars.

3. How are these pieces distinguished?

According to their use as Sea-coast, Garrison, Siege, and Field Artillery.

4. What metals are used in the construction of Artillery?

All heavy artillery, such as that for sea-coast, siege, and garrison equipment, is made of iron; and that for field service, of bronze.

5. What is bronze for cannon?

10]

An ALLOY consisting of 90 parts of copper and 10 of tin, allowing a variation of one part of tin more or less. It is commonly called *brass*.

6. Why is bronze used in preference to iron, for field pieces?

This metal, having greater tenacity and strength than iron, the pieces can be made lighter.

7. In what respect does iron merit a preference?

Iron is less expensive than bronze, and is more capable of sustaining long-continued firing with larger charges; such pieces are, therefore, better calculated for the constant heavy firing of sieges.

NOTE.—In the sieges in Spain, bronze guns could never support a heavier fire than 120 rounds in twenty-four hours, and were never used to batter at distances exceeding 300 yards; whereas, with iron guns, three times that number of rounds were fired with effect, from three times the distance,

for several consecutive days, without any other injury than the enlargement of their vents. The comparative power of conducting heat in iron and copper being respectively as 3.743 to 8.932, taking gold at 10.000, it is evident that in practicing with iron and bronze pieces of the same calibre, it would soon become necessary to reduce the charges in the bronze pieces, and, also to increase the time between the discharges, to prevent their softening and drooping; while with iron, full charges and rapid firing may be kept up.

8. What additional objection has been urged to bronze for cannon?

The difficulty of forming a perfect alloy, in consequence of the difference of fusibility of tin and copper.

9. What iron pieces are used in the land service?

12, 13, and 24-pdr. siege and garrison guns, 32 and 42-pdr. sea-coast guns, 8-in. siege and 24-pdr. garrison howitzers, 8 and 10-in. sea-coast howitzers, 8 and 10-in. columbiads, 8 and 10-in. siege, and 10 and 13-in. sea-coast mortars. 11]

NOTE.—The 24-pdr. eprouvette is also of iron, and used for the proof of powder.

10. What are the kinds of bronze pieces in use at present?

6 and 12-pdr. field guns; 12-pdr. mountain howitzer; 12, 24 and 32-pdr. field howitzers; stone and 24-pdr. Coehorn mortars.

11. What is a battery?

This term is applied to one or more pieces, or to the place where they are served.

12. What regulate the dimensions of cannon?

The tenacity and elasticity of the metals employed in their fabrication. Their thickness must be proportioned to the effect developed by the powder; and the length is determined by experiment, and should not exceed 24 calibres. The exterior surface of a cannon is composed of several surfaces, more or less inclined to the axis of the bore, the forms of which have been determined by experiment.

13. Why is a piece made stronger near the breech than towards the muzzle?

Because the elastic force of the inflamed gunpowder is there greatest, constantly diminishing in intensity as the space increases in which it acts.

14. What is the length of a piece?

The distance from the rear of the base-ring to the face of the piece.

15. What is the extreme length?

12]

From the rear of the cascable to the face.

16. What is the *bore* of a piece?

It includes the part bored out, viz: the cylinder, the chamber (if there is one), and the conical or spherical surface connecting them.

17. What is understood by the *calibre* of a piece?

The diameter of the bore.

18. How do you ascertain the number of calibres in a piece?

Divide the length of the cylinder, in inches, by the number of inches in the calibre.

19. The number of calibres being known, how do you find the length of the cylinder?

Multiply the number of calibres by the calibre in inches.

20. What is meant by the *sights* of a piece?

Artificial marks on the piece for determining the line of fire.

21. How are the sights determined?

Usually by means of the gunner's level, when the trunnions are perfectly horizontal.

22. What is the *line of metal* or the natural line of sight?

It is a line drawn from the highest point of the base-ring to the highest point on the swell of the muzzle.

23. What is the *axis* of a piece?

An imaginary line passing through the centre of the bore.

24. What is the *natural angle of sight*?

It is the angle which the natural line of sight makes with the axis of the piece.

25. What is the *dispart* of a piece?

13]

It is the difference of the semi-diameter of the base-ring and the swell of the muzzle, or the muzzle-band. It is, therefore, the tangent of the natural angle of sight to a radius equal to the distance from the rear of the base-ring to the highest point of the swell of the muzzle, or the front of the muzzle-band, as the case may be, measured parallel to the axis.

26. Give the nomenclature of a piece?

The *CASCABLE* is the part of the gun in rear of the base-ring, and is composed generally of the *knob*, the *neck*, the *fillet*, and the *base of the breech*.

The *BASE OF THE BREECH* is a frustum of a cone, or a spherical segment in rear of the breech.

The *BASE-RING* is a projecting band of metal adjoining the base of the breech, and connected with the body of the gun by a concave moulding.

The *BREECH* is the mass of solid metal behind the bottom of the bore, extending to the rear of the base-ring.

The *REINFORCE* is the thickest part of the body of the gun, in front of the breech; if there be more than one *reinforce*, that which is next the breech is called the *first reinforce*; the other the *second reinforce*.

The *REINFORCE BAND* is at the junction of the first and second reinforces in the heavy howitzers and columbiads.

The *CHASE* is the conical part of the gun in front of the reinforce.

The *ASTRAGAL AND FILLETS* in field guns, and the *chase ring* in other pieces, are the mouldings at the front end of the chase.

The *NECK* is the smallest part of the piece in front of the astragal or the chase ring.

14]

The SWELL OF THE MUZZLE is the largest part of gun in front of the neck. It is terminated by the *muzzle mouldings*, which in field and siege guns, consist of the *lip* and *fillet*. In sea-coast guns and heavy howitzers and columbiads, there is no fillet. In field and siege howitzers, and in mortars, a *muzzle-band* takes the place of the *swell of the muzzle*.

The FACE of the piece is the terminating plane perpendicular to the axis of the bore.

The TRUNNIONS are cylinders, the axes of which are in a line perpendicular to the axis of the bore, and in the same plane with that axis.

The RIMBASES are short cylinders uniting the trunnions with the body of the gun. The ends of the rimbases, or the *shoulders* of the trunnions, are planes perpendicular to the axis of the trunnions.

The BORE of the piece includes all the part bored out, viz.: the cylinder, the chamber (if there is one), and the conical or spherical surface connecting them.

The CHAMBER in howitzers, columbiads, and mortars, is the smallest part of the bore, and contains the charge of powder. In the howitzers and columbiads,^[1] the chamber is cylindrical; and is united with the large cylinder of the bore by a conical surface; the angles of intersection of this conical surface with the cylinders of the bore and chamber, are rounded (in profile) by arcs of circles. In the 8-inch siege howitzer, the chamber is united with the cylinder of the bore by a Spherical surface, in order that the shell may when necessary, be inserted without a sabot.

The BOTTOM OF THE BORE (to facilitate sponging) is a plane 15] perpendicular to the axis, united with the sides (in profile) by an arc of a circle the radius of which is one-fourth of the diameter of the bore at the bottom. In the columbiads, the heavy sea-coast mortars, stone mortar, and eprouvette, the bottom of the bore is hemispherical.

The MUZZLE, or mouth of the bore, is chamfered to a depth of 0.15 inch to 0.5 inch (varying with the size of the bore), in order to prevent abrasion, and to facilitate loading.

The TRUE WINDAGE is the difference between the true diameters of the bore and of the ball.

27. What is the vent?

The aperture through which fire is communicated to the charge.

28. What is to be observed in reference to the diameter of the vent?

It should be as small as the use of the priming wire and tube will allow.

29. Why?

As the velocity of the gases arising from the combustion of the powder is extremely great, a large amount escapes through the vent, which contributes nothing to the velocity of the projectile. It therefore follows, that the effect produced by a given charge will diminish as the diameter of the vent increases. Besides, on account of the increase of power in the current that escapes from them, large vents are more rapidly injured than small ones.

30. What is the diameter of the vent?

16]

0.2 of an inch in all pieces except the eprouvette, in which it is 0.1.

31. What is the position of the axis of the vent?

The axis of the vent, is in a plane passing through the axis of the bore, perpendicular to the axis of the trunnions. In guns, and in howitzers having cylindrical chambers, the vent is placed at an angle of 80° with the axis of the bore, and it enters the bore at a distance from the bottom equal to one-fourth the diameter of the bore. As this inclination renders it easy to pull the friction tube out of the vent, that of the new 12-pdr. field gun, and the new columbiads has been placed perpendicular to the axis.

32. What are the *quarter-sights* of a piece?

Divisions marked on the upper quarters of the base ring, commencing where it would be intersected by a plane parallel to the axis of the piece, and tangent to the upper surface of the trunnions.

NOTE.—Not used in our service.

33. To what use are the quarter-sights applied?

For giving elevations up to three degrees but especially for pointing a piece at a less elevation than the natural angle of sight.

34. What is a *breech-sight*?

An instrument having a graduated scale of tangents, by means of which any elevation may be given to a piece.

35. How are the divisions of the tangent scale found?

17]

By taking the length of the piece, from the rear of the base-ring to the swell of the muzzle, measured on a line parallel to the axis, and multiplying it by the natural tangent of as many degrees as may be required; and then deduct the dispart. Thus, for 5° elevation, and the gun supposed to be 5 feet, or 60 inches long, multiply .08748, which is the natural tangent of 5°, by 60; the product gives 5.2488 inches; supposing the dispart to be 1 inch, the graduating of the tangent scale will be 4.2488 inches.

36. With what pieces are breech-sights used?

Guns and howitzers.

37. What is a *pendulum hausse*?

It is a tangent scale, the graduations of which are the tangents of each quarter of a degree of elevation, to a radius equal to the distance between the muzzle-sight of the piece, and the axis of vibration of the hausse, which is one inch in rear of the base-ring. At the lower end of the scale is a brass bulb filled with lead. The *slider* which marks the divisions on the scale is of thin brass, and is clamped at any desired division on the scale by means of a screw. The scale passes through a slit in a piece of steel, with which it is connected by a screw, forming a pivot on which the scale can vibrate laterally. This piece of steel terminates in pivots, by means of which the pendulum is supported on the *seat* attached to the gun, and is at liberty to vibrate in the direction of the axis of the piece. The *seat* is of metal, and is fastened to the base of the breech by screws, so that the centres of the steel pivots of vibration shall be at a distance from the axis of the piece equal to the radius of the base-ring.

A MUZZLE-SIGHT of iron is screwed into the swell of the muzzle of guns, or into the middle of the muzzle-ring of howitzers. The height of this sight is equal to the dispart of the piece, so that a line joining the muzzle-sight and the pivot of the tangent scale is parallel to the axis of the piece.

18]

38. What is a *gunner's level*, or gunner's perpendicular?

An instrument made of sheet-brass; the lower part is cut in the form of a crescent, the points of which are made of steel; a small spirit-level is fastened to one side of the plate, parallel to the line joining the points of the crescent, and a slider is fastened to the same side of the plate, perpendicular to the axis of the level.

39. What is it used for?

To mark the points of sight on pieces.

40. What is a *plummet*?

A simple *line* and *bob* for pointing mortars.

41. What is a *gunner's quadrant*?

It is a graduated quarter of a circle of sheet-brass, attached to a brass rule 18 inches long. It has a vernier turning on a pivot, to which is attached a spirit-level. To get a required elevation, the vernier is fixed at the indicated degree, the brass rule is then inserted in the bore parallel to the axis of the piece; the gun is then elevated or depressed until the level is horizontal.

There is another graduated *quadrant* of wood, of 6 inches radius, attached to a rule 23.5 inches long. It has a *plumb-line* and *bob*, which are carried, when not in use, in a hole in the end of the rule, covered by a brass plate.

42. What is an *elevating arc*, and its use?

19]

It is an arc attached to the rear part of the cheek of a gun-carriage, having its centre in the axis of the trunnions; the arc is graduated into degrees and parts of a degree. By placing the axis of the piece horizontal, and marking the breech at any one of the divisions on the arc, any elevation or depression required will be noted by the number of degrees below or above this mark. It turns on a pivot which admits of the arc, when not in use, being placed inside the cheek to which it is attached.

43. What is the use of the knob of the cascable?

To facilitate the handling of the piece in mounting and dismounting it, and moving it when off its carriage.

44. Of what use are the trunnions of a piece?

By means of them the piece is attached to its carriage; and by being placed near the centre of gravity, it is easily elevated or depressed.

45. What are the dolphins of a piece?

Two handles placed upon the piece with their centres over the centre of gravity, by which it is mounted or dismounted.

46. Are all pieces provided with dolphins?

Only the 12-pdr. brass guns, and the 24 and 32-pdr. brass howitzers.

47. What is understood by the preponderance of a piece?

It is the excess of weight of the part in rear of the trunnions over that in front; it is measured by the weight which it is necessary to apply in the plane of the muzzle to balance the gun when suspended freely on the axis of the trunnions.

48. Why is this preponderance given?

To prevent the sudden dipping of the muzzle, in firing, and violent concussion on the carriage at the breech.

49. What is bushing a piece of artillery?

201

Inserting a piece of metal about an inch in diameter (near the bottom of the bore), through the centre of which the vent has been previously drilled. It is screwed in.

50. What kind of metal is used for bushing bronze pieces?

Pure copper always, which is not so liable to run from heat as gun metal.

51. What is the object of bushing a piece?

To prevent deterioration of the vent, or provide a new one when this has already occurred.

52. Is all new artillery bushed?

No, only bronze pieces, and iron pieces, only when repeated firing has rendered it absolutely necessary.

53. How is artillery rendered unserviceable?

- I. Drive into the vent a jagged and hardened steel spike with a soft point, or a nail without a head; break it off flush with the outer surface, and clinch the point inside by means of the rammer.
- II. Wedge a shot in the bottom of the bore by wrapping it with felt, or by means of iron wedges, using the rammer or a bar of iron to drive them in.
- III. Cause shells to burst in the bore of bronze guns.
- IV. Fire broken shot from them with large charges.
- V. Fill the piece with sand over the charge, to burst it.
- VI. Fire a piece against another, muzzle to muzzle, or the muzzle of one to the chase of the other.
- VII. Light a fire under the chase of a bronze gun, and strike on it with a sledge, to bend it.
- VIII. Break off the trunnions of iron guns; or burst them by firing them at a high elevation, with heavy charges and full of shot.

54. State how to unspike a piece.

21]

If the spike is not screwed in or clinched, and the bore is not impeded, put in a charge of powder $\frac{1}{3}$ of the weight of the shot, and ram junk wads over it; laying on the bottom of the bore a slip of wood, with a groove on the under side containing a strand of quick-match, by which fire is communicated to the charge. In a brass gun, take out some of the metal at the upper orifice of the vent, and pour sulphuric acid into the groove, and let it stand some hours before firing. If this method, several times repeated, is not successful, unscrew the vent piece if it be a brass gun; and if an iron one, drill out the spike, or drill a new vent.

55. Explain how to drive out a shot wedged in the bore.

Unscrew the vent piece, if there be one, and drive in wedges so as to start the shot forward; then ram it back again in order to seize the wedge with a hook; or pour in powder, and fire it after replacing the vent piece. In

the last resort, bore a hole in the bottom of the breech, drive out the shot, and stop the hole with a screw.

56. What is scaling a piece of artillery?

Flashing off a small quantity of powder to clean out the bore; about 1-12 of the shot's weight. The practice is discontinued.

57. How are cannon in our service marked?

22]

As follows, viz: The *number of the gun* and the *initials of the inspector's name* on the face of the muzzle,—the numbers in a separate series for each kind and calibre at each foundry; the initial letters of the *name of the founder*, and of the foundry, on the end of the right trunnion; *the year of the fabrication* on the end of the left trunnion; *the foundry number* on the end of the right rimbase, above the trunnion; *the weight of the piece in pounds* on the base of the breech; the letters U. S. on the upper surface of the piece, near the end of the reinforce.

58. What marks are used to designate condemned pieces?

Pieces rejected on inspection are marked X C on the face of the muzzle; if condemned for erroneous dimensions which cannot be remedied, add X D; if by powder proof, X P; if by water proof, X W.

59. What are the kinds of proof which artillery must undergo, before being received into the service?

- 1st. They are gauged as to their several dimensions, internal and external; as to justness and position of the bore, the chamber, vent, trunnions, &c.
- 2d. They are fired with a regulated charge of powder and shot, being afterwards searched to discover irregularities or holes produced by the firing.
- 3d. By means of engines, an endeavor is made to force water through them.
- 4th. They are examined internally, by means of light reflected from a mirror.

60. Are brass cannon liable to external injury, caused by service?

23]

They are little subject to such injury, except from the bending of the trunnions sometimes, after long service or heavy charges.

NOTE.—Recent experiments at Fort Monroe show that brass guns, when *rifled*, and fired with large charges and heavy shot, expand so much that the projectile does not take the grooves.

61. What are the causes of internal injury?

Internal injuries are caused by the action of the elastic fluids developed in the combustion of the powder, or by the action of the shot in passing out of the bore.

62. Name the injuries of the first kind.

Enlargement of the bore by the compression of the metal; *corrosion of the metal* at the inner orifice of the vent, or at the mouth of the cylindrical chamber; *cracks*, from the yielding of the cohesion of the metal; *cavities*, cracks enlarged by the action of the gas, and by the melting of the metal, observable especially in the upper surface of the bore.

63. Name those of the second kind.

The *lodgment of the shot*,—a compression of the metal on the lower side of the bore, at the seat of the shot, which is caused by the pressure of the gas in escaping over the top of the shot. There is a corresponding *burr* in front of the lodgment; and the motion thereby given to the shot causes it to strike alternately on the top and bottom of the bore, producing other *enlargements*, generally *three* in number: the first, on the upper side a little in advance of the trunnions; the second, on the lower side about the astragal; the third, in the upper part of the muzzle; it is chiefly from this cause that brass guns become unserviceable. *Scratches*, caused by the fragments of a broken shot, or the roughness of an imperfect one; *enlargement of the muzzle* by the striking of the shot in leaving the bore; *external cracks*, or longitudinal slits, caused by too great a compression of the metal on the inside.

64. When is a piece said to be honeycombed?

When the surface of the bore is full of small holes or cavities.

65. To what is this due?

To the melting and volatilization of a portion of the tin in the alloy; tin being much more fusible than copper.

66. Do *lodgments* cause an inaccuracy of fire?

They do.

67. How may this in a measure be remedied?

By using a wad over the cartridge, in order to change the place of the shot; or by wrapping the shot in woollen cloth or paper, so as to diminish the windage. In field guns, the paper cap which is taken off the cartridge should always be put over the shot.

68. To what injuries are iron cannon subject?

To the above defects in a less degree than brass, except the corrosion of the metal, by which the vent is rendered unserviceable from enlargement. The principal cause of injury to iron cannon is the rusting of the metal, producing a roughness and enlargement of the bore, and an increase of any cavities or honeycombs which may exist in the metal.

69. How may you judge of the service of an iron gun?

Generally by the appearance of the vent.

70. What rules are laid down for the preservation of artillery?

25]

Cannon should be placed together, according to kind and calibre, on skids of stone, iron, or wood, laid on hard ground well rammed and covered with a layer of cinders or of some other material to prevent vegetation. In case of *guns and long howitzers*, the pieces should rest on the skids in front of the base ring and in rear of the astragal, the axis inclined at an angle of 4° or 5° with the horizon, the muzzle lowest, the trunnions touching each other; or the trunnion of one piece may rest on the adjoining piece, so that the axis of the trunnions may be inclined about 45° to the horizon; the vent down, stopped with a greased wooden plug, or with putty or tallow. The pieces may be piled in two tiers, with skids placed between them exactly over those which rest on the ground; the muzzles of both tiers in the same direction and their axes preserving the same inclination. In case of *short howitzers and mortars*, the pieces should stand on their muzzles, resting on thick planks, the trunnions touching, the vents stopped.

71. What additional precautions should be observed in case of iron pieces?

They should be covered on the exterior with a lacker impervious to water; the bore and the vent should be greased with a mixture of *oil* and *tallow*, or of *tallow* and *beeswax* melted together and boiled to expel the water. The lacker should be renewed as often as necessary, and the grease at least once a year. The lacker and grease should be applied in hot weather. The cannon should be frequently inspected, to see that moisture does not collect in the bore.

PART I. SECTION II.

ON GUNS.

1. What are *Guns*?

Long cannon without chambers.

2. How are guns denominated?

By the weight of their respective shot.

3. What are the principal parts of a gun?

The cascable, breech, reinforce, chase, and muzzle.

4. What proportion usually exists between the length and calibre of a gun?

It varies from 15 to 23 calibres.

5. What proportion does the dispart of a gun bear generally to its length?

About a sixtieth part in field guns, about a thirtieth part in sea-coast, and about a thirty-eighth part in siege and garrison guns.

6. What is the natural angle of sight in siege and garrison guns?

One degree and thirty minutes.

7. What is it in field guns?

One degree.

8. Why have sea-coast guns no natural line of sight?

Because the swell of the muzzle is not visible when the eye is on a level with the base ring.

9. Upon what are guns mounted?

On field, siege, barbette or casemate carriages.

10. What projectiles are used with guns?

Solid shot, spherical case, grape, and canister.

11. About what are the weights of the different guns?

6-pdr., 884 lbs.; brass 12-pdr., 1,757 lbs.; iron 12-pdr., 3,590 lbs.; 18-pdr., 4,913 lbs.; 24-pdr., 5,790 lbs.; 32-pdr., 7,200 lbs.; 42-pdr., 8,465 lbs.

12. Give the entire length of the several guns?

6-pdr. field-gun, 65.6 inches; 12-pdr. field-gun, 85 inches; 12-pdr. iron gun, 116 inches; 18-pdr., 123.25 inches; 24-pdr., 124 inches; 32-pdr., 125.2 inches; 42-pdr., 129 inches.

PART I. SECTION III. ON HOWITZERS.

1. What is a *Howitzer*?

A chambered piece, of larger calibre than a gun of like weight, and mounted in a similar manner.

2. What form of chamber is given to howitzers?

That of a cylinder.

3. How is it united with the large cylinder of the bore?

By a conical surface, except in the 8-inch siege howitzer, where it is united with the cylinder of the bore by a *spherical* surface, in order that the shell may—when necessary—be inserted without a sabot.

4. What advantages are gained by the employment of howitzers?

They project larger shells than the guns with which they are associated, are well adapted for ricochet fire, the destruction of field works, breaking down palisades, and setting fire to buildings.

5. What projectiles are used with howitzers?

Shells usually, spherical case, canister, grape and carcasses.

6. Give the entire length of the several howitzers.

29]

Iron 10-inch, 124.25 inches; 8-inch sea-coast, 109 inches; 8-inch siege and garrison, 61.5 inches; 24-pdr. garrison, 69 inches; 32-pdr. field, 82 inches; 24-pdr. field, 71.2 inches; 12-pdr. field, 58.6 inches; mountain, 12-pdr., 37.21 inches.

7. What is the weight of a howitzer of each kind?

10-inch, 9,500 lbs.; 8-inch sea-coast, 5,740 lbs.; 8-inch siege and garrison, 2,614 lbs.; 24-pdr. garrison, 1,476 lbs.;

32-pdr. field, 1,920 lbs.; 24-pdr. field, 1,318 lbs.; 12-pdr. field, 788 lbs.; 12 pdr. mountain, 220 lbs.

8. What is the natural angle of sight in siege and garrison and field howitzers?

One degree.

9. What in mountain howitzers?

Thirty-seven minutes.

10. Why have sea-coast howitzers no natural line of sight?

Because the swell of the muzzle is not visible when the eye is on a level with the base ring.

PART I. SECTION IV.

ON COLUMBIADS.

1. What is a *Columbiad*?

A gun of much larger calibre than the ordinary gun, used for throwing solid shot or shells.

2. What are some of the peculiarities of this gun, when mounted in barbette?

Its carriage gives a vertical field of fire from 5° depression to 39° elevation; and a horizontal field of fire of 360°.

3. Are these pieces chambered?

Those of the old pattern have chambers; but they are now cast without any.

4. Give the weight of this piece?

10-inch, 15,400 lbs.; 8-inch, 9,240 lbs.

5. What is the entire length of this gun?

10-inch, 126 inches; 8-inch, 124 inches.

6. What is the natural angle of sight in this piece?

8-inch, 1° 23'; 10-inch, 1° 21'.

PART I. SECTION V. ON MORTARS.

1. What is a *Mortar*?

The shortest piece in service; the trunnions are placed in rear of the vent at the breech; the bore is very large in proportion to the length, and is provided with a chamber.

2. What are the principal advantages obtained by the employment of mortars?

Reaching objects by their vertical fire—such as a town, battery, or other place—whose destruction or injury cannot be effected by direct or ricochet fire; dismounting the enemy's artillery; setting fire to and overthrowing works; blowing up magazines; breaking through the roofs of barracks, casemates, &c.; and producing havoc and disorder amongst troops.

3. What do you mean by vertical fire?

That produced by firing the mortar at a high elevation.

4. What are its advantages?

The shell having attained a great elevation, descends with great force on the object, in consequence of the constant action of the force of gravity on it.

5. Why are mortars constructed stronger and shorter than other pieces? 321

Because greater resistance is required in consequence of the high elevation under which they are fired; and were they longer, the difficulty experienced in loading them would become too great.

6. Why is a mortar constructed with a chamber?

In consequence of employing various charges, some very small, it becomes necessary to use a chamber to concentrate the charge as much as

possible, so that the shell may be acted on by the entire expansive force of the powder.

7. What form of chamber is given to mortars?

Usually that of a frustum of a cone. The bottom is hemispherical in the sea-coast, stone, and eprouvette mortars. In siege mortars it is a plane surface, the angles of intersection being rounded in profile by arcs of circles.

8. What is this form of chamber called?

Gomer Chamber.

9. What is the advantage of the conical over the cylindrical chamber?

Cylindrical chambers are objectionable, as the projectile is frequently broken in consequence of the small surface exposed to the action of the charge. This defect is obviated by large chambers, and particularly by those that are conical, in which the charge is expended upon nearly a hemisphere.

10. What form of chamber has the eprouvette?

That of a cylinder, it being the only mortar whose chamber is of this shape.

11. How are mortars mounted?

On beds of wood or iron.

12. What is the object of mounting mortars on beds in preference to wheel carriages? 33]

On account of the high elevation at which they are usually fired, when the recoil, instead of forcing the piece backwards, tends to force it downwards, and this tendency becomes so great at the higher angles that no wheel carriage could long sustain the shock.

13. What is the entire length of each mortar?

13-inch, 53 inches; 10-inch sea-coast, 46 inches; 10-inch siege, 28 inches; 8-inch, 32.5 inches; stone mortar, 31.55 inches; coehorn, 16.32 inches.

14. What are the weights of mortars?

13-in., 11,500 lbs.; 10-in. sea-coast, 5,775 lbs.; 10-in. siege, 1,852 lbs.; 8-in., 930 lbs.; stone mortar, 1,500 lbs.; coehorn, 164 lbs.; eprouvette, 220 lbs.

15. What are the weights of the different mortar beds?

8-in. siege, 920 lbs.; 10-in. siege, 1,830 lbs.; coehorn, 132 lbs.; eprouvette, 280 lbs.

16. What are the diameters of the bores of the stone, coehorn, and eprouvette mortars?

Stone mortar, 16 inches; coehorn, 5.82 inches; eprouvette, 5.655 inches.

17. What is the length of the bore, exclusive of the chamber, of the different mortars?

13-in., 26 inches; 10-in. sea-coast, 25 inches; 10-in. siege, 15 inches; 8-in., 12 inches; stone mortar, 19.8 inches; coehorn, 8.82 inches; eprouvette, 11.5 inches.

18. What is the length of the chamber of the different mortars?

13-in., 13 inches; 10-in. sea-coast, 10 inches; 10-in. siege, 5 inches; 8-in., 4 inches; stone mortar, 6.75 inches; coehorn, 4.25 inches; eprouvette, 1.35 inches.

19. For what purpose is the eprouvette used?

34]

For determining the relative strength of gunpowder.

20. To what purpose is a stone mortar applied?

To throw stones a short distance, from 150 to 250 yards; and also 6-pr. shells from 50 to 150 yards.

21. In what manner are the stones disposed in this mortar?

They are put into a basket fitted to the bore, and placed on a wooden bottom which covers the mouth of the chamber.

22. What use is made of coehorn mortars?

They are fired either from behind intrenchments like other mortars, or they may accompany troops in effecting lodgments in towns and fortified places.

23. What kind of projectiles are thrown from mortars?

Shells, fire-balls, carcasses, and stones.

24. How rapidly may siege mortars be fired?

At the rate of twelve rounds per hour continuously; and in case of need with greater rapidity.

PART I. SECTION VI.

SEA-COAST ARTILLERY.

1. How are *Sea-Coast* pieces mounted?

On barbette, casemate, flank-casemate, and columbiad carriages; and the carriage upon which the mortar is mounted—called its *bed*. These carriages do not subserve the purpose of transportation; the barbette carriage may, however, be used for moving its piece for short distances, as from one front of the work to another.

2. What number and kind of pieces are required for the armaments of forts on the sea-board?

In our service they are prescribed by the War Department, according to the character and extent of the work.

3. What disposition should be made of heavy and light pieces in a fortification?

Heavy pieces should be employed on the salients of the work, or for enfilading channels where a long range is required; light pieces, where the range is shorter.

PART I. SECTION VII.

SIEGE ARTILLERY.

1. How are siege-guns mounted?

Usually on travelling-carriages, with limbers.

2. Of what number and kind of pieces is a siege train composed?

This must altogether depend on circumstances; but the following general principles may be observed in assigning the proportion of different kinds and calibres, and the relative quantity of other supplies for a train of 100 pieces:

GUNS.	{	24-pdr., about one-half the whole number,	50
		18-pdr. or 12-pdr., one-tenth,	10
HOWITZERS.		8-in. siege, one-fourth,	25
MORTARS.	{	10-in. siege, one-eighth,	12
		8-in. siege,	3
STONE-MORTARS,	{	in addition to the 100 pieces,	6
COEHORN MORTARS,			6
WALL PIECES.			40

CARRIAGES.

37]

For 24-pdr. guns, and 8-in. howitzers, one-fifth spare,	90
For 18-pdr. and 12-pdr. guns,	12
For 10-in. mortars and stone-mortars, one-sixth spare,	21
For 8-in. mortars,	4
<i>Mortar-wagons</i> , 1 for each 10-in. mortar and bed, for each stone mortar and bed, and for three 8-in. mortar and beds,	38
<i>Wagons</i> for transporting implements, intrenching and miner's tools, laboratory tools and utensils and other stores, each loaded with about 2,700 lbs., say,	140
<i>Carts</i> (carrying balls, &c. on the march),	50
<i>Park battery-wagons</i> , fully equipped,	28
<i>Park forges</i> , " "	8
<i>Sling carts</i> , large,	5
<i>Do.</i> hand,	4

DRAUGHT HORSES.

For each gun and howitzer, with its carriage,	8
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“	Spare gun-carriage,	6
“	Mortar wagon,	8
“	Battery wagon,	6
“	Forge,	6
“	Cart,	2
“	Sling-cart, large,	2
“	Spare horses,	<u>1-10th</u>
Total, about		1,900 horses.

PROJECTILES AND AMMUNITION.

38]

FOR GUNS.	{	Round-shot, 800 to each 24-pdr., 1,000 to each 18 and 12 pdr.
		Grape and canisters strapped, 20 rounds to each piece.
		Spherical case strapped, 20 rounds to each piece.
FOR HOWITZERS.	{	Shells, 800 to each 8-in. howitzer.
		Canisters strapped, 5 to each.
		Spher. case strapped, 20 to each.
FOR MORTARS.	{	600 shells to each 10-inch.
		860 “ “ 8-inch.
		200 “ “ Coehorn.

Gunpowder, in barrels, 500,000 lbs.

Computing for each 24-pdr. round-shot, one-third the weight of shot.

“	“	18 and 12 pdr. round-shot, one-fourth the weight of shot.	
“	“	grape, canister and spherical case, one-sixth the weight of shot.	
“	“	round of howitzer ammunition,	} including charge of shell.
“	“	round 10-in. mortar ammunition,	
“	“	round 8-in. mortar ammunition,	
“	“	round Coehorn mortar ammunition, ½ lb.	
“	“	round stone mortar ammunition, 1 lb.	

3. What is the best position for guns in order to make a breach?

On the glacis, within 15 or 16 feet of its crest; but if the foot of the revetment cannot be seen from thence, the guns must be placed in the covered way, within 15 feet of the counterscarp.

4. In what manner should the fire of siege guns be conducted in order to form a breach?

39]

1st. Make a horizontal section the length of the desired breach along the scarp, at one-third its height from the bottom of the ditch, and to a depth equal to the thickness of the wall.

2d. Make vertical cuts through the wall, not farther than ten yards apart, and not exceeding one to each piece of ordnance, beginning at the horizontal section and

ascending gradually to the top of the wall.

3d. Fire at the most prominent parts of the masonry left standing; beginning always at the bottom and gradually approaching the top.

4th. Fire into the broken mass with howitzers until the breach is practicable.

5. How long would it take to make a breach of 20 yards in length?

Breaches of more than 20 yards in length have been opened by way of experiment, and rendered practicable in less than ten hours, by about two hundred and thirty 24-pdr. balls and forty shells in one case, and by three hundred 18-pdr. balls and forty shells in another.

6. How many discharges can an iron gun sustain?

An iron gun should sustain twelve hundred discharges, at the rate of twelve an hour; but whatever may be the rate of fire, it is deemed unsafe after that number of discharges. As many as twenty an hour have been made for sixteen consecutive hours.

PART I. SECTION VIII.

ON FIELD GUNS AND BATTERIES.

1. What proportion of artillery should be allotted to an army in the field?

The proportion of artillery to other troops varies generally between the limits of one and three pieces to 1,000 men, according to the strength of the army, the character of the troops composing it, the strength and character of the enemy, the nature of the country which is to be the theatre of the war, and the character and objects of the war.

2. What regulates the selection of the kinds of artillery and the proportion of the different kinds in the train?

Similar considerations to those specified in the foregoing answer. The following principles may be observed in ordinary cases:

3 pieces to 1,000 men.	{	$\frac{2}{3}$ guns, of which $\frac{1}{3}$ howitz., of which	{	$\frac{1}{4}$ are 12-pdrs. $\frac{3}{4}$ “ 6-pdrs. $\frac{1}{4}$ “ 24 or 32-pdrs. $\frac{1}{4}$ are 12-pdrs.
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3. What is a field-battery?

A certain number of pieces of artillery so equipped as to be available for attack or defense, and capable of accompanying cavalry or infantry in all their movements in the field.

4. How many pieces are allotted to a field-battery?

Four guns and two howitzers.

5. Are all field batteries alike?

41]

No; field batteries accompanying infantry are composed of the heavier, and those accompanying cavalry of the lighter pieces, the first manned by foot-artillery, and the latter by horse-artillery.

6. In what respect does a battery of horse-artillery differ from one of foot-artillery?

The main difference consists in the cannoneers in a battery of horse-artillery being mounted; in rapid evolutions of foot-artillery they are conveyed on the carriages.

7. What is the composition of a field-battery on the war establishment?

KIND OF BATTERY.		12-PR.	6-PR.		
GUNS.	{ 12-pdrs., mounted,	4			
	{ 6-pdrs., “		4		
HOWITZERS.	{ 24-pdrs., mounted,	2			
	{ 12-pdrs., “		2	6	6
CAISSONS.	{ For guns,	8	4		
	{ For howitzers,	4	2	12	6
TRAVELING FORGES,		1	1		
BATTERY WAGON,		1	1	2	2
Whole No. of carriages with a battery,				20	14

AMMUNITION.	{	For 4 guns,	{	Shot,	560	560	896	800
			{	Spher. case,	224	80		
			{	Canisters,	112	160		
	{	For 2 howitzers,	{	Shells,	168	120	322	312
			{	Spher. case,	112	160		
			{	Canisters,	42	32		
			Total No. rounds with a battery,				1218	1112

KIND OF BATTERY.		12-PR.	6-PR.
DRAUGHT HORSES.	{ 6 to each carriage,	120	84
	{ Spare horses, 1-12,	10	7
Total,		130	91

NOTE.—For two 32-pdr. howitzer carriages and four caissons, the number of rounds of ammunition is:

Shells,	112
Spher. case,	84
Canisters,	14
Total,	210

8. What is the composition of a battery of mountain howitzers?

Howitzers,	6
Gun-carriages,	7
Ammunition-chests,	36
(48 rounds for each howitzer.)	
Forge and tools, in 2 chests,	1
Set of carriage-makers' tools in 2 chests,	1
Pack saddles and harness,	33
Horses or mules,	33

9. What composes the *Field-Park*?

The spare carriages, reserved supplies of ammunition, tools, and materials for extensive repairs, and for making up ammunition, for the service of an army in the field from the *Field-Park*, to which should be attached also the batteries of reserve.

10. What determines the quantity of such supplies?

The quantities of such supplies must depend in a great measure on the particular circumstances of the campaign.

11. How is the ammunition which cannot be transported by the batteries carried?

With the park; in caissons, or in store wagons.

12. Do any other carriages and stores form part of the *Field-Park*?

43]

Yes; spare gun-carriages, one to each field-battery,

<i>Traveling Forges,</i>	}	one or more of each.	}	in store wagons.
<i>Battery-Wagons,</i>				
<i>Spare spokes,</i>	50 to each battery,			
<i>Spare fellies,</i>	20 to each battery,			
<i>Spare harness,</i>	}	in boxes.		
<i>Horse-shoes and nails,</i>				

Gunpowder, saltpetre, sulphur, charcoal, laboratory paper, cannon-primers (percussion and friction), fuzes and plugs for field-service, stuff for cartridge bags, woollen yarn, cotton yarn, glue.

13. Are any other pieces ever used for field service?

Yes: sometimes the 12 and 18-pdr. siege guns, and the 8-in. siege howitzer.

14. For what particular service are these different pieces most suitable?

The siege pieces for batteries of position; the 12-pdr. battery, for following the movements of infantry, and the 6-pdr. battery for those of cavalry.

NOTE.—These siege pieces should be placed on the weakest points of a line, and on heights which either form a key to the position, or from whence the greatest and longest continued effect may be produced.

15. What are the peculiar advantages of Horse-Artillery?

Possessing, from their lighter construction and mounted detachments, much greater locomotive powers than other field batteries, they are especially adapted for following the rapid evolutions of cavalry, for sudden attacks upon particular points, and for supporting the advance or covering the retreat of an army.

16. How is a field gun mounted?

44]

Upon a four-wheel carriage, which answers for its transportation as well as for its service, similar to a siege carriage, but lighter, and the limber carrying an ammunition chest.

17. Where should a battery be placed before the commencement of an action?

As much as possible under cover, by taking advantage of banks, hollow-ways, buildings, woods, &c.

18. Is it advisable to move a battery at once into position in the field?

No; but if unavoidable, it should be masked as much as possible until ordered to open its fire.

19. How should a battery be masked?

If practicable, by covering it with cavalry, in preference to infantry, as the former does it more effectually, and is sooner moved out of the way.

20. In commencing an action, how should the fire of a battery be directed?

When the enemy is in line, the fire should be directed over the whole line, and not upon the real points of attack; but when in column, ready to advance, it should be concentrated upon the real points of attack.

21. How should batteries be placed in relation to the troops with which they are acting?

Upon the flanks of a line, but at such a distance as not to impede its movements, and at the same time to be unfettered in their own; the artillery may thus represent the faces of a bastion, and the line of troops the curtain.

22. Is the front of a line of troops an advantageous position for a field battery? 45]

On the contrary, it is the worst possible, as offering a double object to the enemy's fire, and greatly obstructing the movements of the troops; while a position in rear is nearly as bad, as the fire might seriously injure, or at least, greatly disquiet them.

23. In supporting an attack, what precautions are necessary?

The battery should be carefully kept clear of the intended line of march of our own troops, and such points occupied as may afford the greatest annoyance to the enemy.

24. How should batteries be disposed with regard to the enemy's troops?

Generally so as to secure a cross fire on his position, and on all the ground over which he moves to the attack, endeavoring to take him at all times in the direction of his greatest dimensions; that is, obliquely or in flank when in line, and in front when formed in columns. Moderate heights, commanding as much as possible the surrounding country, should always be taken advantage of, but not such as may prevent operations in advance if required.

25. Is it imperatively necessary to confine positions for field batteries to the flanks of a line?

When, from particular circumstances, the front of the army is too extended, and unavoidably divided into two lines, it may become necessary to place one or more batteries in the centre, if those on the flanks are unable to sweep the whole front; but great care must be taken not to impede the advance or retreat of the troops when required.

26. Should the fire of field batteries be carried on at the same uniform rate? 46]

Certainly not; the destruction of the enemy being the object, it follows that at distant ranges, a greater degree of care is required in pointing the guns; the fire is slow and steady, and increasing in rapidity as the enemy advances, without however, impairing its precision.

27. Should the fire of field batteries be carried on in salvoes or otherwise?

Never in salvoes; but in a regular manner, well sustained, and with distinct intervals between every round, commencing slowly, and increasing in rapidity as the range diminishes.

28. Is the fire of batteries more efficacious when dispersed than when concentrated?

The effects of the fire will be in proportion to the number of guns brought together, and therefore, in order to strike a decisive blow, this should at once be done.

29. What projectiles are used with field guns?

Solid shot, spherical case, and canister.

30. At what distance from the enemy should the several kinds of projectiles be employed with field battery pieces?

Solid shot from 350 yards and upwards; spherical case from 600 up to 1000 yards, although it may be used within the first range; and canister within 350 yards, or up to 400 against extended formations.

31. What number of rounds can be fired from a field gun in one minute?

Two solid shot or spherical case, or three of canister.

32. Why are more rounds of canister fired in a minute than of solid shot or spherical case?

Because the latter are fired at greater distances than canister, and require the piece to be carefully aimed, thus requiring more time.

33. What is the smallest number of guns that may with safety be employed in the face of an enemy?

Never less than two, in order to secure a continuous fire and mutual support.

34. Is the practice of employing field batteries against those of the enemy recommended?

Only under peculiar circumstances; as for instance, when his troops are well covered and his guns exposed, or their fire very destructive.

Their fire should be directed principally against columns of attack, and masses, or upon positions which are intended to be carried.

35. In what time could a battery come into action in the field?

It could come into action and fire one round in 25 seconds, timing from the order "action front," to the discharge of one piece.

36. Suppose cavalry to be advancing to attack infantry, and first observed at the distance of a mile, passing over the first half mile at a trot; the next quarter of a mile at the manœuvring gallop; and the remaining distance at an increased gallop, terminating with the charge; occupying altogether about six minutes: during the last 1500 yards of their advance how many rounds per piece might a battery fire in that time?

Eleven rounds with effect, thus:

From 1500 to 650 yards,	3' 32"—spherical case,	7
“ 650 to 350 “	0' 48"—solid shot,	2
“ 350 to close quarters,	0' 34"—canister,	2

37. What number of rounds could a battery fire against infantry, supposing them to pass over 1500 yards in about 16¼ minutes? 48]

Thirty-six rounds with effect, viz:

From 1500 to 650	quick step,	9' 45"—spher. case,	19
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“	650 to 350,	“	3' 50 —solid shot,	7
“	350 to 100,	“	2' 30"—canister,	8
“	100 to close quarters	{ double quick and the charge. }	0' 40"—can.	2

38. Should the enemy attempt to force the passage of a river, what is the best position for artillery to oppose it?

Wherever the best cross fire can be obtained in order to obstruct and harass him as much as possible, and if he has succeeded in passing over any portion of his troops, it should be directed against their formation.

39. When the enemy is making the passage of a river in retreat, where should your guns be posted?

In such a position as to bear upon the batteries that cover the retreat, and also upon his bridges.

40. In forcing the passage of a river, what is the most advantageous position for artillery?

The bridge being generally laid in a re-entering angle, batteries should be posted on each side of the bridge, and far enough from it to secure a cross fire on the opposite bank.

41. Should the indiscriminate expenditure of ammunition be permitted in the field during action?

Upon no account; ammunition should at all times be carefully husbanded, particularly at the commencement of an action, as the want of it at the close may decide the fate of the day; it should also be sparingly used in skirmishes and minor affairs, especially when at a distance from supplies, or in anticipation of a general action.

42. When should the reserve be employed?

49]

When a particular point of the line requires additional support, a favorable position is to be seized, an impression has been made on the line by the enemy, a forward or retrograde movement is in contemplation, or when a determined attack is to be made on him, then the reserve should come up and take part in the action; and it is of the utmost importance that this should be done as expeditiously as circumstances will permit.

43. Where should the reserve be placed previous to an engagement?

In rear with the second line, out of the range of shot, and as little exposed as circumstances will admit, but always in such a position as to have ready access to the front or rear.

44. Should guns be lightly abandoned before an enemy?

Never until the VERY LAST EXTREMITY. An artilleryman must never forget that his gun is his proper ARM; that here lies his strength; that here is his post of honor and of duty; also, that the LAST DISCHARGES are always THE MOST DESTRUCTIVE, and MAY POSSIBLY INSURE THE SAFETY OF THE WHOLE ARMY, OR TURN THE TIDE OF VICTORY IN THEIR FAVOR.

45. What is the position for cavalry when placed in support of a battery?

On its flank, about the distance of 100 yards, and as much concealed as possible.

46. What is the proper position of field batteries when infantry squares are attacked by cavalry? 50]

When infantry are formed in squares to resist the charge of cavalry, the guns should be placed outside at the angles of the squares, the limbers, horses, &c., inside. Should the detachments be driven from their guns, they will retire into the square, after discharging their pieces, and taking with them the sponges and other equipments; the moment the enemy has retired, they recommence their fire. Supposing the infantry formed in echelon of regimental squares, and that the time, or small extent of the squares would not admit of the limbers, &c., being placed inside, then the wagons and limbers should be brought up with their broadsides to the front, so as to occupy, if possible, the space between the guns, leaving no intervals for the cavalry to cut through: the prolonge or drag ropes might also offer an effectual momentary impediment to them, if properly stretched and secured.

PART II. SECTION I.

POINTING GUNS AND HOWITZERS.

1. What is meant by the term *pointing* a piece?

To point a piece, is to give it such a direction and elevation, or depression, that the shot may strike the object; and the rule (except in case of mortars) is: First give the *direction* and then the *elevation*, or depression.

2. When a shot is fired from a piece, by how many forces is it acted on?

By three.—1st. The impulsive force of the powder, which urges it forward.

2d. The resistance of the air, which tends to stop it.

3d. The force of gravity which causes it to descend.

3. Why is it necessary to give a certain degree of elevation to a piece?

Because a shot describes under the action of the above forces a curve called a trajectory, which is situated below the prolongation of the axis of the piece, the extent of its departure from *this line* increasing with the time of flight. Therefore, the more distant the object, the greater must be the elevation to enable the shot to reach it.

4. How is the direction given to a gun or howitzer?

By directing the line of metal upon the object.

5. How is the elevation or depression given?

52]

The elevation or depression, which depends upon the charge, the distance, and the position of the object above or below the battery, must be ascertained from tables or by experiment, and the proper degree given by means of instruments.

6. When will the object be struck by merely directing the line of metal upon it?

But in one case,—when it is at point-blank distance.

7. How must the line of metal be directed for all ranges less than the point-blank range, in order to strike it?

So as to pass below the object.

8. Give a simple rule for firing at objects within point-blank.

Add to the point-blank range the difference between it and the required range, set the scale to the elevation corresponding to this sum, as shown by tables of firing. Then aim the gun directly at the object; now apply the scale, and observe where the visual ray of the scale strikes the ground, and having noted this point, aim the gun directly at it.

9. How must the line of metal be directed for ranges greater than the point-blank range, in order to strike it?

Above it.

10. When the line of metal passes over the object, what instruments must be employed for giving the proper elevation?

The gunner's quadrant, or the breech-sight.

11. How is the quadrant used?

53]

After the direction has been given, the quadrant is applied, either by its longer branch to the face of the piece, or this branch is run into the bore parallel with the axis, or it may be applied to the upper surface of the lock-piece, making the allowance due to its inclination with the axis of the piece, which ought to be previously determined, and the elevating screw turned, or the quoin adjusted, until the required degree is indicated.

12. How is the breech-sight used?

It is first set to the elevation corresponding to the distance; it is then applied to the highest point of metal on the base-ring, and by the elevating screw, or quoin, the notch of the breech-sight, the highest point on the swell of the muzzle, and the object, are brought in the same line.

13. What is a line thus determined called?

An artificial line of sight.

14. In the absence of instruments, how may the elevation be given?

By placing one or more fingers of the left hand upon the base-ring, perpendicular to the axis, and using them as a breech-sight.

NOTE.—In practice, it is well to fire two or three shot to determine the *range* experimentally, as *it* is affected by divers causes.

15. Should the line of metal be always directed in the vertical plane passing through the object?

No; as in practice there are circumstances (as, for instance, a strong wind blowing across the field of fire) which will cause a ball to deviate from this plane, it follows that to strike the object, in such a case, the line of metal must be directed to its right or left; the gunner judging of the distance by observing the striking of the shot.

16. Is the line of metal a permanent line under all circumstances?

54]

No; in batteries for garrison and sea-coast defense, where the platforms are fixed, the line of metal may be considered as nearly permanent; but with siege guns, which are mounted on traveling carriages, the wheels of which are liable to vary in position from unevenness of ground, or unequal settling in newly constructed platforms, this line is constantly changing. It approximates the higher wheel in proportion to the difference of level between the wheels; and hence, to secure the greatest accuracy of fire, it must be frequently verified; the old marks, if not found correct, should be erased and new ones substituted.

17. When the notches or sights, which are sometimes made upon the base-ring and swell of the muzzle in field guns, for aiming the piece are used, how is the error of direction remedied when the wheels are not on the same level?

The piece must be aimed more or less to that side which corresponds to the higher wheel, according to the inclination.

18. When the elevation or depression has once been ascertained for any given distance, how may the firing at that distance be facilitated?

By noting some point on the elevating screw or quoin; adjusting some fixed measurement from a point on the stock to another point on the under

side of the breech; or by a chalk mark drawn across the face of a trunnion and its corresponding cheek.

19. When firing either within or beyond point-blank range, may remarkable points on the ground be taken advantage of, in order to furnish an object to aim at? 55]

Yes; some fixed object may often present itself which will serve as a point upon which to direct the line of metal. No means should be neglected that may tend to secure accuracy of aim; for the shot that is thrown away by carelessness in pointing, had better not be thrown at all.

20. How may precision of fire be secured at night?

When a fixed object is to be fired at by night, the piece should be directed during the day, and two narrow and well-dressed strips of wood laid on the inside of the wheels, and two others outside of the trail of a siege carriage, and nailed or screwed to the platform. In case of a barbette carriage, the traverse-wheels should be chocked in the proper position. To preserve the elevation, measure the height of the elevating screw above its box, or take the measure between a point on the gun, and another on the stock; cut a stick to this length and adjust the gun on it at each fire.

21. Should night-firing with *guns* be limited?

Yes; it should be limited to a small number of rounds, as it consumes ammunition to little advantage.

PART II. SECTION II. POINTING MORTARS.

1. What is the rule for pointing mortars?

First give the elevation, and then the direction.

2. How is the elevation given?

By applying the quadrant to the face of the piece, and adjusting the quoin until the required number of degrees is indicated.

3. Are the same means employed for giving mortars their direction as those which are used with guns and howitzers?

No; because mortars are usually masked from the object to be struck, by an epaulment or parapet.

4. To what are all the methods employed for giving the direction to mortars reduced?

To determining practically two fixed points, which shall be in line with the piece and the object, and sufficiently near to be readily distinguished by the eye. These points being covered by the plummet, determine a vertical plane, which, when including the line of metal, becomes the plane of fire.

5. What is the simplest manner of directing the mortar?

By means of *pointing-wires*.

6. Describe this method.

57]

The two fixed points required are determined by planting two wires upon the epaulment, one upon its crest, and the other about a yard in advance of it, both as nearly as possible in the vertical plane passing through the centre of the platform and the object. The points being thus established, the direction is given to the mortar, by causing a plummet held in rear of it, to cover the wires and the line of metal.

7. In what respects is this method defective?

Both in accuracy of aim, and the liability of the wires being deranged by the shots of the enemy or by other causes.

8. Give a better method.

By means of *pointing-stakes*, by which one of the fixed points is established upon the crest of the parapet, or at the foot of the interior slope, and another in rear of the piece. Then by a cord called the *pointing-cord*, stretched between these two points, with the plummet suspended from it, a vertical plane is determined with which the line of metal is made to coincide.

9. How are the stakes planted?

A stake, a foot or more in length, is driven into the crest of the epaulment, as nearly as practicable in the vertical plane of fire passing through the centre of the platform; sighting by this stake, another long one is planted, three or four feet in front of it, in line with the object. To this stake the cord is temporarily attached, and stretched by the first stake, just grazing it, to a point on the ground, one yard in rear of the platform. At this point a third stake is driven. The cord is removed from the second stake, which may now be taken away, and permanently attached to the first.

10. How is the mortar directed?

58]

The cord is stretched to the rear stake, and as near the muzzle-band as possible, with the left hand, while the plummet is suspended against it with the right; or the plummet may be attached to the cord, just in rear of the mortar.

11. How does it appear that the mortar is thus *properly* directed?

Because the cord, the plummet, and the line of metal, are evidently in the vertical plane of fire.

12. What is done in case the shell should strike constantly to the right or left of the object?

The pointing-cord is shifted to some notch on the *pointing-board*, to the right or left, until the shell falls at the desired point.

13. Describe the pointing-board.

This is a piece of wood one foot long, two or three inches wide, and one inch thick, having a notch cut in the middle of one side, to fit on the stake and which is graduated into equal divisions from its middle. When not in use, the pointing-cord may be wound on it.

14. Describe another mode of planting the *pointing-stakes*.

59]

The mortar being placed upon the middle of the platform, the gunner mounts upon it, and suspends the plummet in front of the muzzle, covering the object. Where the plummet thus suspended cuts the crest of the epaulment, the first stake is driven. A second stake is then driven in the same line between the mortar and the epaulment. The pointing-cord being attached to the first stake and stretched to the rear, over the point where the plummet touches the top of the mortar, determines the point on the ground at which the rear stake is driven. The first stake is then removed, and the cord attached permanently to the second stake.

When the object cannot be seen from the mortar, owing to the interposition of some obstacle, as a parapet or a hill, two persons in sight of each other, one of whom can see the mortar, and the other the object, must by successive changes of position, place themselves in the vertical plane of fire, and at the points thus determined, stakes must be driven, one of which will serve as the object.

15. How may precision of fire be secured at night with mortars?

The *direction* is preserved by nailing or screwing two boards to the platform outside of the cheeks; the *elevation* is marked on the quoin, or the quoin may be nailed in the proper position.

60]

PART III. CHARGES.

1. What is the charge of a piece of artillery?

The powder with which it is loaded.

2. What is the ordinary service charge of powder for heavy guns?

One-fourth the weight of the shot.

3. What is it for firing *double shot*?

One-sixth the weight of one shot.

4. What is the breaching charge?

One-third the weight of the shot.

5. What kind of charges are used in hot shot firing?

Small charges from *one-fourth* to *one-sixth* the weight of the ball.

6. For what reason?

Because balls fired with small velocities split the wood in a manner which is favorable to its burning; with a great velocity the hole closes, the ball sinks deep, and, deprived of air, chars without setting fire to the surrounding wood.

7. To what depth should hot shot penetrate?

Not deeper than ten or twelve inches.

8. In ricochet firing, what kind of charges are used?

Light charges generally; varying from *two-thirds* to *one-eighth* of the ordinary charge.

9. In what manner are the charges of mortars regulated?

61]

The charges vary with the elevation; or if the elevation be fixed at any particular angle, they must be determined by the range.

10. What are the charges for field guns and field howitzers?

[See Table, page 46.](#)

11. What are the charges for heavy guns, columbiads, and howitzers?

[See Table, page 46.](#)

12. What are the greatest charges of the sea-coast, siege, and coehorn mortars?

[See Table, page 46.](#)

13. What charge is used for projecting fire balls from mortars?

One *twenty-fifth* the weight of the ball.

**CHARGES FOR FIELD GUNS
AND FIELD HOWITZERS.**

62]

KIND	FOR GUNS.		FOR HOWITZERS.			
	12-pr.	6-pr.	32-pr.	24-pr.	12-pr.	Mountain
	lbs.	lbs.	lbs.	lbs.	lbs.	lbs.
For shot,	2.5	1.25				
For spher. case or canister,	1.5	1.0	2.5	1.75	0.75	0.5
For shells, {	Small charge,	—	2.5	2.0	1.00	0.5
	Large charge,	—	3.25	2.50	1.00	0.5

**CHARGES FOR HEAVY GUNS, COLUMBIADS
AND HOWITZERS.**

GUNS					COLUMBIADS		HOWITZERS.			
42-pr.	32-pr.	24-pr.	18-pr.	12-pr.	10-inch.	8-inch.	Siege 8-inch.	24-pr. Garrison	Sea-Coast.	
									10-in.	8-in.
lbs. 10.5	lbs. 8.	lbs. 8.	lbs. 6.	lbs. 4.	lbs. 14.	lbs. 8.	lbs. 4.	lbs. 2.	lbs. 12.	lbs. 8.

**GREATEST CHARGES OF SEA-COAST, SIEGE
AND COEHORN MORTARS.**

SEA-COAST.		SIEGE.		COEHORN.	STONE MORTAR.	
18-inch.	10-inch.	10-inch.	8-inch.		120 pds. of stones	15 6-pr. shells.
lbs. 20.	lbs. 10.	lbs. 4.	lbs. 2.	lbs. 0.5	lbs. 1.5	lbs. 1.

63]

PART IV. RANGES.

1. What is meant by the *range* of a piece of artillery?

The distance from the muzzle to the first graze.

2. How may the range of a projectile be extended?

Either by raising the piece to a higher level, or by giving its axis greater elevation within certain limits.

3. Define *point-blank range*.

The distance from the muzzle of the piece to that point in a shot's trajectory where it cuts the prolongation of the natural line of sight, a second time.

4. In what does the French definition for point-blank range differ from ours?

It requires that the natural line of sight should be horizontal.

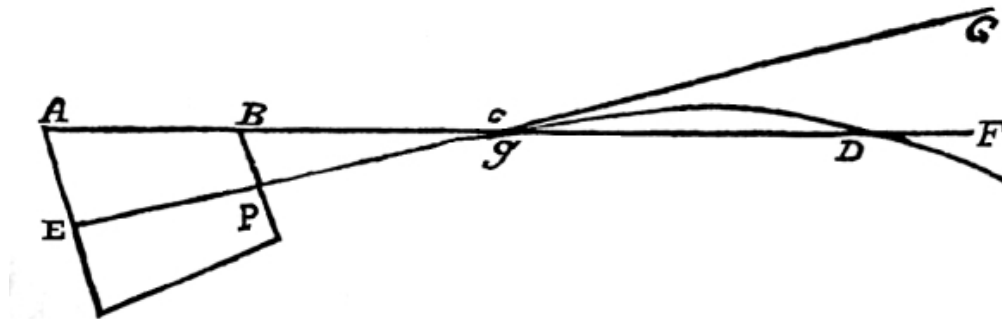
5. What is the British definition for point-blank range?

The distance from the muzzle to the first graze when the axis of the piece is parallel to the plane upon which the carriage stands.

6. Explain by a figure, the position of, and relations existing between the line of sight, the line of fire or axis of the piece, and the trajectory, and also what the point-blank range is.

64]

Fig. 1



$ABcF$, the line passing through the highest points of the base ring and swell of the muzzle, or the muzzle-band, is called, *the natural line of sight*. $EPcG$, is the *axis of the piece* or *line of fire*; the curved line, PgD , described by the projectile, is called the

trajectory, and is entirely below the line of fire, in consequence of the action of the force of gravity giving the projectile a downward tendency. The point D is called the *point-blank*, and its distance from the mouth of the piece, the *point-blank range*.

7. Mention some of the causes which vary the point-blank range.

The form of the cannon; the weight or force of the charge; the diameter and weight of the projectile; and the inclination of the line of sight to the horizon.

8. Why has the form of the cannon an influence on the point-blank range?

Because as the difference between the diameter of the breech and muzzle become greater, the angle of sight, $BcP = GcF$ ([see fig.](#)) increases, and the point-blank D is removed farther off; on the contrary, as the diameters approach to an equality, the point-blank approaches the piece. Within a certain angle, or when there is no angle of sight, as is the case with some old howitzers in which the line of sight is parallel to the axis of the bore, there will be no point-blank, as the trajectory will be constantly below the line of sight.

9. What influence has the charge on the point-blank range?

65]

An increase of the charge determines a more distant point-blank; its diminution produces a contrary effect: but beyond a charge equal to one-third the weight of the ball, the increase of range is inconsiderable, and the force of the recoil becomes very great.

10. How do the diameter and weight of the projectile affect the range?

As the ball increases in size and density, it will overcome with more ease the resistance of the air.

11. Does the inclination of the line of sight to the horizon have much effect on the point-blank range?

Only when this inclination is very considerable. For the ordinary inclination, from 0° to 15° , above or below the horizon, the difference may be wholly neglected.

12. What is the effect on the point-blank range of firing upwards under a large angle?

The action of the weight being nearly directly opposed to the impulsive force, the trajectory becomes compressed and the point-blank distance diminishes. The contrary effect obtains in firing downwards under a similar angle, as the weight and the force then act in nearly the same direction.

13. Why may the point-blank be considered constant for the same calibre?

66]

The dimensions, charges, and weights of projectiles, being constant, and the inclination of the natural line of sight, except in a very few cases, being comprised between 0° and 15° , it follows that for the same calibre, the point-blank may be

considered constant, and may serve as a point of reference in firing at different distances.

14. What is the extreme range of a piece of artillery?

The distance from the piece to where the projectile finally rests.

15. For a given velocity what effect has an increase of the angle of fire on the range?

It increases with the angle of fire up to a certain limit, beyond which it diminishes.

16. What angle gives the greatest range in *vacuo*?

Forty-five degrees.

17. When will this angle give the maximum range in practice?

Only for feeble charges, and very heavy projectiles.

18. How is the angle of greatest range in practice affected by a change in the velocity and size of the projectile?

It seems to diminish as the velocity is increased, and as the ball is reduced. For the musket the angle of maximum range varies from 28° to 30° ; and is nearly 42° for mortars.

19. Under what angle is a mortar usually fired?

Under the constant angle of 45° , and the charge is varied according to the range required.

20. What are the advantages of this practice?

Economy of ammunition; the recoil being inconsiderable, the mortar and its bed receive but little strain; the ranges are more uniform, and the effect of a slight error in the angle of fire is less than with any other.

21. Is the mortar ever fired at any other angle than 45° ?

67]

Yes; sometimes at 60° .

22. When is the mortar fired under an angle of 60° ?

When the battery is situated very near the object assailed, and it is desired that the shells may fall upon the magazines of the besieged. It is evident that projectiles the higher they are thrown up acquire more velocity in falling, besides striking the object more directly and with increased violence.

23. Under what angle are stone-mortars usually fired?

Under an angle of 60° , and sometimes of 75° , that in falling from a great height, the stones may have the maximum force of percussion.

24. Under what angle should grenades be thrown from stone-mortars?

About 33°; otherwise they will be buried in the earth, and their fragments will not be sufficiently destructive.

25. When a gun or howitzer is aimed with the line of metal horizontal, what is the elevation equal to?

The natural angle of sight or dispart.

26. How is the time of flight for siege mortars at an elevation of 45° determined?

It is nearly equal to the square root of the range in feet divided by four.

RANGES OF FIELD GUNS AND HOWITZERS.

68]

KIND OF PIECE.	Powder.	Ball.	Elevation	Range	Remarks.
6-Pounder Field Gun	lbs. 1.25	Shot.	deg. min	yds.	P. B. Range
		“	0	318	
		“	1	647	
		“	2	867	
		“	3	1138	
		“	4	1256	
		“	5	1523	
	1.	Sph. case	2	650	Time of flight 2”
		“	2 30	840	do. 3”
		“	3	1050	do. 4”
12-Pounder Field Gun	2.5	Shot.	0	347	P. B. Range
		“	1	662	
		“	1 30	785	
		“	2	909	
		“	3	1269	
		“	4	1455	
		“	5	1663	
	1.5	Sph. case	1	670	Time 2 seconds.
		“	1 45	950	“ 3 “
		“	2 30	1250	“ 4 “
12-Pounder Field Howitzer.	1.	Shell.	0	195	
		“	1	539	
		“	2	640	
		“	3	847	
		“	4	975	
		“	5	1072	
	0.75	Sph. case	2 15	485	Time 2 seconds.
		“	3 15	715	“ 3 “
		“	3 45	1050	“ 4 “
24-Pounder Field Howitzer.	2.	Shell.	0	295	

KIND OF PIECE.	Powder.	Ball.	Elevation	Range	Remarks.
		“	1	516	
		“	2	793	
		“	3	976	
		“	4	1272	
		“	5	1322	
	1.75	Sph. case	2	600	Time 2 seconds.
		“	3	800	“ 3 “
		“	5 30	1050	“ 4 “
		2.	“	3 30	880
	32-Pounder Field Howitzer.	2.5	Shell.	0	290
“			1	531	
“			2	779	
“			3	1029	
“			4	1203	
“		5	1504		
2.5		Sph. case.	3	800	Time 2¾ sec'ds.
Mountain Howitzer.	0.5	Shell.	0	170	Time 2 seconds.
		“	1	300	
		“	2	392	
		“	2 30	500	
		“	3	637	
		“	4	785	
		“	5	1005	
	0.5	Sph. case.	0	150	Time 2 seconds.
		“	2 30	450	
		“	3	500	
		“	4	700	
		“	4 30	800	
	0.5	Canister.	4 to 5°	250	Time 2¾ sec'ds.

69]

RANGES OF HEAVY ARTILLERY.

KIND OF PIECE.	Powder.	Ball.	Elevation	Range	Remarks.
18-Pdr. Siege and Garrison Gun on Barbette Carriage.	4.5	lbs.	deg. min	yds.	Point-Blank.
		Shot.	1	641	
		“	1 30	800	
		“	2	950	
		“	3	1256	
		“	4	1450	
24-Pdr. Siege and Garrison Gun on Siege Carriage.	6.	“	5	1592	
		Shot.	0	412	
		“	1	842	

KIND OF PIECE.	Powder.	Ball.	Elevation	Range	Remarks.
	8.	“	1 30	953	Point-Blank.
		“	2	1147	
		“	3	1417	
		“	4	1666	
		“	5	1901	
		“	1	883	
		“	2	1170	
		“	3	1454	
		“	4	1639	
		“	5	1834	
32-Pdr. Sea-Coast Gun on Barbette Carriage.	6.	Shot.	1 45	900	
	8.	“	1	713	
		“	1 30	800	
		“	1 35	900	
		“	2	1100	
		“	3	1433	
		“	4	1684	
		“	5	1922	
		“	1	780	
		“	2	1155	
		“	3	1517	
	10.67				

70]

KIND OF PIECE.	Powder.	Ball.	Elevation	Range	Remarks.
42-Pdr. Sea-Coast Gun on Barbette Carriage.	lbs.		deg. min	yds.	
	10.5	Shot.	1	775	
		“	1 30	860	
		“	2	1010	
		“	3	1300	
		“	4	1600	
	14.5	“	5	1955	
		“	1	770	
		“	2	1128	
		“	3	1380	
		“	4	1687	
		“	5	1915	
8-in. Siege Howitzer on Siege Carriage.	4.	45-lb. Shell.	0	251	Time $\frac{2}{3}$ sec'ds.
		“	1	435	“ 1½ “
		“	2	618	“ 2 “
		“	3	720	“ 3 “

KIND OF PIECE.	Powder.	Ball.	Elevation	Range	Remarks.
		“	4	992	“ 4 “
		“	5	1241	“ 5 “
		“	12 30	2280	
24-Pdr. Iron Howitzer on a Flank Casemate Carriage.	2.	17-lb. Shell.	0	295	Time 2 seconds.
		“	1	516	
		“	5	1322	
	1¾	Sph. case.	2	600	
		“	5 30	880	
	2.	“	3 30	1050	“ 3 “
8-in. Sea-Coast Howitzer on a Barbette Carriage.	4.	45-lb. Shell.	1	405	
		“	2	652	
		“	3	875	
		“	4	1110	
		“	5	1300	
	6.	“	1	572	
		“	2	828	
		“	3	947	
		“	4	1168	
		“	5	1463	
	8.	“	1	646	
		“	2	909	
		“	3	1190	
		“	4	1532	
		“	5	1800	
10-in. Sea-Coast Howitzer on Barbette Carriage.	12.	90-lb. Shell.	1	580	Time 3 sec'ds.
		“	2	891	
		“	3	1185	
		“	3 30	1300	
		“	4	1426	
		“	5	1650	“ 5¼ “
					“ 6 “

KIND OF PIECE.	Powder.	Ball.	Elevation	Range	Remarks.
	lbs.		deg. min	yds.	
8-in. Columbiad on Barbette Carriage.	10.	65-lb. Shot.	1	932	Axis of gun 16 feet above the water.
		“	2	1116	
		“	3	1402	
		“	4	1608	
		“	5	1847	
		“	6	2010	Shot ceased to ricochet on
		“	8	2397	
		“	10	2834	

KIND OF PIECE.	Powder.	Ball.	Elevation	Range	Remarks.
	15. 10.	50-lb. Shell.	“ 15	3583	the water.
			“ 20	4322	
			“ 25	4875	
			“ 27	4481	
			“ 27 30	4812	
			1 1	919	
			“ 2	1209	
			“ 3	1409	
			“ 4	1697	
			“ 5	1813	
			“ 6	1985	
			“ 8	2203	
			“ 10	2657	
			“ 15	3556	
			“ 20	3716	
			“ 25	4387	
			“ 27	4171	
			15. “ 27 30	4468	
10-in. Columbiad on Barbette Carriage.	18.	128-lb. Sht.	0	394	Axis of gun 16 feet above the water.
			“ 1	752	
			“ 2	1002	
			“ 3	1230	
			“ 4	1570	
			“ 5	1814	
			“ 6	2037	
			“ 8	2519	
			“ 10	2777	
			“ 15	3525	
			“ 20	4020	
			“ 25	4304	
			“ 30	4761	
			“ 35	5433	
	20.	“	39 15	5654	Shot ceased to ricochet on the water.
	18.	100-lb. Shell.	1	800	
			“ 2	1012	
			“ 3	1184	
			“ 4	1443	
			“ 5	1604	
			“ 0	448	
			“ 1	747	
			“ 2	1100	
			“ 3	1239	

KIND OF PIECE.	Powder.	Ball.	Elevation	Range	Remarks.
		“	4	1611	
		“	5	1865	
		“	6	2209	
		“	8	2489	
		“	10	2848	
		“	15	3200	
		“	20	3885	
		“	25	4150	
		“	30	4651	
		“	35	4828	Time 35 sec's.

72]

KIND OF PIECE.	Powder.	Ball.	Elevation	Range	Remarks.
	lbs.		deg. min	yds.	
13-in. Sea-Coast Mortar.	20.	200-lb. Shell..	45	4325	Time 40 sec's.
10-in. Sea-Coast Mortar.	10.	98-lb. Shell.	45	4250	Time 35 sec's.
10-in. Siege Mortar.	1.	90-lb. Shell.	45	300	Time 6.5 sec's.
	1.5	“	“	700	“ 12. “
	2.	“	“	1000	“ 14. “
	2.5	“	“	1300	“ 16. “
	3.	“	“	1600	“ 18. “
	3.5	“	“	1800	“ 19. “
	4.	“	“	2100	“ 21. “
8-in. Siege Mortar.	lbs oz				
	0 8	45-lb. Shell.	45	209	Time 6.75 sec's.
	1 12	“	“	376	Time 9. sec's.
	1 0	“	“	650	Time 11.5 sec's.
	1 4	“	“	943	Time 14. sec's.
	1 8	“	“	1318	Time 16.5 sec's.
	1 12	“	“	1522	Time 18.5 sec's.
	2 0	“	“	1837	Time 20.5 sec's.
24-Pounder Coehorn Mortar.	oz				
	0.5	17-lb. Shell.	45	25	
	1.	“	“	68	
	1.5	“	“	104	
	1.75	“	“	143	
	2.	“	“	165	
	2.75	“	“	260	
	4.	“	“	422	
	6.	“	“	900	

KIND OF PIECE.	Powder.	Ball.	Elevation	Range	Remarks.
	8.	“	“	1200	
Stone Mortar.	lbs 1.5	Stones. 120 lbs.	60	150 to 250	Fuze 15 sec's.
	1	15 6-pdr. shells.	33	50 to 150	

NOTE.—Fire-balls, according to their size, are fired from mortars of corresponding calibres. With a charge of ONE TWENTY-FIFTH its weight, the ball is thrown 600 to 700 yards.

PART V. RICOCHET.

1. What is understood by *ricochet firing*?

That obtained by firing a piece at very small angles of elevation, by which means the projectile which falls on ground of ordinary firmness at an angle not greater than 10° , or upon water at 4° or 5° , will make one or more bounds. In this case the projectile is said to *ricochet*.

2. What is the object of ricochet firing?

To enfilade a face of the enemy's work, which is effected by causing a projectile to bound along the terreplein of the face with the view of annoying his cannoneers, and dismounting his pieces. It is employed also in harassing an enemy, when formed or in the act of forming behind a rising ground or other obstacle, taking post in a wood, &c.; and in enfilading a line of troops.

3. What are the peculiar advantages of this fire?

In being able to reach objects which cannot be reached by direct fire, on account of intervening obstacles.

4. In enfilading a face of an enemy's work, what is the object to be fired at?

Usually some point of the interior crest of the parapet which covers a flank of the terreplein to be reached.

5. What is the *point of fall*?

The point of the terreplein which is first struck by the projectile, after having grazed the interior crest.

6. What is the *angle of fall*?

74]

It is the angle made at the point of fall by the tangent to the trajectory with a horizontal line in the plane of fire.

7. How does the angle of fall compare with that of *elevation*?

It is greater.

8. Upon what do the charge and elevation depend?

Upon the distance of the object from the battery; upon the difference of level between these points; the distance of the desired point of fall from the parapet; the height of the parapet, &c.

9. If the embrasure be such that the object is masked, how is the piece pointed?

The *direction* must be given, as with the mortar, by the plummet; this is held by the person who points, in such a manner as to cover both the line of metal and the object. The *elevation* is then given by the quadrant.

10. What is the maximum angle of elevation in ricochet firing?

Against troops it should seldom exceed 3° above the surface of the ground occupied by them. Against fortresses, forts, and fortified lines, it varies from 3° to 9° above the horizontal.

11. At what distance from the object should the ricochet battery be placed?

Never at a greater distance than 600 yards.

12. In enfilading a work, how should the ricochet firing be conducted? 75]

The projectile should be made to graze the parapet while in the descending branch of the trajectory; and this must be effected by regulating the charges and elevating or depressing the piece until the shot is seen to fall just over the interior crest of the parapet. Light charges are generally used, varying from *two-thirds* to *one-eighth* of the ordinary charge.

13. What pieces are best adapted for ricochet fire?

Those which throw heavy shells, for, if used to enfilade a work, the shells lodge and explode in the traverses, and render the guns more liable to be dismounted, and their detachments put *hors de combat*.

14. What determines the *nature* of the ricochet?

The angle of fall: it is *flattened* when this angle does not exceed 6° , and *curvated* when it is between 10° and 12° . In the first of these fires, the velocities are great, and in the second small.

15. What are the charges for a *flattened ricochet* for siege guns at an angle of about 3° ?

[See Table, page 58.](#)

16. What are the charges for a *flattened ricochet* for siege howitzers at an angle of about 3°?

[See Table, page 58.](#)

17. What are the charges for a *curvated ricochet* for a siege howitzer at an angle of about 10°?

[See Table, page 58.](#)

76]

**CHARGES FOR A FLATTENED RICOCHET
FOR SIEGE-GUNS.**

DISTANCE.	ELEVATION.	CHARGE.
660 yards.	2° 45'	1-12 weight of ball.
550 “	3°	1-15 “ “
440 “	3° 15'	1-20 “ “
330 “	3° 35'	1-30 “ “

**CHARGES FOR A FLATTENED RICOCHET
FOR SIEGE HOWITZERS.**

DISTANCE.	ELEVATION.	CHARGE.
550 yards.	1° 45'	3 lbs.
440 “	2° 15'	2 lbs. 3 oz.
330 “	2° 15'	1 lb. 12 oz.
220 “	2° 45'	1 lb. 2 oz.

**CHARGES FOR A CURVATED RICOCHET
FOR SIEGE HOWITZERS.**

DISTANCE.	ELEVATION.	CHARGE.	REMARKS.
550 yards.	7° 30'	1 lb. 4 oz.	The height of the object above the level of the battery being supposed to be 20 feet.
440 “	“	1 lb. 1 oz.	
330 “	“	14 oz.	
220 “	“	10 oz.	

77]

PART VI.

RECOIL.

1. What is meant by the *recoil* of a piece of artillery?

The retrograde motion impressed upon cannon by the discharge is termed the *recoil*.

2. What causes the recoil of a piece?

The gas produced by the ignition of the charge in the bore, expanding with equal force in every direction, finds only two ways of escape (the muzzle and vent); the pressure upon these points will therefore cease, while it will be proportionally increased upon the parts directly opposite, that is, the breech and the lower part of the first reinforce, producing in the first case the recoil, and in the other, indirectly, the dipping of the muzzle.

3. How far does a gun usually recoil?

This depends entirely upon the nature and inclination of the ground upon which the carriage stands, the situation of the trunnions, angle of elevation, comparative weight of the gun and carriage, and upon the strength of the charge.

4. What proportion does the velocity of the recoil of a piece bear to that of a ball?

The proportion is inversely as their weights, or their masses.

5. What proportion exists between the pressure acting upon the part of the bore of a piece directly opposite the vent, and that which occasions the recoil? 78]

As the square of the diameter of the vent is to the square of that of the shot.

6. Has the recoil any effect upon the flight of the projectile?

No appreciable effect, the shot being expelled from the gun before it has recoiled a fraction of an inch.

7. What are the principal inconveniences arising from the recoil of guns?

The necessity of running up the gun after every discharge, and consequent fatigue to the men and loss of time; it also necessitates that a greater breadth should be given to the terreplein of a work.

8. What causes the muzzle of a piece of artillery to dip when fired?

The sudden pressure of the gas acting upon the portion of the first reinforce opposite to the vent, causes the piece to strike downwards upon the elevating screw or quoin, and the reaction to make the muzzle dip.

9. What influence has the position of the axis of the trunnions in respect to that of the bore upon the recoil?

If the axis of the trunnions be below that of the piece, the pressure of the breech upon the carriage will increase as the distance between the axes increases; and from this pressure there will arise a friction upon the ground which will diminish the recoil. On the contrary, if the axis of the trunnions, be above that of the piece, the breech will have an upward tendency, the recoil will be increased, but the carriage, and particularly the axle-tree, will be subjected to less strain. Hence, the recoil will be transmitted directly to the trunnions, if their axis (as in our service) be situated in the same plane with the axis of the piece. The size of the trunnions is made proportional to the force of the recoil.

10. Does the position of the trunnions with reference to the centre of gravity of the piece influence the recoil? 79]

Yes; in cannon fired horizontally, or under very small angles, the portion in rear of the trunnions is heavier than that in front; an arrangement which increases the pressure of the trail on the ground so as to diminish the recoil. But in pieces fired under large angles, the trunnions are placed in rear of the center of gravity, for the purpose of increasing the ease of pointing.

PART VII. WINDAGE.

1. What is meant by *windage*?

The difference between the diameter of the projectile and that of the bore.

2. Is it absolutely necessary to allow windage?

Yes, in order to make an allowance for a piece becoming foul, the expansion of shot by heat, the incrustation of rust, and for the tin straps of fixed ammunition.

3. What advantages are derived from reducing the windage?

An increase in the accuracy of fire; a more extensive range, or an equal range with a smaller charge, as there is less loss of gas; and less injury to the surface of the bore.

4. Why should the bore suffer less injury by a diminution of the windage?

Because in proportion to the decrease of windage there will be less space for the reflections of the shot along the bore, and consequently less injurious power exercised upon it.

5. What is the loss of velocity by a given windage proportional to?

It is directly as the windage, and inversely as the diameter of the bore very nearly.

6. What is the loss of velocity by the windage of the ball?

81]

KIND OF GUN.	Charge of powder.	Initial velocity of ball.		Loss of velocity by windage of 1-40th diam.	
		Without windage.	With windage of 1-40th diam.		
32-pdr. Sea-Coast,	lbs.	feet.	feet.	feet.	per cent.
	4	1444	1271	173	12
24-pdr. Siege,	4	1600	1433	167	10
	6	1890	1723	167	9
12-pdr. 25 calibres,	2	1617	1444	173	11
	3	1915	1742	173	9
	4	2124	1951	173	8
12-pdr. Field, 16 calibres,	2	1528	1370	158	10
	3	1793	1635	158	9
	4	1992	1834	158	8
6-pdr. Field,	1.5	1734	1560	174	10

7. What windage is allowed to guns?

IRON.					BRASS.	
Sea-Coast.			Siege and Garrison.		Field.	
42	32	24	18	12	12	6
inches.	inches.	inches.	inches.	inches.	inches.	inches.
0.16	0.15	0.14	0.13	0.10	0.10	0.09

82]

8. What windage is allowed to columbiads and howitzers?

COLUMBIADS.		HOWITZERS.							
Iron.		Iron.				Brass.			
		Sea-Coast.		Siege and Garrison.		Field.		Mountain.	
10-in.	8-in.	10-in.	8-in.	8-in.	24-pdr.	32-pdr.	24-pdr.	12-pdr.	12-pdr.
in.	in.	in.	in.	in.	in.	in.	in.	in.	in.
0.12	0.12	0.12	0.13	0.13	0.14	0.15	0.14	0.10	0.10

9. What amount of windage is allowed to mortars?

IRON.				BRASS.		IRON.
Heavy.		Light.		Stone Mortar.	Coehorn 24-pdr.	Eprouvette.
inches.	inches.	inches.	inches.	inches.	inches.	inches.
0.13	0.13	0.13	0.13	—	0.14	0.025

83]

PART VIII. GUNPOWDER.

1. What are the ingredients in *gunpowder*?

Saltpetre, charcoal and sulphur.

2. What are the proportions?

In the United States, 75 to 76 saltpetre, 14 to 15 charcoal, and 10 sulphur.

England,	75	Saltpetre,	15	Charcoal,	10	Sulphur.
France,	75	“	12½	“	12½	“
Prussia,	75	“	13½	“	11½	“

3. What is the combustible ingredient?

Charcoal.

4. What is the use of the saltpetre?

It furnishes the oxygen necessary to support a rapid combustion, and to change the whole mass into gas.

5. What is the use of sulphur?

It adds consistency to the mixture and intensity to the flame, besides rendering the powder less liable to absorb moisture.

6. On what does the quality of gunpowder depend?

On the intimate mixture and proper proportions and purity of the ingredients.

7. In what does the manufacture of gunpowder consist?

In pulverizing the ingredients, incorporation, compression, granulation, drying, glazing and dusting.

8. Explain the method of making gunpowder by the pounding mill. 84]

The charcoal in small pieces is first placed in the mortars, with a quantity of water, and pounded for half an hour; after which the saltpetre and then the sulphur, previously pulverized and sifted, are put in, and the whole well mixed with the hand; it is then pounded in the mortars, and at the end of each hour, the composition is passed from each mortar into the next. At the sixth or eighth change, add half a pint of water; it is then pounded two hours without changing the mortars, in order that it may form into cake. It is then partially dried, and grained in a graining sieve, or passed between wooden rollers. The grains are then sifted to separate those which are too coarse and too fine, and also to separate from each other the different kinds of grains for *cannon*, *musket*, and *rifle* powder. It is then glazed in large glazing barrels, which make 15 or 20 revolutions in a minute. A charge of 500 lbs. is thus treated for about twenty-four hours. It is then dried either in the open air, or in a drying house. If in the open air, when the sun is too hot, the powder should be covered to prevent the loss of sulphur. It is then *dusted*, by being sifted in fine sieves, or through bolting cloths.

9. What other machines besides the pounding mill are used in pulverizing and incorporating the ingredients of gunpowder?

Rolling barrels, and the cylinder or rolling mill.

10. What advantage is gained by the use of the *rolling barrels*?

85]

It lessens the duration and danger of pounding in the mortars. After the ingredients are pulverized and mixed in the rolling barrels, the mixture is placed under the pestles of the *pounding mill*, 10 per cent. of water is added, and it is beaten for three hours only.

11. Which mill is now generally used?

The CYLINDER MILL, which forms 50 lbs. of composition into cake in from one to three hours.

12. Does powder inflame instantaneously?

No; its inflammation is gradual, and progressive, and in a gun the projectile commences to move before the whole charge is ignited.

13. Why should gunpowder be grained?

In order to facilitate the transmission of the flame. When the powder is very fine, and in large and compact charges, the flame cannot penetrate it, and it burns slowly and in successive layers.

14. Which burns quickest, the small or large grained powder?

Before coming to the limit of dust, the smaller the grain, the more rapid the combustion, and the greater the bursting force of the powder.

15. What is the difference between the ignition and combustion of large and small grained powder?

With the large grained, the ignition is more rapid, but the combustion slower; with small grains, the contrary is the case.

16. Why should the grains be angular?

Because they present a greater surface to the action of the flame, and therefore burn quicker.

17. Why should powder be free from dust?

Because the dust fills up the intervals between the grains, and forming a compact mass, retards combustion.

18. To what special purpose are large and small grained powders applied? 86]

The large for cannon, and the small for small arms.

19. How is the size of the grain for each kind of powder tested?

By means of sieves or gauges.

20. How many grains of powder are in 10 grs. Troy weight?

Cannon, 150; *Musket*, 2,000 to 2,500; and *Rifle*, 12,000 to 15,000.

21. What is the object in glazing powder?

Glazed powder does not absorb moisture, or break up in transportation, so much as unglazed.

22. What is the established mode of proving the strength of powder in the U. S.?

A sample is taken from each barrel, and the strength determined by the eprouvette mortar.

23. What is the least range allowed?

The general *mean-range* of new powder must not be less than 250 yards; but no powder ranging below 225 yards is received.

24. When is powder in magazines considered unserviceable.

When it does not range over 180 yards.

25. What is the range of good powder?

Cannon from 280 to 300 yards. Small grained from 300 to 320 yards.

26. What other means is there for determining the strength of powder?

The GUN AND BALLISTIC PENDULUM, and NAVEZ' ELECTRO-BALLISTIC MACHINE. The latter is considered the best for determining the initial velocity.

27. What is the hygrometric proof of powder?

871

Samples are placed in shallow tin pans, set in a tub, the bottom of which is covered with water; the pans should be about an inch above the water, and the tub covered. Good powder will not absorb more than 2½ per cent. in 24 hours.

28. How can the relative quickness of two kinds of powder be determined?

By burning a train laid in a circular or other groove, which returns into itself, made in a piece of hard wood: one-half of the groove being filled with each kind of powder, and fire communicated at the junction of the two trains, the relative quickness is readily deduced from observation of the point at which the flames meet.

29. What are the qualities of good powder?

It should be perfectly free from dust, uniform in strength and size of grains, angular and irregular in form; in color, brownish black, or slate color; so hard as not to be easily crushed by pressure with the finger; and should leave no beads or foulness when flashed in quantities of 10 grs. on a copper plate.

30. What is the expansive velocity, and pressure of ignited powder?

The expansive velocity is about 5,000 feet per second, and pressure about 2,000 atmospheres.

31. What is the weight of a cubic inch of powder?

About half an ounce; a cubic foot will therefore weigh about 54 pounds, and 32 cubic inches, one pound.

32. How is government powder packed?

88]

In barrels of 100 lbs. each; the barrels being large enough to allow sufficient space for the powder to move when rolled to prevent its caking.

33. How are the barrels marked?

On one head with the *place* and *year* of manufacture, and with the kind of grain, *cannon*, *musket* or *rifle*; on the other head with the *year* in which it was *proved* and the *proof-range*, leaving room for subsequent proofs, which are marked in the same manner.

34. When powder is injured by dampness, can it be restored?

If the water absorbed does not exceed 7 per cent., it can be by drying. If it has absorbed from 7 to 12 per cent., after drying, it remains porous and friable, and is unfit for transportation. In this case it is better to work it over.

35. How is powder stored?

In magazines especially constructed for the purpose. The barrels are generally placed near the sides, three tiers high, or four tiers if necessary; small skids should be placed on the floor and between the several tiers of barrels, in order to steady them, and chocks should be placed at intervals on the lower skid, to prevent the rolling of the barrels.

36. How are the different kinds of powder arranged?

Those barrels of the same kind, place and date of fabrication, and proof range, are piled together.

37. Should it be necessary to pile the barrels more than four tiers high, what is done?

The upper tiers are supported by a frame resting on the floor; or the barrels may be placed on their heads, with boards between the tiers.

38. What is necessary for the preservation of the powder?

89]

The magazine should be opened and aired in clear dry weather, and the ventilators should be kept free.

39. How may the moisture of a magazine be absorbed?

By chloride of lime suspended in a box under the arch, and renewed from time to time.

40. When the magazine is open, what precautions should be observed?

The sentinel or guard should have no fire-arms, and any one who enters it should take off his shoes, or put socks over them. No sword or cane, or anything which might occasion sparks should be carried in.

41. How should powder in barrels be transported?

The barrels should never be rolled; they should be carried in hand-barrows, or slings made of rope or leather. In wagons, the barrels should be packed in straw, and not allowed to rub against each other, and the whole covered with thick canvas.

42. What precaution should be used to prevent powder caking?

The barrels should be taken outside the magazine and rolled on boards.

43. Where should cartridge bags be filled?

In the filling room of the laboratory, or a small magazine, and not in the general magazine.

90]

PART IX. PROJECTILES.

1. What projectiles are made use of in the service?

Solid shot; shells; spherical case, or shrapnel; canister; grape; grenades; stones; carcasses; light and fire balls.

2. What is a solid shot?

A solid sphere of cast-iron, almost exclusively appropriated to guns. The gun derives its denomination from the weight of the shot, as 6-pr., 12-pr., &c.

3. What is a shell, and its use?

A hollow sphere of cast-iron, containing powder, which is ignited by means of a fuze; when fired at troops, it should be prepared to burst over their heads, or, if the ground be favorable, to ricochet a little in front, and plunge into the column. When fired at works or buildings, it should explode after penetration.

4. What is spherical case, and what advantages does it possess?

It is a shell much thinner than the ordinary shell, and filled with leaden bullets and a charge of powder sufficient to burst it, which is done by means of a fuze as with a common shell at any required distance. It is thus calculated to extend all the advantages of canister shot, to distances far beyond the reach of that projectile. It is fired both from guns and howitzers.

5. What are canister shot?

91]

Cylindrical tin cases with iron heads, of calibre suitable for different pieces of ordnance, filled with cast-iron balls arranged in tiers; they are fired at ranges not exceeding 400 yards, but their most destructive effects are from 100 to 200 yards.

6. What are grape shot?

A certain number of iron balls, usually nine, put together by means of two cast-iron plates, two rings, and one pin and nut. Each plate has on the inside three beds for the shot, of a depth equal to half the thickness of the plate, and of the form of a spherical segment, the curvature of which is the same as that of the shot. An iron pin riveted to the bottom iron plate, passes through the centre and also through the top plate, where the whole is secured by a nut and screw.

NOTE.—The use of these shot for field pieces has been discontinued, canister answering the purpose of these shot.

7. How were the balls fixed in the old pattern?

They were placed in tiers around an iron pin attached to an iron tompion at the bottom, and put into a canvas bag, and then quilted around with a strong cord.

8. What is a grenade?

A shell thrown from the hand, or in baskets from the stone-mortar, and ignited as other shells by means of a fuze.

9. How many kinds of grenades are made use of?

Hand-grenades and *rampart-grenades*; six-pounder spherical case may be used for the former, and shells of any calibre for the latter.

10. To what purposes are grenades applied?

92]

They are useful in the defense of works, the smaller, thrown by hand into the head of a sap, trenches, covered way, or upon the besiegers mounting a breach: the larger kinds are rolled over the parapet in a trough.

11. What is a carcass, and its use?

It is a spherical shell having three additional holes, of the same dimensions as the fuze-hole, pierced at equal distances apart in the upper hemisphere of the shell, and filled with a composition which burns with intense power from eight to ten minutes, and the flame issuing from the holes, sets fire to everything combustible within its reach; it is used in bombardments, setting fire to shipping, &c.; and is projected from cannon like a common shell.

12. What is a substitute for a carcass?

Common shells loaded in the following manner: The bursting charge is placed in the bottom of the shell in a flannel bag, over which carcass composition is driven until the shell is nearly filled; then insert four or five strands of quick-match, which must be secured by driving more composition upon it. These shells, after burning as a carcass, explode.

13. What is a fire-ball, and its use?

It is a projectile of an oval shape, formed of sacks of canvas filled with combustible composition, which emits a bright flame. Its use is to light up the enemy's works, and it is loaded with a shell to prevent it from being approached.

14. What is a light ball?

Light balls are the same as fire balls, except that there is no shell in them, as they are used for lighting up our own works.

15. What is a smoke ball?

93]

A hollow paper sphere similar to a light ball, and filled with a composition which emits a dense, nauseous smoke; it is employed to suffocate the enemy's miners

when at work, or to conceal one's own operations; it burns from twenty-five to thirty minutes.

16. In field pieces to what is the projectile attached?

To a block of wood called a sabot.

17. Are the projectile and cartridge ever attached to the same sabot?

Yes, in field guns, and the 12-pr. field-howitzer; the whole then constitutes a *round of fixed ammunition*.

18. What is the arrangement in case of the 32 and 24-pdr. field howitzers?

The projectile is separate from the charge, and the *cartridge* is attached to a block of wood called the *cartridge-block*, the object of which is to give a finish to the cartridge and fill the chamber.

19. What difference is there in sabots for field service?

Sabots for shot, and spherical case for guns, have one *groove* for attaching the cartridge—those for gun canisters and for the 12-pdr. howitzer shells, spherical case, and canisters, have *two grooves*. Those for the 32 and 24-pdr. howitzers have no grooves; but are furnished with handles made of cord, passing through two holes in the sabot, and fastened by knots on the inside.

20. How are projectiles for field service fastened to the sabot?

By straps of sheet-tin, or of *strong canvas*, when tin or sheet iron cannot be procured.

21. How many straps are employed, and how are they fastened?

94]

For *shot*, there are two straps crossing at right angles, one passing through a slit in the middle of the other. For *shells*, there are four straps soldered to a ring of tin, or fastened to it by cutting four slits in the ring, into which the upper ends of the strap are hooked, and turned down on the inside of the ring. The sabots for 32 and 24-pdr. field howitzers having no groove, each strap is fastened by one nail on the side, and two under the bottom of the sabot.

22. What is a *canister for field-service*?

It consists of a tin cylinder attached to a sabot and filled with cast-iron shot.

23. How is it made?

The cylinder is fastened to the sabot by six or eight nails, and a plate of rolled iron is placed at the bottom on the sabot. It is closed with a sheet-iron cover after being filled, the top of the cylinder being cut into strips $\frac{1}{2}$ an inch long, and turned down over the cover.

24. In case of heavy guns are the shot attached to the sabot?

They are generally without a sabot.

25. How is it with shells?

They are strapped to sabots made of thick plank, with strips of tin, as in case of strapping shot for field-service.

26. How is it with canister for siege and sea-coast guns?

They have no sabot; the tin is turned over the iron bottom.

27. How is it with canisters for the 8-in. siege and sea-coast howitzers?

95]

They are attached to sabots in the same way as the field-howitzer canisters. The sabot for the siege howitzer has a hemispherical bottom and the sea-coast a conical one, to suit the connecting surface between the cylinder of the bore and the chamber in these pieces.

28. Are sabots used with grape shot?

Yes, in the 8-inch sea-coast howitzer.

29. What is its form, and how fastened?

It is conical; and may be fastened to the lower plate with screws, or the pin may be made long enough to pass through it; or else the sabot maybe inserted into the piece separately from the stand of grape.

30. What is the object of fixing shot or shells to wooden bottoms?

To prevent injury to brass cannon; and to insure the fuze of a shell being retained in the axis of the piece.

31. What proportion does the weight of one shot bear to that of another?

The proportion is, as the cubes of their diameters.

32. How is the weight of a cast-iron shot or shell determined?

Multiply the cube of the diameter of the shot in inches, or the difference of the cubes of the exterior and interior diameters of the shell by 0.134 for the weight in pounds. In case of lead balls, the multiplier is 0.214.

33. How is the diameter of a cast-iron shot of a given weight found?

Divide the weight in pounds by 0.134, and extract the cube root of the quotient, which will be the diameter in inches.

34. How is the quantity of powder which a shell will contain found?

96]

Multiply the cube of the interior diameter of the shell in inches, by 0.01744 for the weight of powder in pounds.

NOTE.—The above multipliers are found as follows: Suppose W to represent the weight of a body, D its density, V its volume, and g the weight of the unit of mass, then $W = DVg$. Now, if a cubic inch be taken as the unit of volume, then g will be numerically

$$\frac{62.5}{1728} \text{ pounds.}$$

$$\text{Hence, } W = DV \frac{62.5}{1728} = 0.036201;$$

$$DV = 0.03620 D = \frac{\pi}{6} d^3$$

(supposing d to be the diameter, and the body to be spherical)

$$0.036201 \times 0.5236 D d^3 = 0.018955 D d^3.$$

If we now substitute for D the specific gravity of cast-iron shot or shells = 7.000, we have,

$$W = 7 \times 0.018955 d^3 = 0.134 d^3;$$

and if for D we substitute the specific gravity of lead, $W = 0.2142 d^3$;
and in case of powder, $W = 0.01744 d^3$.

For diameters, weights and charges, [see Tables, pages 80-81](#).

35. When shot are heated to a white heat, what expansion takes place?

Calibre.	8-in.	42	32	24	18	12
Expansion, inches,,	0.149	0.11	0.10	0.08	0.06	0.04

36. Do heated shot retain a permanent enlargement?

Yes; in case of the 8-in. shot, for example, after the first cooling the enlargement is 0.054 in.; and, after the second, 0.099 in.

37. Are the igniting powers of a hot shot destroyed by ricochetting upon the water?

No; a shot, properly heated, will ignite wood after having struck the water several times.

38. What is the peculiarity of cartridges for hot shot?

97]

There are two cartridge bags, one being inserted, choke foremost in another of the next higher calibre, and the end of the latter folded under.

39. Explain the process of loading with hot shot.

The piece should be sponged with great care, and the worm frequently passed into the bore. As a precaution, it is well to insert a wet sponge just before putting in the ball. The muzzle is sufficiently elevated to allow the ball to roll down the bore, the cartridge is inserted, the mouth of the outer bag foremost, the fold down, and carefully pushed home without breaking it; a dry hay-wad is placed upon it, and rammed once; then a clay or wet hay wad, and rammed twice; and finally, if firing at angles of depression, a wad of clay a half-calibre in length, or a wet hay-wad is put on the ball.

40. May the ball cool in the gun without igniting the charge?

Yes, with proper precaution in loading. The piece, however, should be fired with as little delay as possible, as the vapor, which arises from the action of the hot ball on the water contained in the wad, diminishes the strength of the powder.

41. What means are afforded at the sea-board forts for heating shot?

Furnaces for this purpose are erected, which hold 60 or more shot.

42. What length of time is required to heat them to a red heat?

98]

The shot being placed and the furnace cold, it requires one hour and fifteen minutes; but after the furnace is once heated, a 24-pdr. shot is brought to a red heat in twenty-five minutes; the 32-pdr. and 42-pdr. shot require a few minutes longer.

43. Describe grates for heating shot.

In siege and other batteries, where there are no furnaces, a grate is used. It consists of four bars 1.75 inches square, three feet long, placed four inches apart on three iron stands, one foot in height. It is placed in an excavation one foot in depth, of the width of the grate, perpendicular at the back and side, open in front, the legs resting on bricks or stones rising about four or five inches from the bottom. A roof is made over it with hoops of flat iron, covered with sods and eighteen inches of earth, having in the back part a chimney six inches square. The shot are placed on the back part of the grate, leaving one-fourth of the front part free; and under and on the front part the wood is put, cut in pieces about fourteen inches long and two inches thick. A thick sod is used as a register, to regulate the draught of the chimney, so that no flame can issue from the front. This grate, which will contain about fifteen 24-pdr. balls, heats them to a red heat in an hour, and will supply three guns.

44. How are wads for firing hot shot made?

Of hay; by twisting from the hay a rope of an inch or an inch and a half in diameter, and then commencing at one end, and doubling it up about one calibre in length, twisting it all the time until it becomes nearly large enough, when the rope is wound around the wad perpendicular to its axis, and fastened with a hitch. Or the hay may be rammed in a *form* of proper calibre, and then bound with spun yarn, and afterwards rammed a second time.

45. Have hot shot been almost entirely superseded?

99]

Yes, since the adoption of the method of throwing large hollow shot from long pieces. These require but little preparation, can be used at once, and are more terrible in their effects.

46. What are *ring* or *grommet* wads, and their use?

They consist of a ring of rope yarn, about 0.7 in. thick, with two pieces of strong twine tied across at right angles to each other. The size of the ring is the full diameter of the bore, in order that it may fit tight, and stop the windage. They increase the accuracy

of fire, and are to be preferred when the object of the wad is to retain the ball in its place, as in firing at a depression. They stop the windage best when placed behind the ball. They may be attached to the straps, or to the ball by twine, or may be inserted like other wads after the ball.

47. How are *junk-wads* made; and for what are they used?

100]

Wad-moulds for each calibre,—consisting of two cast-iron cylinders of different diameters set in oak, or of two strong pieces of oak, strapped with iron, and joined by a hinge,—are employed in their manufacture. The junk, after having been picked, is compressed by being beaten in the smaller mould with a *maul* and *cylindrical drift*—the latter nearly of the size of the mould—until it assumes the requisite dimensions; it is then taken out by raising the upper part of the mould, and closely wrapped with rope yarn, passed over it in the direction of the axis of the cylinder, and fastened by a few turns around the middle of the wad. It is then placed in the large mould, and again beaten with the maul and drift, until its diameter is increased to that of the mould, when it is taken out and its diameter verified by a wooden gauge corresponding to the large shot-gauge of the calibre. These wads are used for proving cannon.

48. Describe the process of loading field shells.

They are set up on their sabots, the charges measured out in the proper powder measure, and poured in through a copper funnel. The fuze-plugs are then driven in with a mallet, allowing the tops to project about 0.1 in., care being taken not to split them. The holes in the plugs are then carefully reamed out, and stopped with tow-wads, which are pressed in firmly with a round stick.

49. Describe the process of loading spherical case shot.

101]

The shot having been cleaned, the balls are put in. A stick with a less diameter than the fuze-hole, and having a groove on each side of it, is inserted and pushed to the bottom of the chamber by working the balls aside. The shot is then placed in a sand-bath or oven, and brought to a proper temperature to receive the sulphur, which in a melted state is poured in to fill up the interstices between the balls; the shot is allowed to cool, and the sulphur to harden, when the stick is withdrawn, and the sulphur adhering to the sides of the eye and the surface of the shot is removed. If a fuze-plug and paper-fuze are to be used, the charge is poured in and the plug inserted exactly as in case of a shell; but, if the Bormann fuze is to be used the charge is inserted and the stopper and fuze screwed into their places, care being taken before placing the fuze in position to puncture the covering of the magazine, so that the fire can communicate with the charge.

Spherical case are now usually loaded by putting in the bullets, and pouring melted sulphur in until the case is full. After the sulphur has cooled, the space for the powder is bored out by a cutter, which removes both the sulphur and portions of the bullets from the space. This is a quicker method, and gives a more compact projectile.

50. What advantages does this mode of loading possess over the old one?

In the old mode there was a liability to accidents, and, if the powder remained in for any length of time before being used, it was ground up and became impaired. By the new mode the powder can be placed in the small chamber, and allowed to remain without fear of damage or danger, and be ready for use when required. Being, besides, in a compact mass, instead of scattered among the bullets, its power is much greater and it acts more effectively in throwing the bullets outward from the centre.

51. Describe the process of filling *Mortar shells*.

102]

Having been inspected to see that they are clean, dry, and in good order, place them on a block made for the purpose, or on rings of rope, or in indentations in the floor of the magazine, or on the ground, with the eyes up. The charge measured out in a powder-measure is poured in through a funnel, and any incendiary composition, such as pieces of port-fire, rock-fire, &c. is inserted. In the mean time the fuze is cut to the proper length according to the range, by resting it in a groove made in the block, or inserting it in a hole made in a block, or in a post, and sawing it across with the fuze-saw; or the fuze may be bored through with a gimlet perpendicularly to the axis, at the proper point.

The fuze is then tried in the eye, and should enter $\frac{3}{4}$ of its length. If it does not, it may be reduced by rasping. The head of it is covered with tow to prevent the breaking of the composition, the fuze-setter placed on, and the fuze driven with the mallet until the head projects not more than 0.2 in. to 0.4 in. above the surface of the shell. These shells are generally filled and the fuzes driven in the battery magazines, as they are required.

52. How are shells for *columbiads* and *heavy guns* loaded?

In the same way as Mortar shells; but as paper fuzes inserted in wooden or bronze fuze-plugs are used instead of wooden fuzes, the plug only is driven into its place, and stopped with tow after the bursting charge has been poured through it into the shell.

53. How are condemned shot and shell marked?

With an X, made with the cold chisel.

54. How should balls be preserved?

They should be carefully lacquered as soon as possible after they are received. When it becomes necessary to renew the lacquer, the old lacquer should be removed by rolling or scraping the balls, which should never be heated for that purpose.

55. How should grape and canister shot be preserved?

103]

They should be oiled or lacquered, put in piles, or in strong boxes on the ground floor, or in dry cellars; each parcel marked with its kind, calibre, and number.

56. How are balls piled?

Balls are piled according to kind and calibre, under cover if practicable, in a place where there is a free circulation of air, to facilitate which the piles should be made

narrow, if the locality permits; the width of the bottom tier may be from 12 to 14 balls according to calibre.

Prepare the ground for the base of the pile by raising it above the surrounding ground so as to throw off the water; level it, ram it well, and cover it with a layer of screened sand. Make the bottom of the pile with a tier of unserviceable balls buried about two-thirds of their diameter in the sand; this base may be made permanent: clean the base well and form the pile, putting the *fuze-holes* of shells *downwards* in the *intervals*, and not resting on the shells below. Each pile is marked with the number of serviceable balls it contains. The base may be made of bricks, concrete, stone, wood, or with borders and braces of iron.

57. How should fixed ammunition for cannon be stored?

104]

Either in boxes or placed in piles, formed of two parallel rows of cartridges, with the sabots together; in 4 tiers for 12-pdr. and 5 for 6-pdr.; chock the lower tier with strips of wood fastened with small nails; put a layer of tow 2 in. thick between the shot; let the piles rest on planks, if there is no floor, and cover them with tarpaulins; have the place swept, and the cartridge bags brushed off. Leave a passage of 18 in. between the double rows, and keep them 2 feet from the walls. Fixed ammunition should not be put into powder-magazines, if it can be avoided; it should be kept in a dry place above the ground floor if practicable; the store-rooms should be always aired in fine weather, the piles should be taken down, and made up again every six months at most, the bags examined and repaired, and the damaged cartridges broken up. A ticket on each pile should show the number and kind of cartridges, the additions to the pile, and the issues.

53. How should canisters be piled?

Like fixed ammunition, in 4 tiers for 24's and 18's; and 5 for 12's and 6's. Empty canisters in 10 or 12 tiers; the bottoms and covers separately.

59. How should *cartridge bags filled* be piled?

Like fixed ammunition, or packed in boxes or barrels.

60. How should *loaded shells* be piled?

On the ground floor of a secure building on planks, if the floor is not boarded; in 6 tiers at most; the fuzes of the lower tier in the vacant spaces between the shells; those of the other tiers turn downwards, like the fuze-holes of empty shells; the piles should be covered with a tarpaulin. Loaded shells should never be put into magazines, except from absolute necessity.

61. How should *fire-balls* be preserved?

In a cool place, separated from each other by shavings or straw, if they are piled up.

62. How is the number of shots or shells in a pile computed, of whatever form the pile may be?

105]

By multiplying the sum of the three parallel edges, by one-third of the number of balls in a triangular face.

63. What is meant by the three parallel edges of the pile?

Of the rectangular or long pile, they consist of the two largest bottom rows and top-row; of the square pile, of two bottom-rows and top shot; and of the triangular pile, of one bottom-row, the shot at the opposite angle, and that at the top.

64. How is the number of shot in a triangular face computed?

Multiply the number in the bottom row, plus one, by half the number in the bottom row, for the number required.

65. How is the shot contained in the top row of a rectangular pile calculated?

One added to the difference between the long and short bottom rows will be the number required.

66. How is the shot in an incomplete pile calculated?

By first computing the number in the pile considered as complete, then the number of what the upper part ought to consist; and the difference of these piles will be the number contained in the frustum or incomplete portion.

DIAMETERS OF SHOT, SHELLS AND SPHERICAL CASE.

106]

13-in.	10-in.	8-in.	42	32	24	18	12	6
in.	in.	in.	in.	in.	in.	in.	in.	in.
12.87	9.87	7.88	6.84	6.25	5.68	5.17	4.52	3.58

WEIGHTS OF SHOT, SHELLS AND SPHERICAL CASE.

	Columbiads and Sea-Coast Howitzers.		Mortars			Guns and Howitzers.					
	10-in.	8-in.	13-in.	10-in.	8-in.	42	32	24	18	12	6
	lbs.	lbs.	lbs.	lbs.	lbs.	lbs.	lbs.	lbs.	lbs.	lbs.	lbs.
	Shot, 128 Shells, 101 Spher. case, —	65 50.5 30	— 197 —	— 87.5 —	— 44.5 —	42.7 31 20.3	32.6 22.5 16	24.4 17 11.86	18.5 13.4 8.7	12.3 8.4 6.1	6.1 — 3.06

The 8-inch Mortar Shell is used for the Siege Howitzer.

WEIGHT OF CANISTER SHOT.

42	32	24-pr. Gun and 8-in. Sieg. Howit.	18	12-pr. Gun and 32-pr. Howitzer.	24-pr. Howitzer.	6	12-pr. Howitzer.	
							Field.	Mountain.
lbs. 1.5	lbs. 1.14	lbs. 0.86	lbs. 0.64	lbs. 0.43	lbs. 0.32	lbs. 0.21	lbs. 0.16	Musket ball.

**WEIGHTS OF FINISHED CANISTERS
AND NUMBER OF SHOT.**

	Siege and Garrison Guns.					8-in. Howitzer.	
	42	32	24	18	12	Siege.	Sea-Coast.
	lbs.	lbs.	lbs.	lbs.	lbs.	lbs.	lbs.
Weights,	48	37	29	23	15	53.5	54.5
No. of shot,	27	27	27	27	27	48	48

107]

**WEIGHT OF GRAPE SHOT AND
GRAPE SHOT STANDS.**

	8-in.	42	32	24	18	12
	lbs.	lbs.	lbs.	lbs.	lbs.	lbs.
Grape shot,	6.1	4.2	3.15	2.4	1.8	1.14
Stands,	74.5	51.2	39.7	30.6	22.1	14.8

WEIGHTS OF FIXED AMMUNITION.

WEIGHTS.		For Guns.		For Howitzers.		
		12	6	32	24	12
Cartridge, including { large charge, Cartridge Block. { small “		lbs.	lbs.	lbs.	lbs.	lbs.
		2.56	1.30	3.88	2.70	
		2.06	1.05	3.10	2.34	1.05
Shot strapped,		12.75	6.28			
Shell strapped and charged,		—	—	24.60	18.80	9.35
Spherical case, strapped and charged,		11.43	5.75	31.00	23.00	11.30
Canister, with Sabot,		14.80	7.32	28.50	21.25	10.80
Round of Ammunition { Shot, complete. { Shell with small charge, { Spherical case, Canister,		15.40	7.60			
		—	—	27.70	21.15	10.50
		13.50	6.82	34.10	25.34	12.50
		16.91	8.40	31.60	23.60	11.85

CHARGES FOR MORTAR SHELLS.

	13-in.	10-in.	8-in.	Coehorn.
	lbs. oz.	lbs. oz.	lbs. oz.	lbs. oz.

		13-in.	10-in.	8-in.	Coehorn.
Charge	of the shell filled with powder,	11.0	5.0	2.9	1.0
	to burst the shell,	6.0	2.0	1.0	0.8
	to blow out the fuze,	0.6	0.5	0.4	0.2
Ordinary service charge.	Cannon Powder,	7.0	3.0	1.12	
	Incendiary match or other composition,	0.8	0.6	0.6	

108]

CHARGES FOR FIELD SHELLS.

		32-pdr.	24-pdr.	12-pdr.	Remarks.
		lbs. oz.	lbs. oz.	lbs. oz.	
Powder required	to fill the shell,	1 5	1 0	0 8	Rifle or musket powder is used in pref'nce to cannon.
	to burst do.	0 11	0 8	0 5	
	to blow out the fuse plug,	0 2	0 2	0 1	
	for service charge,	1 0	0 12	0 7	

CHARGES FOR SPHERICAL CASE SHOT.

CHARGE.	8-in.	42	32	24	18	12	6
No. of musket balls,	486	306	225	175	120	78	38
Bursting charge of powder, oz.	15	9	8	6	5	4.5	2.5
Weight of shot loaded, lbs.	59.5	39	30.13	22.75	16.3	11	5.5

CHARGES FOR SHELLS FOR COLUMBIADS AND HEAVY GUNS.

Charge of Powder.	COLUMBIADS.		FOR GUNS.					
	10-in.	8-in.	42	32	24	18	12	
	lbs. oz.	lbs. oz.	lbs. oz.	lbs. oz.	lbs. oz.	lbs. oz.	lbs. oz.	lbs. oz.
To fill the shell,	3 4	1 12	1 8	1 5	1 0	0 11	0 8	
To burst “	1 6	1 0	0 12	0 1	0 8	0 7	0 5	
To blow out the fuze-plug,	0 10	0 8	0 6	0 2	0 2½	0 1½	0 1	
For ordinary service,	3 0	1 8	1 4	1 0	0 12	0 10	0 7	

109]

PART X.

LABORATORY STORES.

1. What is a *fuze*?

The contrivance by which fire is communicated to the charge in a shell. It consists, essentially, of a highly inflammable composition inclosed in a wood, metal, or paper-case.

2. What fuzes are used in the U. S. service?

Wooden, paper, the Bormann and the United States sea-coast fuzes.

3. Describe the *wooden fuze*.

It consists of a conical plug of wood, of the proper size for the fuze-hole of the shell with which it is to be fired. The axis of this plug is bored out cylindrically, from the large down to within a short distance of the small end, which is left solid. At the large end a cup is hollowed out, and the outside of the plug is divided into inches and parts, generally tenths, commencing at the bottom of the cup. The cylindrical space is filled with composition, pounded hard, and as regularly as possible, and the cup filled with mealed powder moistened with whisky or alcohol. The rate of burning is determined by experiment, and marked on a water proof cap, which is tied over the cup. Knowing the time any shell is to occupy in its flight, the fuze is cut off with a saw at the proper division, and firmly set in the fuze-hole with a fuze-set and mallet. Say the fuze burns 5" to the inch. If a shell be 10" in reaching the mark two inches of fuze will burst it as it strikes. If it takes 8" to reach the mark, $1\frac{6}{10}$ in. will be cut off, &c.

4. What is the disadvantage of this fuze?

110]

Its irregularity, it being very difficult to pound the composition so that equal lengths will burn in equal times. The shell may either burst too soon, and a great part of its effect be lost; or it may burst after burying itself in the ground; or it may burst after passing the proper point. This irregularity of burning is common to all fuzes where the composition is driven in successive layers in a column which burns in the same direction.

5. What is the composition for Mortar fuzes?

No.	Nitre.	Sulphur.	Mealed Powder.	Time of burning 1 in.	Remarks.
1	2	1	3	3.8 sec.	For Siege Mortars.
2	2	1	2¼	5. “	“ Sea-Coast “
3	—	—	1	2.2 “	“ 8-in. Howitzers.

6. Is the wooden fuze used?

Yes, for Mortars.

7. Are these fuzes always cut before being inserted in the shell?

Generally they are; but they are sometimes *bored* through at the proper positions instead of being sawed.

8. Are they ever cut obliquely?

[111]

Yes, when the fuze is so long as to render it likely that it will reach the bottom of the shell; for by cutting it perpendicular to the axis, the whole base of the wood might be driven in contact with the bottom of the shell, and prevent the lighted composition from setting fire to the bursting charge.

9. Describe the *paper-fuze*.

It consists of a conical paper-case, containing the composition, whose rate of burning is shown by the color of the case, as follows:

Black,	burns	2"	to the inch
Red,	“	3"	“
Green,	“	4"	“
Yellow,	“	5"	“

Each fuze is made two inches long, and the yellow consequently burns 10". For any shorter time, the fuze is cut with a sharp knife. With this fuze is used a fuze-plug having a conical opening, which is reamed out to fit the paper-case when the shell is loaded, and the fuze is then pressed in with the thumb.

10. What is the great advantage of this fuze?

Its simplicity, and the little trouble required to place it in the shell, which renders unnecessary the numerous and complicated instruments such as saws, fuze-setter, and extractor, files, &c., which were formerly used in field artillery.

11. What is the composition of paper fuzes made of?

	Mealed Powder.	Sulphur.
Black,	1	0
Red,	8	3
Green,	8	3.5
Yellow,	8	4.0

12. Describe the *Belgian* or *Bormann* fuze?

[112]

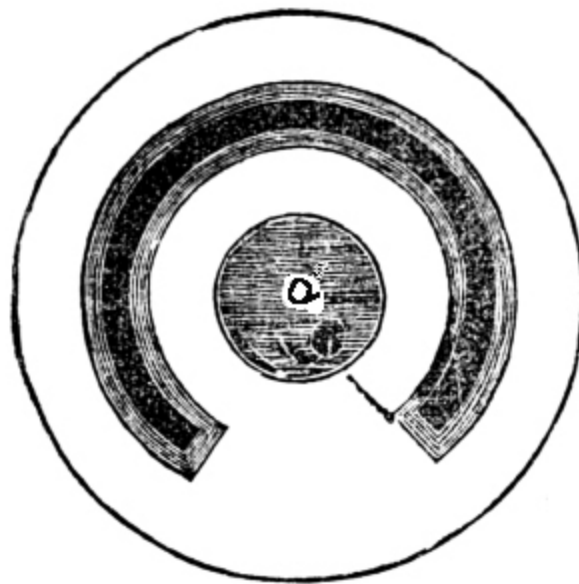


Fig. 2.

The fuze-case is made of metal (a composition of lead and tin), and consists [Fig. 2](#), first, of a short cylinder, having at one end a horse-shoe shaped indentation; *one* end only of which communicates with the magazine of the fuze placed in the centre.

TOPVIEW



SECTION



Fig. 3.

This horse-shoe indentation extends nearly to the other end of the cylinder, a thin layer of the metal only intervening. This is graduated on the outside into equal parts representing seconds and quarter seconds ([see Fig. 4](#)). In the bottom of this channel a smooth layer of the composition is placed, with a piece of wick or yarn underneath it. On this is placed a piece of metal, the cross section of which is wedge shaped ([see Fig. 3](#)); and this, by machinery, is pressed down upon the composition, sealing it hermetically. The cylindrical opening represented at [a Fig. 2](#), is filled with fine powder and covered with a sheet of tin, which is soldered, closing the magazine from the external air.

Before using the fuze, several holes are punched through this sheet of tin, to allow the flame to enter the shell. On the side of the fuze the thread of a screw is cut which fits into one cut on the inside of the fuze-hole, and the fuze is screwed into the shell with a wrench.

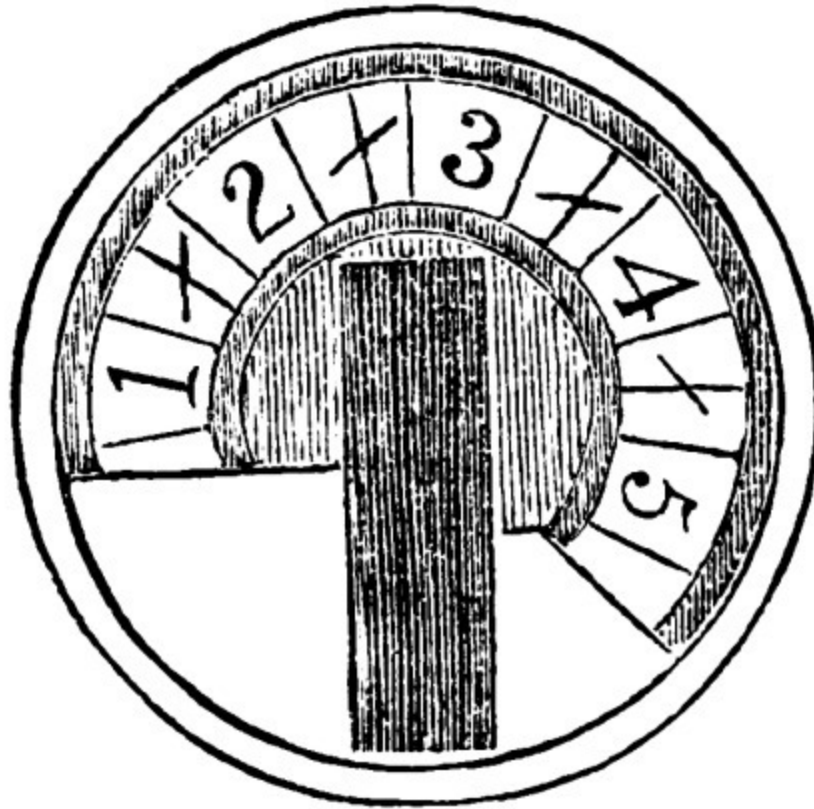


Fig. 4.

The thin layer of metal over the composition is cut through with a gouge or chisel, or even a pen-knife, at the interval marked with the number of seconds which we wish the fuze to burn. To prevent the metal of this fuze, which is soft, from being driven into the shell by the explosive force of the charge, a circular piece of iron, with a hole through its centre, and the thread of a screw on the outside [Fig. 5](#) is screwed into the fuze-hole before the fuze is inserted.

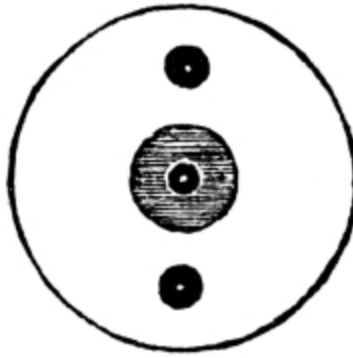


Fig. 5.

13. To what kind of artillery has this fuze been confined?

Principally to light artillery, in firing shells and particularly spherical case, where regularity and certainty are essential requisites.

14. Mention one important advantage of this fuze.

114]

The shells can be loaded, all ready for use, and remain so any length of time, perfectly safe from explosion, as the fuze can be screwed into its place, and the composition never exposed to external fire until the metal is cut through.

15. What is the only operation under fire required?

To gouge through the metal at the proper point, with any kind of chisel, knife, or other instrument.

16. Describe the United States sea-coast fuze.

In the United States, a bronze fuze-plug has been adopted for heavy shells instead of the wooden one. It fits the eye in the same way, and is retained by friction.

It having been found that ricochets, especially over water, were apt to extinguish these fuzes, a safety cap and primer combined have been adopted in the navy. A recess in the top is filled with priming composition and covered, until the fuze is required for use, with a leaden disk which fits accurately the opening. A crooked passage filled with priming, conveys the fire to the fuze composition beneath, and prevents water from being forced in in sufficient quantity to extinguish the fuze.

17. When are paper fuzes for field shells and spherical case inserted?

At the moment of loading the gun, and into wooden *fuze-plugs* previously driven into the shell.

18. What is a *port-fire*?

It consists of a small paper-case filled with a highly inflammable but slow burning composition, the flame of which is very intense and penetrating, and cannot be extinguished by water.

19. What is it used for?

115]

Principally as an incendiary material in loading shells, and for communicating fire to the priming of guns when proving them.

20. What does port-fire composition consist of?

Of nitre, sulphur, and mealed powder, in different proportions. One kind is composed of

Nitre,	65	parts.
Sulphur,	22.5	“
Mealed powder,	12.5	“

A port-fire case, eighteen inches in length filled with this composition, burns ten minutes.

21. What are *priming-tubes*, and their use?

Small pipes having a cap at one end, and filled with a composition for firing cannon.

22. What tube is in general use in our service?

The friction primer.

23. Describe it.

It consists of a short tube of metal inserted into a hole near the top of a longer tube, and soldered in that position. The short tube, is lined with a composition made by mixing together two parts of chlorate of potassa and one of sulphurate of antimony, moistened with gum water. A serrated wire passes through the short tube and a hole opposite to it in the side of the long one, the open end of the short tube being compressed with nippers, and the

wire at the end of the serrated part doubled under to prevent any displacement. The other end of the wire is doubled and twisted by machinery. The long tube is filled with musket powder, its upper end being closed with shellac-varnish, and its lower with shoemakers-wax.

24. What advantage does the friction tube possess?

116]

It gives an enemy at night no clue to the position of your piece, as does the lighted port-fire, or slow-match.

25. What is *slow-match*?

A slow burning match prepared from hemp or flax slightly twisted, soaked in a strong lye, or in water holding in solution sugar of lead. Cotton rope well twisted, forms a good match without any preparation.

26. How long does slow-match prepared from hemp or flax burn?

Four to five inches to the hour.

27. What is the use of slow-match?

It is used principally for the purpose of retaining fire in the shape of a hard-pointed coal, to be used in firing cannon, fire-works, &c. It was formerly used in field batteries for lighting the port fires with which the pieces were discharged; but both are now entirely superseded by the friction tube.

28. What is quick-match?

It is a match made of threads of cotton, or cotton wick, steeped in gummed brandy or whisky, then soaked in a paste of mealed powder and gummed spirits, and afterwards strewed over with mealed powder.

29. How long does it burn?

One yard burns in the open air thirteen seconds.

30. What is the use of quick-match?

To fire stone and heavy mortars, and sometimes in proving pieces. It is extensively used in priming all kinds of fire-works, such as fire and light balls, carcasses, rockets, priming-tubes, &c., and in conveying fire very rapidly from one portion of a piece of fire-work to another.

31. When used for discharging cannon, how is the quick-match set fire to? 117]

By a slow match, port-fire, or any other convenient material.

32. When used to prime carcasses, &c., how is it set on fire?

By the flame from the piece.

33. What is *Valenciennes* composition?

A compound of 50 parts of nitre, 28 of sulphur, 18 of antimony, and 6 of rosin.

34. What is its use?

As an incendiary composition, in charging shells for the purpose of increasing their destructive property, by setting fire to buildings, shipping, &c.

PART XI. PLATFORMS.

1. What is a *platform*?

A strong flooring upon which a piece of ordnance, mounted on its carriage, is manœuvred when in battery.

2. What is the object of a platform?

To facilitate the service of heavy guns and mortars, and to insure accuracy of fire.

3. Mention the kinds of platforms in general use in the service.

Fixed platforms for casemate and barbette batteries in fortifications, which are constructed with the works; the siege-platform for guns and howitzers; and the siege-platform for mortars; the rail-platform; and the ricochet-platform.

4. What properties should wooden platforms possess?

Strength and portability.

5. Are the pieces composing siege-platforms of the same or different dimensions? 119]

All of the same dimensions, viz: 9 feet long, 5 inches wide, and 3½ inches thick; except the sleepers, which in the mortar-platform are one foot less in length.

6. What is the weight of each piece?

About fifty pounds.

7. What is the number of pieces in the siege-platform for guns and howitzers?

Forty-nine in all, one being used as a *hurter* on the front part of the platform to prevent the carriage from running too far forward; and twelve for sleepers.

8. Describe the method of laying a platform for a siege-gun or howitzer.

First establish the centre line of the embrasure, and stretch a cord on this line from the middle of the embrasure to the rear. This is the *directrix* of the platform.

Lay the two outside sleepers parallel to this *directrix*, their outside edges being fifty-four inches distant from it. The four other sleepers are laid parallel to these, the edge of each fifteen and a half inches from the edge of the next. The upper surface of the front ends of these sleepers to be fifty inches on a vertical line below the sole of the embrasure.

They are laid with an elevation to the rear, of one and a half inches to the yard, or four and a half inches in their whole length. This elevation may be determined by placing a block four and a half inches high on the front end of the sleeper, and laying a straight-edge, with a gunner's level on it from this block to the rear end, then so arrange the earth as to bring the level true in this position. The next set of sleepers are laid against and inside of the first, overlapping them three feet, having the rear ends inclined outwards, so that the outer edges of the exterior ones shall be each fifty-four inches from the *directrix*, and the spaces between the edges of the others the same as in the first set, viz: fifteen and a half inches from the edge of one to the edge of the next, all having the elevation to the rear of one and a half inches to the yard, and perfectly level across. The earth is then rammed firmly around these sleepers, and made even with their upper surface. The first deck-plank, with a hole through each end for the eye-bolts, is laid in place perpendicular to the *directrix*, its holes corresponding with those in the sleepers. The hurter is placed on it, and the bolts driven through the corresponding holes in these pieces. The hurter should be so placed as to prevent the wheels from striking against the epaulment when the piece is in battery. If the interior slope has a base of two-sevenths of its height, the inner edge of the hurter should be two and a half inches from the foot of the slope. The other planks are then laid, each one forced against the preceding, the last plank having holes for the rear eye-bolts. By drawing out or driving in the outside sleepers, the holes through their rear ends are made to correspond with those in the last deck-plank, and the bolts are put in.

Drive stakes in the rear of each sleeper, leaving their tops level with the upper surface of the platform. Raise, ram, and level the earth in rear of the

platform, so as to have a plain, hard surface to support the trail when the recoil is great. The earth at the sides should be raised nearly as high as the platform, and well rammed, giving it a slight inclination outwards to allow the water to run off.

9. What are the dimensions of this platform?

121]

Fifteen feet by nine feet.

10. Why is the elevation to the rear given to this platform?

To diminish the recoil and to permit the water to run off.

11. Describe the platform for a mortar.

The mortar-platform is composed of only half the number of sleepers and deck-planks required for the gun or howitzer platform. It is laid level, and the front and rear deck-planks are connected by eye-bolts to every sleeper. Its depth is one-half that of the previous platform.

12. Describe the method of laying the rail-platform.

The rail-platform for siege-mortars consists of three sleepers and two rails for the cheeks of the mortar-bed to slide on, instead of the deck-plank, and is very strong, and easily constructed and laid.

The pieces being notched to fit, are driven together at the battery, the distance between the centre lines of the rails being equal to that between the centre lines of the cheeks. The earth is excavated eight and a half inches, the depth of the sleepers, and the bottom made perfectly level. The directrix being exactly marked by stakes, the platform is placed in position, its centre line coinciding with a cord stretched between the stakes marking the line of fire. The earth is filled in as high as the upper surface of the sleepers, and firmly rammed; and stakes are driven in the rear angles formed by the sleepers and rails, and one at the rear end of each rail.

13. Mention the parts of the *ricochet-platform*.

122]

1 Hurter,	8	ft. long,	8	in. wide, and	8	in. thick.
3 Sleepers,	9	ft. “	5½	“ “ “	5½	“
2 planks,	10	ft. 8 in. l’g,	13	“ “ “	2¼	“
1 plank,	7	ft. long,	13	“ “ “	2¼	“
1 piece plank,	2½	ft. long,	13	“ “ “	2¼	“

And some stakes.

14. Describe the method of laying this platform.

To lay this platform, place the hurter perpendicular to the line of fire, and secure it by four stakes, one at each end and two in front, $31\frac{1}{2}$ inches from the middle towards each end: lay the three sleepers parallel to the hurter, the first 16 inches from the rear edge of the hurter, the second $43\frac{1}{2}$ inches from the rear edge of the first, and the third $43\frac{1}{2}$ inches from the rear edge of the second. Lay the plank $31\frac{1}{2}$ inches from the directrix of the platform to the centre of the plank. Place the piece of plank 60 inches from the rear edge of the last sleeper, and bed it in the ground. Place on the last sleeper and this piece of plank, *the* plank (7 feet long), its front edge 106 inches from the rear edge of the hurter.

PART XII.

ARTILLERY CARRIAGES AND MACHINES.

1. What is meant by artillery carriages?

Carriages of every description employed in the artillery service.

2. How are such carriages classified?

Into two general divisions; first, those carriages on which artillery are mounted, either for firing or travelling; and secondly, such as are especially used for the transportation of artillery ammunition and stores.

3. What is a gun-carriage?

It is the machine on which a piece is mounted for manœuvring and firing.

4. Into what classes may gun-carriages be divided?

Into *movable* and *stationary* carriages.

5. What is the use of *movable* carriages?

They are used for the transportation of the pieces as well as for firing them, and are mounted on large wheels. They are furnished with limbers.

6. Describe the *movable* carriage.

It consists of two cheeks, connected together and with a stock by assembling bolts. The front part supports the piece, and rests upon an axle-tree furnished with wheels, the rear end of the stock or trail resting on the ground.

7. What are the *cheeks*?

124]

The parts of the carriage between which the piece is placed, and upon which the trunnions are supported.

8. What is the wheel composed of?

Of a nave into which the axle-tree enters; of a certain number of spokes fastened in the nave; and a circumference which is composed of a number of fellyes equal to half the number of spokes.

9. What is the *dish* of a wheel?

The inclination outward of the spokes, when fastened in the nave.

10. What is the advantage of this obliquity of the spokes?

It gives elasticity to the wheel, and protects it from the effect of shocks, which would destroy it, if the spokes were in the same plane.

11. What is the object of giving dish to a wheel?

For the purpose of making the body of the carriage wider; to diminish the length of the axle-tree, thus increasing its strength; to throw the mud and water outside the wheels; and to keep the wheel close against the carriage, and prevent any tendency to run off the axle.

12. How are movable gun-carriages distinguished?

As field, mountain, and siege-carriages.

13. What are the principal considerations to be kept in view in the construction of movable carriages? 125]

In firing, the carriage should yield to the recoil. Were it fixed immovably, it would soon be destroyed, no matter how great its solidity. Its weight should be proportional to that of the piece. If too heavy it would soon be destroyed by the shocks of the piece. If too light, the recoil would be immoderate. Its weight should always be less than that of the piece. A heavy piece upon too light a carriage will perform better service than the reverse arrangement, since the effort exerted by a piece depends upon its mass multiplied into the square of the velocity.

14. What are the principal considerations to be kept in view in the construction of field carriages?

Lightness and strength combined, great mobility and flexibility, and a low centre of gravity, in order to surmount all difficulties in the field which must frequently arise while artillery is acting with other troops,—to resist the concussion in firing, and the severe jolting produced when moving rapidly over uneven ground.

15. How many kinds of field gun-carriages have we?

Three, viz: One for the 6 pdr. gun and 12-pdr. howitzer; another for the 24-pdr. howitzer; and the third for the 12-pdr. gun and 32-pdr. howitzer.

16. In what respect are these carriages similar?

In all having the same kind of limber and the same-sized wheels, so that any limber or wheel may be used with any carriage; though, if possible, the heaviest wheel (No. 2) should be used on the carriage of the three heaviest pieces, 12-pdr. gun and 24 and 32-pdr. howitzers.

17. Describe these gun-carriages.

They consist of two short cheeks of wood, bolted upon a stock and wooden axle-body, in a recess of which fits the iron axle on which the wheels are placed. The stock terminates in a *trail* and *trail-plate* which rests on the ground, and has on the end a strong ring called the *lunette*, which is placed on the pintle hook when the piece is limbered. In the stock is placed an elevating screw-box of bronze in which the elevating screw fits.

18. Mention other parts of a field-carriage.

126]

Cap-squares, ear-plates, trunnion-plates, under-strap, elevating screw, wheel guard plate, axle-tree, trail-plate, trail-handles, prolong-hooks, pointing-rings, washer-hooks, lock-chain, sponge-chain, sponge and rammer stop, bolts, rings, bands, hooks, keys, straps, nuts, and nails.

19. What is the limber?

It consists of a similar axle-body, axle, and two wheels, and on these rests a frame-work to receive the tongue. On top of the whole is an ammunition box, the top of which forms a seat for three cannoneers. In rear of the axle-tree is a *pintle-hook* to receive the lunette of the trail. Connected with the frame-work in front, is a fixed *splinter-bar* with four hooks, to which are attached the traces of the wheel-horses. At the extremity of the tongue are placed two pole-chains, by which the tongue or pole is held up, and a pole-yoke with two movable branches, to prevent, as much as possible, the pole from oscillating and striking the horses.

20. What is the use of the limber?

To facilitate the movements of the carriage. By means of it a considerable portion of ammunition and stores may be conveyed for the immediate use of the piece, some of the cannoneers may be seated on the boxes, and by the simple manner in which it is attached to the carriage, the greatest facility is afforded for coming into action, or in retiring.

21. Are there any other advantages from the manner in which the gun-carriage and limber are connected? 127]

These two parts thus possess all the advantages of a four-wheel carriage, and the *freedom* of motion peculiar to each admits of their passing over ground uninjured, or without being overturned or strained, where any other four-wheel carriage would invariably fail.

22. Describe the *mountain artillery* gun-carriage.

It is formed like the field-gun carriage, but much smaller, the cheeks not being formed of pieces distinct from the stock, but all three made of two pieces bolted together. The axle-tree is of wood, which lessens the recoil, and gives an elasticity to the whole carriage, better adapted to resist the shocks of firing. The wheels are but thirty-eight inches high. Ordinarily, over rough ground, the carriage is transported on the backs of mules; but where it is possible, a pair of shafts is attached to the trail to keep it from the ground, and the piece is drawn on its carriage by harnessing one of the pack mules to it. The ammunition is carried in ammunition boxes on the backs of mules.

23. Describe the *prairie carriage*.

The necessity for a small carriage for the mountain howitzer when used on our western prairies, has led to the adoption of a special carriage for that service, with a limber attached as in a field-carriage. This renders the carriage less liable to overturn, and preferable in every respect to the two-wheeled one. The limber is furnished with two ammunition boxes, placed over the axle-tree, and parallel to it, and just wide enough for one row of shells and their cartridges.

24. How many kinds of *siege-gun* carriages are used in our service? 128]

Three; one for the 12-pound gun; another for the 18-pdr.; and the third for the 24-pound gun and 8-in. howitzer.

25. In what respect are they similar?

They are all constructed in the same manner, differing only in their dimensions. All the limbers and wheels are the same, so that they can be used in common.

26. Describe this gun-carriage.

It is similar in its construction to the field-carriage, but is joined to the limber in a different way. Projecting upwards from the limber and in rear of the axle-tree, is placed a pintle, which enters a hole made in the trail from the underside, and a lashing-chain and hook keep the two parts together when once in position. The weight of the trail resting on the rear end of the tongue keeps this nearly horizontal, and relieves the horses of the weight of it, which, as it must be both long and heavy, is too much for the horses to carry.

The splinter-bar is, as in field carriages, stationary, but the traces of the next team are attached to a movable bar which is connected with the end of the tongue. The tongue is furnished with pole-chains, but no yoke, and the rest of the teams are harnessed as in field-artillery. The axle-trees are of iron, with axle-bodies of wood; which last, by its elasticity, renders the shock from the piece less direct and violent.

On the upper surface of the cheeks, near the rear ends, are placed two projecting bolts which, with the curve of the cheeks, form resting places for the trunnions, when the piece is in position for transportation. They are called *traveling trunnion-beds*. When the piece is in this position, its breech rests upon the bolster, which is a curved block of wood, bolted to the upper side of the stock. On each side of the trail, and perpendicular to it, a strong manœuvring bolt is placed to serve as places to apply the handspikes in manœuvering the carriage.

[129]

27. What is the object of the traveling trunnion-beds?

For the purpose of distributing the load more equally over the carriage.

28. Mention the parts composing the limber.

The fork, the splinter-bar, the hounds, the sweep-bar, the tongue, the pintle, the lashing-chain, the axle-tree (iron). The sweep-bar is of iron, and on it rests the trail, which by its weight keeps up the tongue.

29. Why is it unnecessary for siege-carriages to have the same degree of mobility and flexibility as field carriages?

Because siege-carriages are, properly speaking, transportation wagons for use on roads, and never intended for manœuvring with troops.

30. How many horses does the transportation of siege-guns require?

A 24-pdr. requires ten horses (five drivers); a 12 or 18-pdr., eight horses (four drivers).

31. What are stationary gun-carriages used for?

To fire the piece from, and not to transport it except for short distances.

32. For what service are these carriages used?

130]

For garrison and sea-coast pieces; although the siege-gun carriages just described may also be used in a fortification or garrison. Mortar-beds, to be described hereafter, are used either for siege or garrison service.

33. What are the chief requisites for garrison and sea-coast carriages?

Strength, durability, and facility in serving the guns, as they are intended only for the works of a place, coast-batteries, and situations where they are permanently fixed.

34. Why should these carriages be required to possess great strength and durability?

Unless made strong they would soon be shaken by the continued and rapid fire which the defense of a work may demand; and from their constant exposure to the weather they would soon decay if made of a very perishable material.

35. Is the weight of garrison carriages a matter of great importance?

It is of less importance in this class of carriages than in any other, as they are seldom removed from their situations: their weight adds but little to the labor of running them up.

36. Mention the different *stationary* carriages.

The carriage from which a mortar is fired, called its *bed*; the barbette-carriage; the columbiad-carriage; the casemate-carriage; and that for the 24-pdr. iron howitzer, called the flank-casemate carriage.

37. How many kinds of *siege mortar* beds have we?

Four; the 8-in., 10-in., the stone, the coehorn.

38. Which of these are alike?

[131]

The first three, differing only in dimensions. They are made of cast-iron, which has very little elasticity.

39. Describe these beds.

They consist of two cheeks, joined by two transoms, all cast together in the same piece. The manœuvring bolts, placed on each side, one near each end of the cheeks, are made of wrought iron, and set in the mould when the bed is cast.

On the front transom is fastened a wooden *bolster*, grooved to receive the elevating quoin. Notches, on the underside of the front and rear of the cheeks, give hold to the handspikes in throwing the carriage to the right or left.

40. Describe the *coehorn* mortar-bed.

It is made of a block of oak wood, in one piece, or two pieces joined together with bolts. A recess, for the trunnions and part of the breech is made in the top of the bed; and the trunnions are kept in their places by plates of iron bolted down over them. Two iron handles are bolted to the bed on each side, by which four men can carry the bed with the mortar in its place.

41. Describe the *eprouvette* mortar-bed.

It consists of a block of wood, on the top of which is countersunk and bolted the bed-plate, which is a heavy circular plate of cast-iron having a rectangular recess with sloping sides, so as to make it longest at the bottom. Into this recess the sole of the mortar slides. The wooden block is bolted to a stone block of the same size, which is firmly placed in the ground on a masonry foundation.

42. Describe the *heavy sea-coast* mortar bed.

The bed for the heavy ten-inch mortar is the only one which has yet been adopted. The cheeks are of cast-iron, and somewhat similar in form to those in the beds of siege-mortars; but in the front, the cheeks turn up to receive between them the front transom, which has, countersunk in and bolted to it, an elevating screw-bed, through which works an inclined elevating screw, which rises or falls by turning the nut, fitted on it by means of a lever inserted into mortises cut in the direction of the radii of the circular nut. 132]

Both the transoms are made of wood, connected with the cheeks by mortises and tenons, and secured by bolts running through, and nuts on the outside. One of these bolts at each end, is longer than the others, and the projecting ends are made use of as manœuvring bolts. Directly behind and underneath the position for the trunnions, a bronze bed-piece is placed to receive the shock of the piece. It consists of a large beam of bronze, with each end well let in to the face of the cheek. The use of the elevating screw instead of the quoin, is rendered necessary by the great mass of metal to be raised or lowered in sighting the piece.

43. What is a *barbette* carriage?

It is a carriage belonging to the class denominated immovable, on which a gun is mounted to fire *over* a parapet; and a barbette gun is any gun mounted on a barbette-carriage.

44. How many forms of the barbette-carriage are in use in the service?

Two: one for *iron guns* and sea-coast howitzers (12, 18, 24, 32, 42-pdrs., and 8 and 10-in.); and one for the columbiads.

45. Of how many parts are barbette-carriages composed?

133]

Of a *gun-carriage* and a *chassis*.

46. Describe the gun-carriage.

It is formed of two upright pieces of timber, nearly vertical, behind which are placed two inclined braces, mortised into the uprights, and designed to receive the force of the recoil, the whole forming the cheeks, which are firmly connected and braced by transoms and assembling bolts, thus forming a *triangular* frame-work, *which* is less liable than any other form to become deformed from the shocks of the gun. A horizontal piece

(the transom and axle-tie) runs from front to rear between the cheeks, connecting the axle-body and rear transom. The trunnion-bed is at the top of the upright, where it is joined to the brace; and the breech of the gun is supported on an elevating screw, working into a screw-box placed in the rear end of the transom and axle-tie. The front transom is just under the gun; the middle transom is between the braces; and the rear transom is at the lower end of the braces, and under the transom and axle-tie, into which it is notched; the lower part of this transom is notched to receive the tongue of the chassis on which it slides. Between this transom and the transom and axle-tie, the end of a lunette is placed projecting to the rear, and fastened by a bolt for the purpose of attaching a limber to the carriage.

The feet of the uprights and front end of the transom and axle-tie are joined to an axle-body, in which an iron axle is placed. On the ends of the axle are fitted cast-iron rollers, which rest on the rails of the chassis, and support the front of the carriage. On the outside of the roller is placed an octagonal projection, on which the cast-iron nave of the wheel fits, secured by a washer and linch-pin. The spokes of the wheels are wood, inclosed within heavy iron tires. Manœuvring bolts are inserted in front of the feet of the uprights, and in the carriages, for pieces heavier than a 24-pdr., in rear of these feet also. These bolts and the spokes of the wheels form the points of application for the handspikes, in manœuvering the piece. Manœuvering staples are placed in front of the feet of the braces, for the purpose of using handspikes to raise the rear of the carriage from the tongue of the chassis in running to and from battery. 134]

47. What pieces go on the same carriage?

The 32-pound gun, and 8-inch howitzer. All other pieces have separate carriages.

48. How many sizes of rollers are used?

Two: one for the carriages of the 12, 18, and 24-pounders; the other for the remaining carriages.

49. Are cap-squares used with these carriages?

No.

50. What other purposes do the wheels subserve besides assisting in manœuvering the gun-carriage?

In transporting the piece on its carriage for short distances, as from one front of a fort to another.

51. Describe the chassis.

135]

It consists of two *rails* and a *tongue*, joined by three transoms. The tongue is in the middle, and projects considerably beyond the rails, to the rear. At each end of the rails on top, a hard piece of wood is notched in, and bolted. They are called hurters and counter-hurters, and their use is to prevent the gun carriage from running off the chassis. Rail-plates of iron to protect the wooden rails are let in to the outside of the rails. At the rear end of the tongue, a swinging prop is placed to support the end of the tongue when the piece is run back. The lower side of the end of the tongue is notched out, and a manœuvering loop fixed there, with a bolt and screws, to assist in handling the chassis. On the under side of each rail, opposite the rear transom, a mortise is formed, for the reception of a socket of iron which receives the *handle* of the traverse-wheel fork. Each of these forks receives a traverse-wheel, joined to it by an axle-bolt, and these support the rear end of the chassis. The front end is supported on a pintle-plate of iron; through which, and up into the middle of the front transom, passes a pintle or bolt, which serves as a pivot around which the whole system moves.

52. In permanent batteries, how are the pintle and traverse circle fixed?

The pintle is fixed in a block of stone, and the traverse circle is an iron plate set also in stone.

53. In temporary batteries, how is the pintle attached?

To a wooden bolster which is covered by a circular cast-iron plate, and attached by bolts to a wooden cross picketed firmly into the ground.

54. How may a temporary traverse circle be made?

Of plank, pinned to sleepers, and fastened to pickets, or secured to string-pieces, which connect the traverse circle with the pintle-cross.

55. What retains the traverse-wheels and their forks in their places? 136]

The weight of the carriage and gun, and the form of the socket and handle of the fork.

56. Where are the handspikes applied in traversing the carriage?

To the pivot-bolts of the traverse-wheels, which project to the rear; or under the traverse-wheels.

57. Why does the chassis slope towards the front?

In order to diminish the recoil, and aid in running the piece into battery.

58. Describe the *columbiad* gun-carriage.

It is a triangular frame-work, consisting on each side of an upright, a horizontal rail or tie, and a brace, firmly mortised and bolted together, forming the cheeks, which are joined by a transom at each end. These project below the lower surfaces of the ties, and fit in between the rails of the chassis, serving, like the flanges on the rollers in the other barbette-carriages, to prevent the gun-carriage from slipping sideways off the chassis.

Through the front transom, and near the front ends of the ties, an iron axle-tree is passed, working in iron boxes fitting in the ties. On the projecting ends of this axle-tree the rollers or manœuvring wheels are fixed (the axes of which are *eccentric* with the axis of the axle-tree) the extreme ends of the axle, being octagonal in shape, to fit the wrench of the iron handspike.

These *eccentrics* are so arranged that when the centres of the wheels are at their lowest points, the surfaces of the wheels bear on the rails of the chassis and raise the gun-carriage tie from it; and when the centres are at the highest points, the surfaces of the wheels do not touch the rails, and the ties are in contact with them.^[2] A similar arrangement is made for the rear part of the carriage, except that the axle does not extend all the way through, but the wheel on each side has a projecting piece of axle which works into a box placed near the end of the tie.

The wheel is thrown into or out of gear, that is, made to bear on the rail of the chassis or relieved from it, by turning the axle with a wrench placed on the octagonal end. In the direction of the radii of the wheels, but inclined outwards, mortises are placed for the reception of the end of the iron handspikes, by acting on which while inserted the wheels are turned, and the carriage moved back and forth on the chassis. Ordinarily, when the

wheels are thrown into gear, the carriage being back, it will run into battery of itself.

The elevating arrangement consist of an elevating screw, working into a screw-bed, which slides in a vertical box, and carries on the top of it a movable *pawl* to fit into the notches cut in the breech of the gun, in order to give considerable elevations. For the purpose of transferring the pawl from one notch to the next, it has a slit in it, through which the elevating bar is passed, and the gun supported by making use of the edge of the elevating-box as a fulcrum. This arrangement is over the rear transom.

59. Describe the chassis of the columbiad-carriage.

[138]

It consists, like those used with other barbette-carriages, of two rails connected by three transoms; but the tops of the rails are shod with iron plates, and the rear hurters are the large heads of heavy bolts which pass entirely through the rails. The front hurters are fixed to the front transom by a heavy plate and bolt.

Traverse-wheels are placed under both front and rear transoms, and the chassis moves on a pintle passing through the middle transom. Two of these wheels are placed under each end of the chassis, their axes being kept in place by straps bolted to the transoms. Recesses are cut in the underside of the transom, for the wheels to turn in. This chassis has no tongue.

60. Of how many parts are casemate carriages composed?

Like barbette-carriages, of a *gun-carriage* and *chassis*.

61. Describe the gun-carriage.

It consists of two cheeks, joined together by as many transoms, and supported in front by an axle-tree on truck wheels, and in rear on the rear transom, which is notched to fit the tongue of the chassis. Each cheek is formed of two pieces, one on top of the other, and connected by dowels and bolts. On the underside, near the front, a notch is cut for the reception of the axle-tree, which is of oak; and nearly over the axle, on the upper side of the cheek, the trunnion-bed is placed. The rear of the upper piece of the cheek is cut into steps, which give a better hold for the assembling-bolts, than a uniform slope, and give purchases for the handspikes, in elevating the piece. On the inside of each cheek, just in rear of the axle, a vertical guide is fixed to keep the carriage on the chassis. It is

[139]

of wood, and bolted to the front transom and axle-tree. The top of the front transom is hollowed out, to admit the depression of the piece. Behind the rear transom and at the notch cut in it, there is an eccentric-roller, so arranged as to bear the weight of the rear part of the carriage, or not, according as it is thrown in or out of gear.

Near the rear end of each cheek, and outside, a heavy trail-handle of iron is placed, and used in manœuvring the piece. On the ends of the axle truck wheels are placed, with mortises sloping outwards in the direction of the radii, for the insertion of the handspikes in running from battery.

The elevating apparatus consists of a cast-iron bed-plate, secured to the rear transom; an elevating screw and brass nut; the nut being acted on by an oblique-toothed wheel, turned by a handle placed outside the right cheek.

62. Describe the chassis.

It consists of two rails and a tongue, joined by two transoms, and supported on traverse-wheels in front and rear. The track on each rail is curved up at each end, and provided with hurters to prevent the carriage from running off the chassis. A prop fastened under the rear end of the tongue prevents the chassis from upsetting backwards in firing heavy charges, and may be used as a point of support in raising the chassis. An iron fork is bolted to the under side of the front end of the tongue, to which is bolted an iron tongue. An opening in the masonry below the embrasure, is left for this tongue, and it is secured in its place by dropping the pintle from the embrasure down through the eye of the tongue.

63. Where is the *flank-casemate* carriage employed?

140]

It is especially adapted to the mounting of the 24-pdr. iron howitzer in the flanks of casemate batteries, for defending the ditch; and both the gun-carriage and chassis are narrower and lighter than the other casemate carriages.

64. Describe the gun-carriage.

The cheeks are made of white oak, and connected by two iron transoms, the front one projecting below the cheeks, and resting on the chassis with a projection on the bottom of it, fitting in between the rails. The bottom of the trail has the same slope as the upper surface of the chassis on which it rests; so that when its eccentric-roller is out of gear, the rear parts of the cheeks fit

the rails. The remaining portion of the bottom of the cheek makes an angle with the rail, and has in front a fork, and a roller which runs on the rail of the chassis when the eccentric is in gear. Each cheek has on the side a trail-handle and a manœuvring-ring. In rear of the rear transom is placed an eccentric-roller, having a projection in the middle of it, just large enough to fit in between the rails of the chassis, and guide the trail of the carriage. When this *roller* is in gear, the weight of the trail rests upon *it*, while that of the front part of the carriage is thrown upon the front rollers, and the piece is then easily run in and out of battery; but the roller being out of gear, as when the piece is about to be fired, the weight rests upon the rear part of the cheeks and the front transom, and friction is brought into play to diminish the recoil. Cap-squares are used with this carriage.

65. Describe the chassis.

141]

It consists simply of two rails 3 in. apart, and joined by four transoms and assembling-bolts. Hurters on the rear ends of the rails only are used, as the bottom projection of the front transom prevents the carriage running too far into battery. The front end of the chassis rests on the sole of the embrasure. The end is provided with a pintle-plate and a strap of half-in. iron through which the pintle passes to the masonry beneath. The rear of the chassis is supported by an iron prop, the lower end of which is attached to two traverse-wheels.

66. What kind of carriages have been recommended for use in garrisons, instead of wooden ones?

Wrought iron carriages, for all except the flank-casemate; all made in a similar manner, differing only in weight and dimensions.

67. What is the objection to cast-iron?

Its weight, and its great liability to splinter when struck by shot.

68. Of how many parts is this wrought iron carriage composed?

Like the wooden ones, of a gun-carriage and chassis.

69. Describe the gun-carriage.

It consists of two cheeks of thick sheet-iron, each one of which is strengthened by three flanged iron plates bolted to the cheeks. Along the bottom of each cheek, an iron shoe is fixed with the end bent upwards.

In front, this bent end is bolted to the flange of the front strengthening plate. In rear the bent portion is longer, and terminated at top by another bend, which serves as a point of application for a lever on a wheel, when running to and from battery. The trunnion-plates fit over the top ends of the strengthening plates, which meet around the bed, and are fastened to the flanges of the latter by movable bolts and nuts. [142]

The *cheeks* are joined together by transoms made of bar-iron. They are parallel to each other, and in order that the *base ring* on the gun may not interfere with giving it the full elevation by striking against them, it is proposed to dispense with the *base ring* in sea-coast and garrison pieces, and to retain the preponderance by reducing the swell of the muzzle.

The front of the carriage is mounted on an axle-tree, with truck wheels similar to the wooden casemate carriages.

70. Describe the chassis.

It consists of two rails of wrought iron, the cross section of each being in form of a T, the flat surface on top being for the reception of the shoe-rail of the gun-carriage. The rails are parallel to each other, and connected by iron transoms and braces. The chassis is supported on traverse-wheels.

A prop is placed under the middle transom of the chassis, to provide against sagging.

71. What carriage is used for conveying ammunition for a field battery?

The CAISSON.

72. Describe it.

A four-wheel carriage, consisting of two parts, one of which is a limber similar to that of the gun carriage, and connected in a similar way by a wooden stock and lunette. [143]

On the axle-body of the rear part, and parallel to the stock, are placed three rails upon which are fastened two ammunition boxes, one behind the other, and similar to the one on the limber; so that the caisson has three ammunition boxes which will seat nine cannoneers. The interior compartments of the ammunition boxes vary according to the nature of the ammunition with which they are loaded. In rear of the last box is placed a spare wheel, axle of iron, with a chain and toggle at the end of it. On the

rear end of the middle rail is placed a carriage hook, similar to a pintle-hook, to which the lunette of a gun carriage whose limber has become disabled, may be attached, and the gun carried off the field.

The caisson has the same turning capacity and mobility as the gun carriage, so that it can follow the piece in all its manœuvres, if necessary. It also carries a spare wheel, spare pole, &c.

73. What provision is made for repairing the carriages of a field battery when required?

Every field battery is provided with a FORGE.

74. Describe this wagon.

It consists, besides the limber, of a frame-work on which is fixed the bellows, fire-place, &c. Behind the bellows is placed a coal-box, which has to be removed before the bellows can be put in position. In the limber box are placed the smith's tools, horse-shoes, nails, and spare parts (iron) of carriages, harness, &c.

75. Describe the *battery wagon*.

144]

It consists, besides the limber, of a long-bodied cart with a round top, which is connected with the limber in the same way as all other field carriages. The lid opens on hinges placed at the side; and in rear is fixed a movable forage rack for carrying long forage. One of these wagons accompanies each field-battery, for the purpose of transporting carriage-maker's and saddler's tools, spare parts of carriages, harness and equipments, and rough materials for replacing different parts.

Both this and the forge are made of equal mobility with the other field carriages, in order to accompany them wherever they may be required to go.

76. How many kinds of wheels are employed for field carriages?

Two: No. 1 for the 6-pdr. gun carriage, the caisson, the forge, the battery-wagon, and for the limbers of all field-carriages. No. 2 for the 24-pdr. howitzer and the 12-pdr. gun carriages.

77. In what respects are these wheels similar?

They are of the same form and height, and they fit on the same axle-tree arm. The height is 57 inches, and each wheel is composed of 14 spokes and 7 fellies.

78. How do they differ?

In the dimensions of their parts, and in strength and weight.

79. What is the weight of these wheels?

No. 1, 180 lbs., No. 2, 196 lbs.

80. What are the weight and height of a wheel of siege-gun carriages and limbers?

Weight 404 lbs., and height 60 inches.

81. What is the *portable forge* designed for?

Service in a mountainous country, where wheeled vehicles cannot travel, for the purpose of making repairs, not only for the artillery but for all other arms of service taken on such expeditions.

82. What is the *mortar wagon* designed for?

145]

The transportation of siege mortars and their beds, or of guns or large shot and shells.

83. Describe this wagon.

The limber and wheels are the same as those of the siege-gun carriage. The body consists of a platform of rails and transoms, resting on an axle-tree, the two middle rails being prolonged to form the stock; six stakes or standards are inserted in sockets on the side of this platform and used to secure the load.

The side-rails are prolonged to the rear, and furnish pivots for a roller placed immediately in rear of the platform. This roller has holes for the insertion of handspikes, and is used in loading the wagon; the guns, mortars, &c., being drawn up on the stock.

A muzzle bolster on the stock near the limber, and a breech-hurter near the hind part of the wagon, are provided and used when long ordnance is transported on it.

Mortars are usually carried mounted on their beds.

84. What is the use of the *hand-cart*?

For the transportation of light stores in siege and garrison service.

85. Describe it.

It consists of a light body with shafts, mounted on two wheels. The shafts are joined together at the ends, and supported immediately in front of the body by iron legs.

86. What is the use of the *hand sling-cart*?

146]

It is used in siege and garrison service for transporting artillery short distances.

87. Describe it.

It is a two-wheeled carriage made entirely of iron, except the pole, which is of oak. The axle-tree is arched to make it stronger, and connected with the pole by strong wrought iron straps and braces. In the rear of the axle a projection is welded to receive the end of a strong hook. The end of the pole terminates in a ferule and an eye. The eye is for the purpose of attaching to the cart when necessary, a limber or a horse.

88. How great weights can be transported by this cart?

It should not be used with heavier weights than about 4000 lbs., but in case of necessity a 24 or 32-pd. gun may be transported on it. For heavier guns or material, the large *sling-cart* drawn by horses or oxen should be used.

89. What is the *field and siege gin* and its use?

It consists, like all gins, of two legs and a pry-pole, a windlass, sheaves, pulleys, and a fall or rope, and is used for mounting or handling guns, or other heavy bodies, in the field or in the trenches of a siege. The legs are about 14½ feet long and the height of the gun about 12 feet.

90. How does the *garrison gin* differ from the field and siege gin?

It is heavier and stronger, as it is used for mounting heavier guns, and has not to be transported like the other with an army in the field. The legs are longer and the gin higher than the other.

91. Describe the *casemate gin*.

It does not differ from the garrison gin except in its height (which is about that of the field and siege gin) and the thickness and strength of the parts.

PART XIII.

PRACTICAL GUNNERY.

1. How may the velocity of a shot or shell be ascertained?

Approximately by the empirical formula,

$$V = 1600\sqrt{(ac/w)}.$$

Where V = initial velocity.

a = a coefficient, whose value depends on the windage.

c = charge

w = weight of ball } in lbs.

The values of a are:

Windage.	Values of a .
0.175	3.6
0.125	4.4
0.090	5.0

2. Does a shot or shell continue at the same uniform velocity during its flight?

The velocity decreases as the distance increases, in a proportion a little higher than the squares of the velocities throughout.

3. What causes a decrease in the velocity of a shot?

The resistance of the air, which varies as the square of the velocity of the shot.

4. With balls of different diameters, and equal velocities, to what is the resistance of the air proportional? 148]

Their surfaces, or the squares of their diameters.

5. Would the velocity of the shot be increased by lengthening the gun?

Only up to a certain point; in a proportion which is nearly the mean ratio between the square and cube roots of the length of the bore. It is found

that the velocity given by long guns is reduced to an equality with that of short guns within a short distance from the muzzle when fired with similar charges.

6. Would the velocity of a shot be increased by entirely preventing the recoil, or by adding greatly to the weight of the gun?

In neither case would any sensible effect be produced on the velocity.

7. Would the velocity of the shot be increased by using a larger charge of powder?

Only to a certain point, peculiar to each gun; by further increasing the charge the velocity would be gradually diminished; yet the recoil is always increased by an increase of charge.

8. What is the ratio of the velocities of shot, when of different weights, but fired with similar charges?

The velocities are inversely as the square roots of their weights.

9. What is the ratio of the velocities of shot of equal weights when fired with different charges of powder?

The velocities are directly as the square roots of the charges.

10. How may the velocity be increased without augmenting the charge of powder? 149]

By decreasing the windage; the loss of velocity by a given windage being directly as the windage. From 1-8 to 1-12 is lost by a windage of 1-40 diameter.

11. What is meant by the time of flight of a shot or shell?

The time during which it is passing through the air from the piece to the first graze.

12. When firing with common shells at 45° elevation, how is the time of flight found?

Extract the square root of the range in feet and divide by 4, or divide the range in feet by 16 and extract the square root of this quotient.

NOTE.—Range in feet = $\frac{1}{2}gt^2 \times \cotangent \text{ elevation.}$

$$= 16t^2 \times \text{cotangent elevation.}$$

$$= 16t^2 \text{ where the elevation is } 45^\circ.$$

$$\text{Or } t = \frac{1}{4}\sqrt{\text{range in feet for elevation } 45^\circ}.$$

13. Having the time of flight, how is the range ascertained?

Multiply the square of the time of flight by 16 for the range in *feet* (the elevation being 45°).

14. What is meant by the penetration of projectiles?

The depth to which they are forced when fired into any resisting medium.

15. What depth do shot penetrate?

The penetration of balls of the same size, with different velocities or charges, is nearly as the squares of the velocities; where the balls are of different sizes the penetration will be proportionate to their diameters multiplied by the density, and inversely as the tenacity of the medium.

16. Mention the depth of penetration in case of the 24-pdr. siege-gun. 150]

At 100 yards a 24-pdr. ball with a charge of one-third of its weight will penetrate as follows:

	Feet.	Inches.
In earth of old parapets,	8	6
“ “ recently thrown up,	15	0
“ Oak wood, sound and hard,	4	6
“ Rubble stone masonry,	1	10
“ Brick,	3	0

17. What is the depth of penetration of field pieces?

Fired at the distance of 500 or 600 yards, the penetration will be from $4\frac{1}{2}$ to 6 feet in parapets recently constructed, and will traverse walls of ordinary construction; but a 12-pounder is necessary to make a breach in walls of good masonry and of 4 feet in thickness, and in this case the position of the battery must be favorable, and the operation a slow one.

18. In attacking a post, or fortified position, in what manner should the fire from artillery be carried on?

Previous to an assault, the artillery ought to support the other troops by a combined fire of guns, howitzers, and small mortars, so that, if possible, the fire may be simultaneous, as such a diversity of projectiles would tend to distract the defenders and prevent them from extinguishing any fire among buildings, besides throwing them into confusion at the moment of assault. In cases of surprise, when immediate action is required, the above method cannot, of course, be practicable.

19. When firing guns of different calibres at long ranges, what are the probabilities of hitting the object?

As the squares of the diameters of their respective shot, when of equal density, and fired with proportional charges.

PART XIV. MISCELLANEOUS.

1. What is the velocity of sound in the air?

At the temperature of 33° the mean velocity of sound is 1100 feet in a second. It is increased or diminished *half a foot* for each degree of temperature above or below 33° .

2. How can the distance of an object be ascertained by the report of fire-arms?

By observing the number of seconds that elapse between the flash and the report of a gun, and multiplying the number by the velocity of sound in the air.

3. What is momentum?

The force possessed by a body in motion; and is measured by the product of the mass of the body into its velocity.

4. When equal masses are in motion, what proportion do their momenta bear to their velocities?

They are proportional to their velocities.

5. When velocities are equal, what proportion do their momenta bear to their masses?

They are proportional to their masses.

6. What proportion do the momenta bear to each other when neither the masses nor velocities are equal?

They are to each other as the products of their masses into their velocities respectively.

7. What is the average weight of a horse?

About 1000 pounds.

8. What space does a horse occupy in the ranks; in a stall; and at a picket? 152]

In the ranks a front of 40 in., a depth of 10 feet; in a stall; from 3½ to 4½ feet front; at picket 3 feet by 9.

9. What are the comparative effects of the labor of a man, and that of a horse or mule?

Taking the useful effect of a man's daily labor as unity, a horse can carry a load on a horizontal plane, 4.8 to 6.1 times; and a mule, 7.6 times greater than a man. Taking a man with a wheel-barrow as unity, a horse in a four-wheel wagon can draw 17.5, and in a cart 24.3; and a mule in a cart, 23.3 times greater burden.

10. What weight is an artillery horse required to draw?

Not more than 700 lbs., the weight of the carriage included.

11. What weight can a team of four horses or more, draw with useful effect?

Including the weight of carriage, 4 horses can draw 24 cwt., or 6 each; 6 horses, 30 cwt., 5 each; 8 horses, 36 cwt., 4½ each; and 12 horses, 48 cwt., or 4 each. It is usual to estimate the weight of a carriage exceeding 12 cwt. as part of the load.

12. What weights are carried by the riding, pack, and draught horses respectively?

A horse carrying a soldier and his equipments, (say 225 lbs.) travels 25 miles in a day (8 hours); a pack-horse can carry 250 to 300 lbs., 20 miles a day; and a draught-horse, 1600 lbs. 23 miles a day, weight of carriage included.

13. What are the usual paces for horses in the artillery?

Walk, trot and gallop; the last is seldom necessary.

14. What is considered an ordinary day's march for field artillery, and rate of motion? 153]

An ordinary march is about 15 miles at 2½ miles per hour for 6 hours; this must depend upon the condition of the horses, state of the roads, and

various other circumstances. Horses starting fresh, and resting after their work, may, on tolerable roads, perform 2 miles in half an hour; 4 miles in 1½ hours; 8 in 4, and 16 in 10 hours.

15. What is the rate of march of horse-artillery and cavalry?

Walk, 3¾ miles per hour, or 1 mile in 16 minutes; trot, 7½ per hour, or 1 mile in 8 minutes; manœuvring gallop, at the rate of 11 miles per hour, or 1 mile in 3 minutes; cavalry charge, 24 miles an hour, or at the rate of 1 mile in 2½ minutes.

16. At what rate does infantry march?

In common time, 90 steps = 70 yards in 1 minute, or 2 miles 680 yards in an hour; in quick time, 110 steps = 86 yards in 1 minute, or 2 miles 1613 yards in an hour; in double quick, 140 steps = 109 yards in 1 minute, or 3 miles 1253 yards in an hour.

17. What space does a foot soldier occupy in the ranks, and what is his average weight?

A front of 20 in., and a depth of 13 in., without the knapsack; the interval between the ranks is 13 in.; 5 men can stand in a space of 1 square yard. Average weight of men, 150 lbs. each.

18. What is the daily allowance of water for a man?

154]

One gallon, for all purposes.

19. What is it for a horse?

Four gallons.

20. What is the weight of a bushel of oats; or of wheat; and the weight of hay?

40 lbs., or 32.14 lbs. to the cubic foot, in case of oats; 60 lbs. to the bushel, or 48.21 lbs. to the cubic foot, in case of wheat; hay pressed in bundles, weighs 11 lbs. per cubic foot.

21. What weight does an infantry soldier carry when in marching order?

About 45 lbs. in all. His knapsack when packed weighs 24 lbs.; canteen when filled, and one day's provisions in haversack, 5 lbs.; rifle, musket,

sling, and bayonet, 10½ lbs.; belts, complete, including 20 rounds of ammunition, 6 lbs.

22. How is the area of a circle found;

Square the diameter, and multiply by .7854 for the area; or square the circumference, and multiply by .07958 for the same result.

23. How is the content of a conical frustum found?

Add into one sum, the areas of the two ends and the mean proportional between them; take one-third of that sum for the mean area, and multiply it by the perpendicular height of the frustum, for its content.

24. How is the mean proportional found for the above?

155]

By multiplying the areas of the two ends together and extracting the square root of their product. A more simple rule is the following: As the diameter of the large end is to that of the small end, so is area of base to mean proportional required.

25. How is the content of a spherical segment found?

From three times the diameter of the sphere, take double the height of the segment, then multiply the remainder by the square of the height, and this product by .5236; or, to three times the square of the radius of the segment's base, add the square of its height, then multiply the sum by the height, and this product by .5236, for the content.

26. How is the capacity or content of a Gomer chamber computed?

This chamber being the frustum of a cone with a hemispherical bottom, its capacity will be found by applying the foregoing rules, viz: first find the content of the frustum, then that of the spherical segment or bottom, and add their contents into one sum for the capacity.

27. How is the content of a rectangular box ascertained?

Multiply the length by the breadth, and this product by the depth.

28. How is the capacity of a cylinder calculated?

Multiply the area of the base by the height.

29. How is the content of a barrel found?

Multiply half the sum of the areas of the two interior circles, taken at the head and bung, by the interior length; or, to the area of the head, add twice the area at the bung, multiply that sum by the length, and take one-third of the product for the content.

30. What is meant by the term, enfilade?

156]

Sweeping the whole extent of a work, line of troops, deck of a ship, &c., with shot or shells.

31. What does defilade mean?

The art of disposing guns, troops, or works in such a manner, that they shall be protected from a plunging-fire from adjoining heights.

32. What are the dimensions required for an earthen parapet to resist the fire of field or siege guns?

6 feet for 6-pdrs.; 14 feet for 12-pdrs.; 18 feet for 24 or 18-pdrs.; four feet of oak or brick will resist cannon shot.

33. What thickness of ice will admit the passage of infantry, cavalry, and artillery?

Ice 3 inches thick, will bear infantry marching in file; from $4\frac{1}{2}$ to $6\frac{1}{2}$ inches, cavalry and light artillery; and beyond that the heaviest gun carriages may pass in safety. Ice 8 inches thick will bear nearly 10 cwt. upon a square foot without danger.

34. How is the size of a rope designated?

By its circumference: thus, a two-inch rope is a rope two inches in circumference.

35. How is the strength of a hemp rope, or the weight it will support, ascertained?

Square the circumference in inches, and divide by 5, for the weight in tons, that it will bear suspended from it.

36. How can the breadth of a river be ascertained without instruments?

As follows:

157]

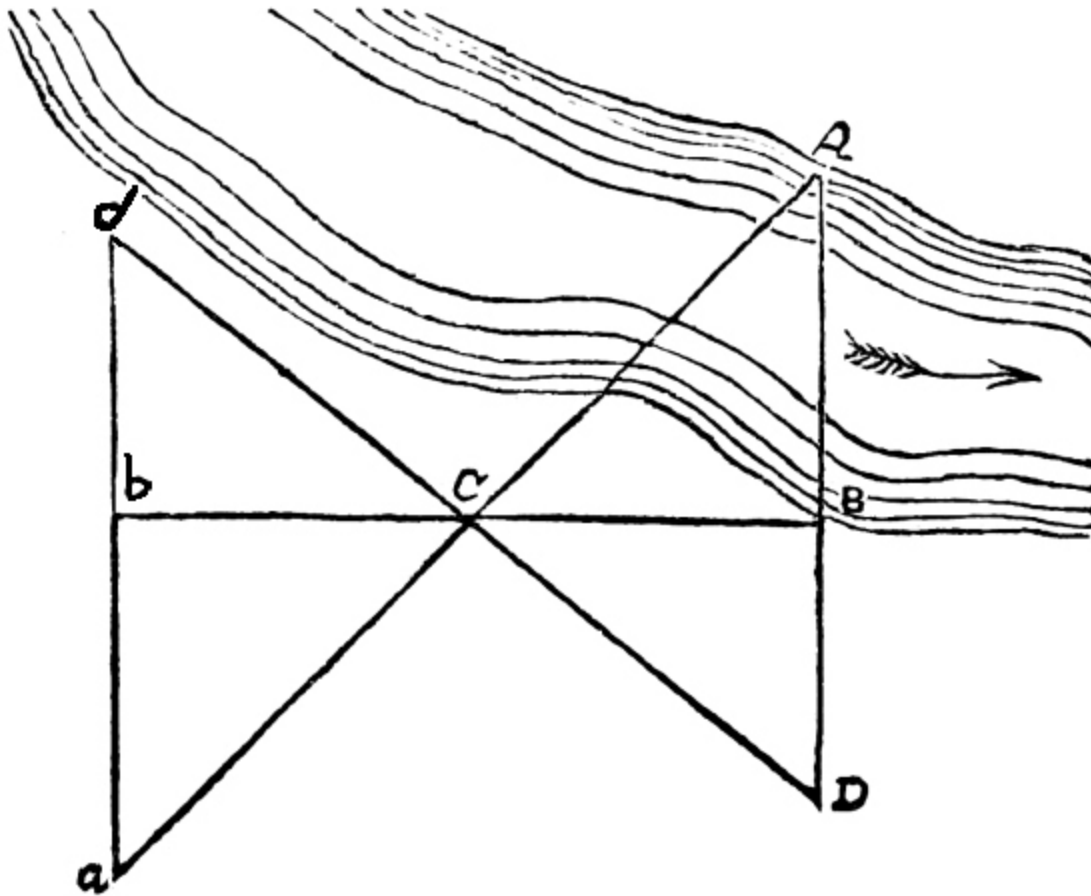


Fig. 6.

1st. The line AB (the distance to be determined) is extended upon the bank to D , from which point, after having marked it, lay off equal distances DC and Cd ; produce BC to b , making $Cb = CB$; then extend the line db until it intersects the prolongation of the line CA at a . The distance ab is equal to AB or the width of the river.

2d. Lay off any convenient distance, BC , perpendicular to AB , erect a perpendicular DC to AC , note the point D where it intersects AB produced; measure BD ; then

$$AB = \frac{\overline{BC}^2}{BD} \cdot [3]$$

37. How can the breadth of a river be ascertained by the means of the peak of a cap, or cocked hat?

Place yourself at the edge of one bank, and lower the peak of the cap, or point of the hat till the edge cut the other bank, then steady your head, by placing your hand under your chin, and turn gently around to some level spot of ground on your own side of the river, and observe where your peak or point of your hat again meets the ground; measure this distance, which will be nearly the breadth of the river. [158]

38. How do you ascertain the distance of an object by means of the tangent scale of a gun, the height of the object at the required distance being known?

Direct the line of metal of the gun on the top of the object; then raise the tangent slide till the top of it and notch on the muzzle are in line with the foot of the object, and note what length of scale is required; then, by similar triangles, as the length of the raised part of the tangent scale is to the length of the gun, so is the height of the distant object to the distance required.

39. What composition may be used for greasing the axle-trees of artillery carriages?

Hog's lard softened by working it. If this cannot be procured, tallow or other grease may be used; if hard, it should be melted with fish-oil.

40. What is the simplest method of bursting open strong gates?

Suspend a bag of gunpowder containing 50 or 60 lbs., near the middle of the gate, upon a nail or gimlet, having a small piece of port-fire inserted at the bottom, and well secured with twine.

41. What is the length of a pendulum to vibrate seconds, half and quarter seconds respectively?

Seconds, 39.1 inches; half-seconds, 9.8 inches; and quarter-seconds, 2.45 inches.

42. Give a formula for determining the length of the seconds pendulum in any latitude. [159]

$$l = \frac{1}{9.8696044} [32.1803 \text{ feet} - 0.0821 \cos. 2 \text{ lat.}]$$

43. How are the times of a single oscillation of two pendulums to each other?

As the square roots of their lengths.

44. Repeat the table of measures.

10 tenths,	1 inch.
4 inches,	1 hand.
12 inches,	1 foot.
28 inches,	1 pace.
3 feet,	1 yard.
2 yards,	1 fathom.
220 yards,	1 furlong.
1760 yards,	1 mile.

45. Repeat the table of avoirdupois weight.

47.34735 grains,	1 dram.
16 drams,	1 ounce.
16 ounces,	1 pound.
28 pounds,	1 quarter.
4 qrs. or 112 lbs.	1 cwt.
20 cwt.	1 ton.

In some of our States the ton is estimated at 2000 lbs.

46. What is the force of gravity?

It is that force of attraction exerted by the earth upon all particles of matter which tends to urge them towards its centre.

47. What is the specific gravity of a body?

The ratio of the weight of a body to that of an equal volume of some other body assumed as a standard, usually pure distilled water at a certain temperature.

48. What is the law of descent of falling bodies?

The spaces fallen through from the commencement of the descent are proportional to the squares of the times elapsed.

49. What compositions are made use of for preserving iron cannon?

1. Black lead, pulverized,	12
Red lead,	12
Litharge,	5
Lampblack,	5
Linseed Oil,	66

Boil it gently about twenty minutes,
during which time it must be
constantly stirred.

2. Umber, ground,	3.75
Gum Shellac, pulverized,	3.75
Ivory black,	3.75
Litharge,	3.75
Linseed Oil,	78
Spirits of turpentine.	7.25

The oil must be first boiled half an hour;
the mixture is then boiled 24 hours,
poured off from the sediment, and
put in jugs, corked.

3. Coal tar (of good quality),	2 gals.
Spirits of turpentine,	1 pint.

In applying lacker, the surface of the iron must be first cleaned with a scraper and a wire brush, if necessary, and the lacker applied hot, in two thin coats, with a paint brush. It is better to do it in summer. Old lacker should be removed with a scraper, or by scouring, and not by heating the guns or balls, by which the metal is injured.

About 5 gallons of lacker are required for 100 field guns and 1000 shot; about 1 quart for a sea-coast gun. Before the lacker is applied, every particle of rust is removed from the gun, and the vent cleared out.

50. How many gallons does a cubic foot contain?

7.48 gallons.

51. What is the weight of a gallon of distilled water?

161]

At the maximum density (39°.83 Fahr.), the barometer being at 30 inches, it weighs 8.33888 avoirdupois pounds, or 58373 Troy grains.

52. What are the different lengths of plummets for regulating the march of infantry?

Common time,	99	steps	in a	minute,	17.37	inches.
Quick time,	110	“	“	“	11.6	“
Double quick,	140	“	“	“	7.18	“

53. How is a plummet made?

By means of a musket ball, suspended by a silk string, upon which the required lengths are marked; the length is measured from the point of suspension to the centre of the ball.

54. Explain how to embark and disembark artillery and its stores.

1. Divide the total quantity to be transported among the vessels, and place in each vessel every thing necessary for the service required at the moment of disembarkation, so that there will be no inconvenience should other vessels be delayed.

2. If a siege is to be undertaken, place in each vessel with each piece of artillery its implements, ammunition, and the carriages necessary to transport the whole or a part; the platforms, tools, instruments, and materials for constructing batteries; skids, rollers, scantling, and plank.

3. If a particular calibre of gun is necessary for any operation, do not place all of one kind in one vessel, to avoid being entirely deprived of them by any accident.

4. Dismount the carriages, wagons, and limbers, by taking off the wheels and boxes, and, if absolutely necessary, the axle-trees. Place in the boxes the linch-pins, washers, &c., with the tools required for putting the carriage together again. Number each carriage, and mark each detached article with the number of the carriage to which it belongs.

162]

5. The contents of each box, barrel, or bundle, should be marked distinctly upon it. The boxes should be made small for the convenience of handling, and have rope handles to lift them by.

6. Place the heaviest articles below, beginning with the shot and shells (empty), then the guns, platforms, carriages, wagons, limbers, ammunition boxes, &c.; boxes of small arms and ammunition in the dryest and least exposed part of the vessel. Articles required to be disembarked first should be put in last, or so placed that they can be readily got at.

If the disembarkation is to be performed in front of the enemy, some of the field pieces should be so placed that they can be disembarked immediately, with their carriages, implements and ammunition; also the tools and materials for throwing up temporary intrenchments on landing.

7. Some vessels should be laden solely with such powder and ammunition as may not be required for the immediate service of the pieces.

8. On a smooth, sandy beach, heavy pieces, &c., may be landed by rolling them overboard as soon as the boats ground, and hauling them up with sling-carts.

APPENDIX.

RIFLE CANNON.

A *rifle* is a firearm which has spiral grooves cut into the surface of its bore, for the purpose of communicating a rotary motion to a projectile around an axis coinciding with the direction of its flight.

The object of this rotation is to increase the range of a projectile, by causing it to move through the air in the direction of its least resistance, and to correct the cause of deviation by distributing it uniformly around the line of flight.

Various plans have been tried to secure the safest and surest means of causing the projectile to follow the spiral grooves as it passes along the bore of a rifled piece. Those projectiles, which promise to be the most successful for heavy guns, may be ranged under two heads, viz:

1st. Those which have flanges or projections on them to fit into the grooves of the gun in loading.

The flanges are made of softer metal than the body of the projectile.

2d. Those which are constructed on an expanding principle.

The body is generally made of cast-iron; and the expanding portion is a band or cup of some softer metal, as *pewter*, *copper*, or *wrought iron*, which enters the bore of the piece freely when it is loaded, but which is forced into the grooves by the discharge. 164]

The grooves are of different forms, determined by the angle made by the tangent line at any point with the corresponding element of the bore. If the angles be equal at all points, the groove is said to be *uniform*. If they increase from the breech to the muzzle, the grooves are called *increasing*; if the reverse, *decreasing* grooves. The practical method of cutting grooves consists in moving a rod armed with a cutter, back and forth in the bore, and at the same time revolving it around its axis. If the velocities of translation and rotation be both uniform, the grooves will have a uniform twist; if one of the velocities be variable, the grooves will be either increasing or decreasing, depending on the relative velocities in the two directions.

Twist is the term employed by gunmakers to express the inclination of a groove at any point, and is measured by the tangent of the angle which the groove makes with the axis of the bore; and this is *always equal to the circumference of the bore divided by the length of a single revolution of the spiral measured in the direction of the axis*.

The most suitable inclination of grooves for a rifle cannon has not yet been determined experimentally; and consequently a wide diversity of twists is employed by

different experimenters.

The following table^[4] presents a synopsis of the results in case of some rifle cannon tested at Fort Monroe, Va., in 1859, by a Board composed of ordnance and artillery officers.

The following is extracted from the report of the Board: 165]

“The method of obtaining rifle motion in these different guns is of two kinds:

“1. Flanged projectiles entering into the grooves of the gun.

“2. Expanding projectiles, which are forced into the grooves by the action of the charge. Although the flanged projectile, when made with great precision, has given good results, as shown by the tables of firing, the extreme nicety in its fabrication, and the care and trouble to load the gun, particularly when it becomes foul by firing, seems to render it not as suitable for service as the expanding projectile.

“From the results obtained, the conclusion is inevitable that the era of smooth-bore field artillery has passed away, and that the period of the adoption of rifle cannon for siege and garrison service cannot be remote. The superiority of elongated projectiles, whether solid or hollow, with the rifle rotation, as regards economy of ammunition, extent of range, and uniformity and accuracy of effect, over the present system, is decided and unquestionable.”

The ARMSTRONG GUN, of which so much has been said, belongs to the class of breech-loading rifle-cannon. Its projectile is made of cast-iron, surrounded by two leaden rings, placed at the extremities of the cylindrical part, for the purpose of engaging the grooves, when it is forced through the bore. The great range and accuracy claimed for this projectile, are probably derived from its great length compared with its diameter; but a gun of great strength would be required to project it.

TARGET 40' BY 20'. 166]

NAME.	CALIBRE.	BORE.		GROOVES.			TWIST.
		Diameter.	Length.	No.	Width.	Depth.	
Sawyer,	24-pdr.	in. 5.862	in. 110	6	in. 1.5	0.25	Uniform, one turn in 34½ feet.
Dimick,	32-pdr	6.4	101	6	2.0	0.2 rectan.	Increasing from 0 to one turn in 62½ feet at muzzle; twist to the right
Dr. Reed,	12-pdr. Siege	4.854	109	7	1/14th circum.	.03 to 0.8	Increasing from 0 at commencem't to one turn in 50

NAME.	CALIBRE.	BORE.		GROOVES.			TWIST.
		Diameter.	Length.	No.	Width.	Depth.	
Do.	12-pr. Field	4.636	74	7	do.	do.	feet at muzzle.
Do.	32-pr.	6.425	110	3	$\frac{1}{6}$ th circum.	.085 to .012 circular	Do. do. Uniform, one turn in 40 feet.
Do.	6-pr.	3.69	103.4	3	do.	.077 to .111 circular	Uniform, to the right, one turn in 25 feet.
Capt. Dyer,	3-pr.	2.9	44.5	8	0.4	.05	Uniform, one turn in 16 feet.
Do.	6-pr. bronze	3.67	57.5	16	0.5	.025	Uniform, one turn in 19 feet.

LEGEND:

167]

- (A) = Weight of Projectile.
 (B) = Weight of Charge.
 (C) = No. of shots fired.
 (D) = No. of direct hits.
 (E) = No. of ricochet hits.
 (F) = Angle of Elevation.
 (G) = Time of flight.

NAME.	Weight of Gun.	(A)	(B)	1000 YARDS.				
				(C)	(D)	(E)	(F)	(G)
	lbs.	lbs.	lbs.				° ' "	"
Sawyer,	8822	45	5½	15	13	2	2	
Dimick,	9300	51	6	7	5	1	2 15	
Dr. Reed,	5000	22	3	26	14	9	2 15	
Do.	1900	15	2	48	16	3	2	3
Do.	8500	50	6	10	8	2	2 15	3
Do.	1200	12	1½	28	18	4	2 10	
Capt. Dyer,	250	9	1	28	16	5	2 25	
Do.	880	14	1½	22	11	4	2 15	

LEGEND:

- (A) = Weight of Projectile.
 (B) = Weight of Charge.
 (C) = No. of shots fired.
 (D) = No. of direct hits.

(E) = No. of ricochet hits.
(F) = Angle of Elevation.
(G) = Time of flight.
(H) = Corresponding Elevation.

NAME.	Weight of Gun.	(A)	(B)	2000 YARDS.					Average Range.	(H)	(G)
				(C)	(D)	(E)	(F)	(G)			
	lbs.	lbs.	lbs.				° ' "			° ' "	
Sawyer,	8822	45	5½	119	32	17	4 30	6	4359	13½	
Dimick,	9300	51	6	58	21	6	5	6½			
Dr. Reed,	5000	22	3	30	5	8	4 30				
Do.	1900	15	2								
Do.	8500	50	6	84	19	8	5	6½	3665	11 30	
Do.	1200	12	1½	52	9	5	4 45				
Capt. Dyer,	250	9	1	18	4	2	5½	7	3270	13 30	15
Do.	880	14	1½								

The following is a description of the several projectiles, viz:

SAWYER'S.—Flanged projectile; elongated; entire shell coated with an alloy chiefly of lead, and has a percussion cap on small end.

DIMICK'S.—Expanding shell; elongated; cup of soft metal cast on rear end of projectile.

REED'S.—The body is of cast-iron and the expanding portion is a cup of wrought iron, which is fastened to the body by inserting it in the mould and pouring the melted metal around it.

DYER'S.—Description nearly the same as that of Dimick's.

INDEX.

The pages refer to the sidenotes in the margin of the text.

AIR: Resistance of, [148](#).

AMMUNITION: For field battery, [41](#);

for seige train, [37-8](#);

storage [103-4](#);

preparation, [93-4](#), [100-1-2](#);

weights of fixed, [107](#).

ANGLE: Of greatest range, [66](#);

of fall, [74](#);

of least elevation for mortars, [67](#);

of elevation for stone-mortars, [67](#)

—Natural angle of sight, [12](#);

of guns, [26](#);

of howitzers, [29](#);

of columbiads, [30](#).

ANIMAL POWER, [152](#).

ARC, elevating, [18](#).

AREA of a circle, [154](#).

ARMSTRONG GUN, [165](#).

ARTIFICIAL LINE OF SIGHT, [53](#).

ARTILLERY: Definition, [9](#)

—Method of embarking and disembarking, [161-2](#)

—Carriages (see *carriages*)

—Kinds of, [9](#);

how distinguished, [9](#)

—Proportion of, to Infantry, [40](#)

—Proportion of different kinds in a field train, [40](#);

in seige train, [36](#);

for mountain service, [42](#);

for armament of forts, [35](#)

—How rendered unserviceable, [20-1](#).

ASTRAGAL AND FILLETS: Definition, [13](#).

ATTACK of a post, [150](#).

AVOIRDUPOIS WEIGHT, [159](#).

AXIS OF A PIECE: Definition, [12](#).

BALLS: Diameters and weights, [105-6](#);
computation of weight and diameter, [95](#);
piling, [103](#);
number in a pile, [104-5](#);
fire, [92](#);
light, [92](#);
smoke, [93](#);
penetration, [149-50](#).

BARBETTE CARRIAGE: Kinds, [132](#)

—Parts composing, [133](#)

—Description, [133-4](#).

BARRELS: For gunpowder, marking, [88](#);

piling, [88](#);

transportation, [89](#).

BATTERY: Definition of, [11](#)

—Of field artillery, composition, [41](#);

tactics, [44-5-6-7-8-9](#), [50](#)

—Mountain Artillery, [42](#)

—Ammunition for field battery, [41](#).

BATTERY-WAGON, [143-4](#).

BASE-RING: Definition, [13](#).

BASE OF THE BREECH: Definition of, [13](#).

BEDS, mortar: Weights, [33](#)

—Trunnion-beds, [129](#)

—Siege mortar, [130-1](#)

—Coehorn, [131](#)

—Eprouvette, [131](#)

—Heavy sea-coast, [131-2](#).

BOARD, Pointing, [58](#).

BORE: Definition, [14](#); bottom of, [15](#).

BORMANN FUZE, [112-13](#).

BRASS CANNON: External injury, [23](#).

BREADTH of a river ascertained, [157-8](#).

BREACHING: Battery, best position for, [38](#);
manner of, [38-9](#);
time required, [39](#).

BREECH: Definition, [13](#)

—Sights, [16](#);
how used, [53](#);
construction, [17](#);
pieces supplied with, [17](#).

BRONZE: Objections to, for cannon, [10](#);
why used for field pieces, [10](#);
kinds of bronze pieces used, [11](#).

BURNING gunpowder: Quickness of, [87](#).

BURSTING OPEN gates, [158](#).

BUSHING a piece, [20](#);
metal used for, in bronze pieces, [20](#);
object of, [20](#);
all new artillery not bushed, [20](#).

CAISSON: Description of, [142-3](#);
number with a field battery, [41](#).

CAKING of powder prevented, [89](#).

CALIBRE: Definition, [12](#);
number in a piece ascertained, [12](#).

CANISTERS, 91: For field service, how made, [94](#);
for siege and sea-coast service, how made, [95](#)
—How piled, [104](#).

CANNON: Bore, [12](#)
—Brass, external and internal injuries, [23](#)
—Dimensions, how regulated, [11](#)
—For siege train, [36](#)
—Iron, injuries, [24](#)
—preservation of, [25](#);
service of, how judged, [24](#)
—How marked, [21-2](#)
—Condemned cannon, how marked, [22](#)
—Proof of, [22](#)

- Kinds, [9](#)
- Length of, [27-8](#), [30](#), [33](#)
- Rifle cannon, [163-7](#).

CARCASSES, [92](#).

CARRIAGES: Classification, [123](#)

- Movable, [123](#);
 - field gun, [125-6](#);
 - mountain artillery, [127](#);
 - prairie, [127](#);
 - limbers, [126-9](#);
 - siege-gun, [128](#)
- Stationary, [129-30](#);
 - barbette, [132-3-4](#);
 - casemate, [138-9](#);
 - columbiad, [136-7](#);
 - flank-casemate, [140](#)
- Mortar beds, [131](#)
- Wrought iron, [141-2](#)
- Number in a field battery, [41](#).

CARTRIDGE BAGS: Where filled, [89](#).

CARTRIDGES: For hot shot, [97](#).

CASCABLE, [13](#).

CASEMATE: Carriage, [138-9](#)

- Gin, [146](#).

CHAMBER: Definition, [14](#);

- object of, [32](#);
- form for mortars, [32](#);
- for howitzers, [28](#);
- for eprouvette, [32](#);
- gomer, [32](#).

CHARGES: Definition, [60](#)

- For breeching, [60](#)
- For double shot, [60](#)
- For field guns and howitzers, [62](#)
- For fire-balls, [61](#)
- For heavy guns, columbiads and howitzers, [62](#)
- Service charge for heavy guns, [60](#)

- For hot shot, [60](#)
- Greatest charges for mortars, [62](#)
- For mortars, how regulated, [61](#)
- For ricochet firing, [60](#)
- For shells fired from columbiads and heavy guns, [108](#)
- For field shells, [108](#)
- For mortar shells, [107](#)
- For spherical case shot, [108](#).

CHASE: Definition, [13](#).

CHASSIS: For barbette carriage, [134-5](#)

- For casemate, carriage, [139-40](#)
- For columbiad, [138](#)
- For flank-casemate, [141](#)
- For wrought iron carriage, [142](#).

CHEEKS, [124](#).

COEHORN MORTAR: Diameter, [33](#)

- Weight of bed, [33](#)
- Length of, [33](#)
- Length of bore, [33](#)
- Length of chamber, [33](#)
- Use, [34](#)
- Greatest charge for, [62](#)
- Bed, description of, [131](#).

COLUMBIADS: Definition, [30](#)

- Windage, [82](#)
- Charges, [62](#)
- Chamber, [30](#)
- Peculiarities, [30](#)
- Weights, [30](#)
- Length, [30](#)
- Natural angle of sight, [30](#)
- Gun carriage, [136-7](#);
chassis, [137-8](#)
- Shells, charges for, [108](#);
method of loading, [102](#).

COMPOSITION, for preserving iron pieces, [160](#);
application, [160](#);

for axle-trees of carriages, [158](#).
CONDEMNED cannon, how marked, [22](#);
shot, how marked, [102](#).
CONTENT: Of a barrel, [155](#);
box, [155](#);
conic frustum, [154](#);
gomer chamber, [155](#);
spherical segment, [155](#);
cylinder, [155](#).

DAY'S MARCH: Of field artillery, [153](#).
DEFILADE: Definition, [156](#).
DEPTH OF PENETRATION of balls, [149-50](#).
DESCENT of falling bodies: Law of, [159](#).
DIAMETER: Of cohorn mortar, [33](#);
of eprouvette, [33](#);
of stone-mortar, [33](#);
of cast-iron shot, how found, [95](#);
of shot, shells, and spherical case, [105](#);
of vent, [16](#).
DIMENSIONS: Of cannon, how regulated, [11](#);
of a parapet to resist field artillery, [156](#).
DIPPING OF THE MUZZLE, [77-8](#).
DISCHARGES: Number an iron gun can sustain, [39](#).
DISH, of a wheel, [124](#).
DIRECTION, how given: To guns and howitzers, [51-2](#);
to mortars, [56-8](#)
—At night, [55](#), [59](#)
—When wheels are not on same level, [54](#).
DISTANCE: For firing field pieces, [46](#)
—Ascertained by sound, [151](#)
—Determined by a tangent scale, [158](#)
—Of recoil, [77](#)
—Of ricochet battery from object, [74](#).
DISPART: Definition, [13](#).
DOLPHINS: Definition, [19](#);
pieces furnished with, [19](#).

DRIVING OUT shot wedged in the bore, [21](#).

ELEVATION: Necessity for, [51](#)

—How given to guns and howitzers, [52](#);

to mortars, [56](#);

instruments for, [52](#)

—Angle of, for mortars [67](#);

greatest angle *in vacuo*, [66](#);

angle of, for ricochet fire, [74](#).

ELEVATING ARC, [18](#).

EMBARKING: Artillery and its stores, [161-2](#).

ENFILADE: Definition, [156](#).

ENFILADING a work, [73-4-5](#)

—Object to be fired at, [73](#).

EPROUVETTE, [11](#);

form of chamber, [32](#)

—Calibre, [33](#)

—Use of, [34](#)

—Bed, [33](#), [131](#)

—Length of bore, [33](#).

EXPANSION of hot shot, [96](#).

EXTERNAL injury to cannon, [23-4](#).

FACE of the piece: Definition of, [14](#).

FALL: Point of, [73](#)

—Angle of, [74](#).

FALLING BODIES, law of descent, [159](#).

FIELD ARTILLERY: Charges for, [62](#)

—Kinds, [40](#)

—Tactics, [44-9](#), [50](#).

FIELD BATTERY: Number of pieces, [40](#)

—Battery of horse-artillery, [41](#)

—Composition of, on a war establishment, [41](#)

—Composition of mountain howitzer battery, [42](#)

—Ammunition, [41](#)

—Draught horses, [42](#).

FIELD GUN, how mounted, [44](#)

—Charges for, [62](#).
FIELD CARRIAGES: Kinds of, [125](#)
—Description, [125-6](#).
FIELD AND SIEGE GIN, [146](#).
FIELD SHELLS: Loading, [100](#)
—Charges, [108](#).
FIELD-PARK, [42](#);
quantity of supplies for, [42](#);
carriages, [43](#).
FILLING: Mortar shells, [101](#);
columbiad shells, &c., [102](#).
FIRE BALLS: Definition, [92](#)
—Charges, [61](#)
—How preserved, [104](#).
FIRING: Field pieces, [46-7-8](#)
—Rapidity of, for mortars, [34](#);
for field pieces, [46](#)
—Within point-blank range, rule for, [52](#)
—At night with guns and howitzers, [55](#);
with mortars, [59](#)
—Mode of facilitating firing for any given distance, [54](#);
use of remarkable points on the ground, [55](#)
—Ricochet firing, [73](#)
—Effect of firing upwards under a large angle, [65](#).
FIXED AMMUNITION: Storing, [103-4](#)
—Weights of, [107](#).
FLIGHT OF PROJECTILES: Time of, [149](#).
FLANK-CASEMATE carriage, [140-41](#).
FOOT, number of gallons in a cubic, [160](#).
FOOT SOLDIER, space occupied by, in ranks, [153](#).
FORGE, [143](#)
—Portable, [144](#)
—Number with a field battery, [41](#);
with field-park, [43](#).
FORCES acting on a projectile, [51](#).
FORCE of gravity, [159](#).
FRICTION PRIMER: Description, [115](#);

advantages of, [116](#).
FURNACES for hot shot, [97](#).
FUZES: Definition, [109](#)
—Wooden, [109-10](#)
—Paper, [111](#)
—Bormann, [112-13](#)
—U. S. sea-coast, [114](#)
—Composition for mortar fuzes, [110](#);
for paper fuzes, [111](#).

GINS: Field and siege, [146](#);
garrison, [146](#);
casemate, [146](#).
GOMER CHAMBER, [32](#).
GRAPE SHOT, [91](#);
weight of, [107](#).
GRATES for heating shot, [98](#).
GRAVITY: Specific, [159](#)
—Force of, [159](#).
GREASE for wheels, [158](#).
GROOVES for rifle cannon, [163-4](#).
GRENADES: [91](#)
—Angle of elevation for, when thrown
from stone-mortars, [67](#).
GROMMETS, [99](#).
GUNS: Definition, [26](#)
—Lengths, [27](#)
—Weights, [27](#)
—Proof, [22](#)
—Ranges, [68-9](#), [70-1-2](#)
—Nomenclature, [13](#), [14](#), [15](#)
—Principal parts of, [26](#)
—Projectiles used with, [27](#)
—How mounted, [26](#)
—Natural angle of sight of, [26](#).
GUN METAL: Bronze, [9](#), [10](#)
—cast-iron, [9](#), [10](#).

GUNNERS' IMPLEMENTS: Level, [18](#)

—Quadrant, [18](#)

—How used, [52-3](#).

GUN-CARRIAGES: Field, [125-6](#);

siege, [128](#);

barbette, [132-4](#);

casemate, [138-9](#);

flank-casemate, [140](#);

mountain howitzer, [127](#);

wrought iron, [141](#);

prairie, [127-8](#)

—Columbiad, [136-7](#).

GUNPOWDER: Materials, [83](#)

—Proportions, [83](#)

—Manufacture, [84](#)

—Qualities of, [87](#)

—Packing, [87-8](#)

—Proving, [86](#)

—Expansive velocity and pressure, [87](#)

—Hygrometric proof, [87](#)

—Relative quickness, [87](#)

—Preservation and storage, [88-9](#)

—Transportation, [89](#).

HAND-CART, [145](#).

HAND SLING-CART, [145-6](#).

HAUSSE: Pendulum, [17](#).

HAY: Weight of, [154](#).

HORSES: Number required for a field battery, [42](#);

for siege train, [37](#)

—Power of, [152](#)

—Space occupied by, [152](#)

—Number required for siege-gun, [129](#)

—Weight, [152](#).

HORSE-ARTILLERY: Peculiar advantages of, [43](#).

HOT SHOT, [97-8](#)

—Loading with, [97](#)

—Expansion of, [96](#).
HOWITZERS: Definition, [28](#)
—Kinds of, and weights, [29](#);
—Lengths, [28-9](#)
—Number in field battery, [41](#);
 in siege train, [36](#)
—Chamber, form of, [28](#)
—Advantages of, [28](#)
—Projectiles used with, [28](#)
—Natural angle of sight of, [29](#)
—Charges for, [62](#)
—Pointing, [51](#).

ICE: Strength of, [156](#).

IMPLEMENTS: Quadrant, [18](#);
 breech-sight, [16](#);
 pendulum hausse, [17](#);
 gunner's perpendicular, [18](#);
 pointing stakes, [57-8](#);
 pointing-wires, [56](#);
 plummet, [18](#), [58](#);
 pointing-cord, [57](#).

INCENDIARY COMPOSITION, [117](#).

INJURIES to cannon, [23](#).

IRON preferred to bronze, [10](#).

IRON CANNON used in land service, [10](#), [11](#).

JUNK-WADS, [99](#).

KNOB of cascable, [13](#);
 use of, [19](#).

LACQUER, for iron guns, [160](#).

LENGTH of cannon: Definition of, [11](#)

—Extreme length, [12](#).

LINE: Of fire, [64](#)

—Of metal, [12](#);

 how directed, [52-3](#);

not permanent, [54](#)
—Artificial line of sight, [53](#).
LIGHT BALLS, [92](#).
LIMBERS: For field carriages, [126](#);
for siege-carriages, [129](#).
LOADING: With hot shot, [97](#)
—Field shells, [100](#)
—Spherical case, [101-2](#)
—Mortar shells, [101-2](#)
—Shells for columbiads
and other heavy guns, [102](#).

MAGAZINES: Moisture of, how absorbed, [89](#);
powder stored in, [88](#);
precautions to be observed when open, [89](#).
MARKING: Cannon, [21-2](#);
condemned shot and shell, [102](#);
powder barrels, [88](#).
MARCHES: Horse-artillery, field-artillery,
cavalry, and infantry, [153](#).
MATCH: Quick, [116](#); slow, [116](#).
METALS for artillery, [9](#).
MOMENTUM, [151](#).
MORTAR WAGON, [145](#).
MORTARS, [31](#)
—Advantages of, [31](#)
—Lengths and weights, [33](#)
—Kinds, [11](#)
—Beds, weights of, [33](#)
—Platform, [121](#)
—Form of chamber, [32](#)
—Length of chamber and of bore, [33](#)
—Kinds of projectiles used with, [34](#)
—Rapidity of fire of siege mortars, [34](#)
—Pointing, [56](#), [58](#);
Greatest charges for, [62](#)
—Angles of elevation for, [66-7](#)

- Siege mortar beds, [130-1](#)
- Coehorn mortar bed, [131](#)
- Eprouvette bed, [131](#)
- Sea-coast mortar bed, [132](#).

MOUNTAIN artillery: Dimension and weights of, [29](#)

- Composition of a battery, [42](#)
- Ranges, [69](#).

MULES: Strength of, [152](#).

MUZZLE: Definition, [15](#)

- Sight, [17](#).

NATURAL ANGLE of sight, [12](#).

NECK: Definition, [14](#).

NOMENCLATURE of a piece, [13](#), [14](#), [15](#).

NIGHT firing: With guns and howitzers, [55](#);
with mortars, [59](#).

OATS: Weight of, [154](#).

PACK horses, [152](#).

PENDULUM HAUSSE, [17](#).

PENDULUMS: Length of, [158](#).

PENETRATION of balls: In masonry, [150](#);
in earth, [150](#).

PERPENDICULAR, gunner's, [18](#).

PILING: Balls, [103](#)

- Canisters, [104](#)
- Loaded shells, [104](#)
- Powder-barrels, [88](#)
- Number of shot in a pile, [104-5](#).

PLATFORMS, [118](#)

- Siege, [119-20](#)
- Mortar, [121](#)
- Rail, [121](#)
- Ricochet, [122](#).

PLUMMET: For mortar service, [18](#), [58](#)

- For regulating march of infantry, [161](#).

POINT-BLANK RANGE, [63-4](#)

—Causes which vary it, [64](#)

—Effect on it of firing upwards under a large angle, [65](#).

POINTING: Guns and howitzers, [51](#)

—Mortars, [56](#)

—Stakes, [57-8](#)

—Wires, [56](#)

—Cord, [57](#)

—Board, [58](#).

PORT FIRES, [114](#)

—Composition for, [115](#).

POINT of fall, [73-4](#).

PRAIRIE CARRIAGE, [127-8](#).

PREPONDERANCE: Definition, [19](#)

—Why given, [19](#).

PRIMERS: Friction, [115](#).

PRIMING-TUBES, [115](#).

PRESERVATION: Of cannon, [25](#)

—Fixed ammunition, [103-4](#)

—Balls, [102](#)

—Grape and canister, [103](#)

—Fire balls, [104](#).

PROJECTILES: Solid shot, [90](#)

—Shell, [90](#)

—Spherical case, [90](#)

—Canister, [91](#)

—Grape, [91](#)

—Grenades, [91](#)

—Carcasses, [92](#)

—Fire-balls, [92](#)

—Light balls, [92](#)

—Smoke balls, [93](#)

—Hot shot, [97-8](#)

—Forces acting on, when fired from a piece, [51](#)

—Kind used with field pieces, and distance at which they should be employed, [46](#).

QUADRANT, gunners: How used, [52-3](#).

QUARTER-SIGHTS, [16](#).

QUICK-MATCH, [116](#)

—How set fire to, [117](#).

RANGES: Definition, [63](#)

—Point-blank, [63](#)

—British point-blank, [63](#)

—Causes which vary point-blank, [64](#)

—Extreme range, [66](#)

—Angle of greatest range *in vacuo*, [66](#)

—Tables of, [68-9](#), [70-1-2](#)

—How ascertained, [149](#).

RATE OF MARCH of horse-artillery, cavalry, and infantry, [153](#).

RECOIL: Definition, [77](#)

—Cause of, [77](#)

—Amount, [77](#)

—Has no appreciable effect on flight of projectile, [78](#)

—Influence of position of axis of trunnions on, [78-9](#).

REINFORCE, [13](#)

—Band, [13](#).

RESISTANCE OF AIR to projectiles, [147-8](#).

RICOCHET: Definition, [73](#)

—Object of, [73](#)

—How conducted, [74-5](#)

—Advantages of, [73](#)

—Nature of, [75](#)

—Charges for a *flattened* ricochet, [76](#);
for *curvated*, [76](#)

—Tables of ricochet firing, [76](#)

—Pieces best adapted for, [75](#)

—Distance from object of ricochet battery, [74](#)

—Greatest angle of elevation for ricochet firing, [74](#).

RIFLE-CANNON: Experiments at Fort Monroe, [166-7](#)

—Armstrong gun, [165](#).

RIMBASES, [14](#).

RING WADS, [94](#).

RIVERS: Breadth, [156-7](#).

ROPES: size and strength of, [156](#).

SABOTS: Difference in, for field service, [93](#)

—Arrangement for field guns and 12-pdr. field howitzer, [93](#);
in 24 and 32-pdr. field howitzers, [93](#)

—Mode of fastening sabots to projectiles for field-service, [93-](#)

[4](#);

for heavy shells, [94](#);

for canisters, [94-5](#);

for grape shot, [95](#).

SEA-COAST PIECES, how mounted, [35](#)

—Number and kind required for sea-board forts, [35](#)

—Heavy sea-coast mortar-bed, [131-2](#).

SCALING a piece, [21](#).

SHOT: Solid, [90](#)

—Hollow, [90](#)

—Rule for finding weights and diameters of cast-iron shot, [95](#)

—Condemned shot, how marked, [102](#)

—Piling, [103-4](#)

—Preservation, [102-3](#)

—Forces acting on a shot, [51](#)

—Penetration, [149-50](#)

—Time required to heat, [97-8](#)

—Expansion of by heat, [96](#)

—Ranges of, [68-72](#)

—Method of driving out shot wedged in the bore, [21](#)

—Velocity of, [147](#).

SHELLS, [90](#);

—Dimensions and weights, [106-7](#)

—Mode of computing weight of, [95](#)

—Quantity of powder to fill, [96](#)

—Strapping, [93](#)

—Loading, [100-1-2](#)

—Ranges, [68-72](#)

—Condemned, how marked, [102](#)

—Velocity, [147](#).

SIGHTS of a piece: Definition, [12](#)
—How determined, [12](#)
—Quarter, [16](#).

SIEGE ARTILLERY: Kinds, [36](#)
—Proportions in a siege train, [36](#);
 of carriages, [36-7](#);
 draught horses, [37](#);
 projectiles and ammunition, [37-8](#)
—Siege mortar-beds, [130-1](#).

SLING-CART: Hand, [146](#).

SLOW MATCH, [116](#).

SMOKE BALLS, [93](#).

SOUND: Velocity, [151](#)
—Distance determined by, [151](#).

SPECIFIC GRAVITY, [159](#).

SPHERICAL CASE, [90](#)
—Loading, [100-1](#).

SPIKING cannon, [20](#).

STAKES, pointing: How planted, [57-8](#).

STONE MORTAR: Length, [33](#)
—Weight, [33](#)
—Calibre, [33](#)
—Length of bore, [33](#)
—Length of chamber, [33](#)
—Use of, [34](#)
—Stones, how disposed, [34](#).

STORING of fixed ammunition, [103-4](#).

STRAPPING SHOT and shells, [93-4](#).

STRENGTH: Of ice, [156](#)
—Of rope, [156](#).

SWELL of the muzzle, [14](#).

TABLES: Of charges, [62](#)
—Of ranges, [68-9](#), [70-1-2](#)
—Of windage, [81-2](#)
—Of weights of projectiles, [106-7](#)
—Of measures, [159](#)

—Of avoirdupois weight, [159](#).
TACTICS of field artillery, [44-5-6-7-8-9-50](#).
TANGENT SCALE, [16](#).
TIME OF FLIGHT for siege-mortars, [67](#)
—How found, [149](#).
TRAJECTORY, [64](#).
TRANSPORTATION: Of artillery by sea, [161-2](#)
—Of siege-guns, [129](#).
TRAVERSE circles, [135](#).
TRUNNIONS: Definition, [14](#)
—Use, [19](#)
—Position in mortars, [31](#)
—Beds, [129](#).
TRUE WINDAGE: Definition, [15](#).

UNSPIKING cannon, [21](#).

VALENCIENNES composition, [117](#).
VELOCITY: Of balls, [147](#)
—Loss of, by resistance of air, [147](#)
—Of sound, [151](#)
—Loss of, by windage, [81](#).
VENT: Definition, [15](#)
—Position and diameter of, [16](#).
VERTICAL FIRE, [31](#).

WADS: Grommet, [99](#)
—Junk, [99](#)
—Hay, for firing hot shot, [98](#).
WATER: Weight of, [161](#)
—Allowance for a man and a horse, [154](#).
WEIGHTS: Guns, [27](#)
—Columbiads, [30](#)
—Howitzers, [29](#)
—Mortars, [33](#)
—Projectiles, [106-7](#)
—Of wheels for field carriages, [144](#);

- and for siege-carriages, [144](#)
- Proportion between weights of shot, [95](#)
- Of cast-iron shot or shell, how determined, [95](#)
- Quantity of powder to fill a shell, how found, [96](#)
- Carried by horses, [152](#)
- Carried by an infantry soldier, [154](#).
- WHEELS: Field-carriage, size and weight, [144](#)
- Siege carriage, size and weight, [144](#)
- Parts of, [124](#).
- WINDAGE: Definition, [80](#)
- Amount, [81-2](#)
- Loss of velocity by, [81](#)
- Advantage of a reduction of, [81](#).
- WIRES, pointing, [56](#).

Footnotes:

[1] The new columbiad is made without a chamber.

[2] A couple of notches or indentations are made on the ends of the eccentric axles. When these notches are in a vertical line, the wheels rest on the *rails*; but when they are in a horizontal or inclined line, the ties rest on *them*.

[3] The 2d method was suggested to me by Captain Vogdes, 1st Artillery, U. S. Army.

[4] See Table, pages 126, 127.

Transcriber's Notes:

Antiquated spellings and words were not corrected.

The illustrations have been moved so that they do not break up paragraphs and so that they are next to the text they illustrate.

Typographical and punctuation errors have been silently corrected.

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